What a Solution Remembers About Its Equation: Groups Are Factors, Terms Interact, and Where Exactness Ends

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Abstract. Given one exact solution $u(t,x)$ of an unknown linear equation with constant coefficients, what does u remember about the equation? We prove that the right unit is not the additive term but the irreducible *factor* of the symbol. (i) u determines its operator P exactly when no partial quotient $(P/P_i)(\partial)u$ is an exponential-polynomial; no finite amount of data can certify this, and a black-box test is impossible, yet it becomes decidable under a modest extra hypothesis. (ii) On a simply connected region u splits into one component per factor of the symbol, uniquely up to an exponential-polynomial overlap of known finite dimension; a split per additive term is in general impossible; the split is stable exactly when the sum of the factors' solution spaces is closed, which for two factors means a positive Friedrichs angle. (iii) Inside a factor, for evolution equations $u_t = \sum_n A_n u$ with commuting terms, the part of the solution due to the interplay of the terms is $\prod_n(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_n \Delta_n$ applied to the datum; it costs $O(N)$ per value but $\Theta(2^N)$ as a table, and it is not continuous in Sobolev norms. (iv) Nonlinear equations such as Burgers admit analogous statements, with sharp asymptotics across a shock. Every statement carries a verification status. Each concept is illustrated by a worked example, and a primer for readers new to ideals and companion matrices is included.

Key words. differential operators, annihilator ideal, factorization, exponential-polynomials, interaction term, Friedrichs angle, Burgers equation

MSC codes. 35E20, 35A25, 13P25, 35Q53

1 Introduction

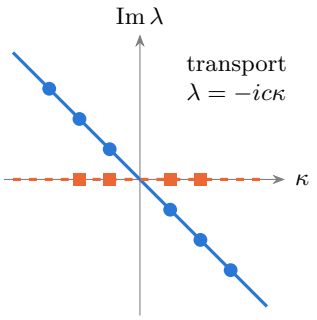
Consider a quantity that is carried by a flow and diffuses at the same time. A natural model is

$$(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)\,u = 0. \tag{1}$$

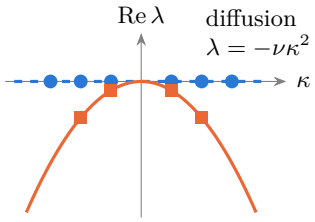
Its solutions are sums of a travelling profile and a diffusing one. Suppose you are handed one such solution $u(t,x)$, known exactly on a region of the plane, and are not told the equation. Three questions arise, and this paper answers all three.

- **Q1. Identification.** Can the operator be reconstructed from u, and when is it unique?
- **Q2. Decomposition.** If the operator is built from pieces, can u be split into the parts each piece generates? For (1): is there a travelling part and a diffusing part, and what are they?
- **Q3. Interaction.** If the pieces are the *terms* of the operator (advection, diffusion, ...), what part of u is due to no single term but to their interplay, and can it be computed exactly and cheaply?

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(a) oscillation



(b) damping

Figure 1: The idea of the paper on the equation $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)u = 0$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor of the symbol $(\tau + c\xi)(\tau - \nu\xi^2)$: circles on the line of the transport factor, squares on the parabola of the diffusion factor (schematic). Identifying the equation means finding these two groups.

The setting is deliberately analytic: no noise, no statistical model. The aim is an *exact* statement and a precise account of what is impossible.

The idea in one paragraph In Fourier variables a constant-coefficient operator is a polynomial in (τ, ξ): that of (1) is $(\tau + c\,\xi)(\tau - \nu\,\xi^2)$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor: the modes of the first factor on the line $\lambda = -ic\kappa$, those of the second on the parabola $\lambda = -\nu\kappa^2$ (Fig. 1). The modes therefore fall into groups, and the groups are the *factors*, not the terms $(\partial_t^2,\ c\nu\,\partial_x^3,\ \dots)$ of the expanded operator. Everything that follows makes this precise.

How to read the status labels Each statement carries one of: [proved] (complete proof, independently reviewed), [proved mod. X] (complete except for a cited published theorem X whose source was checked), [conditional] (depends on an unverified lemma), [open], or [false] (a counterexample exists). The reviewers were independent AI agents, not human referees; fifteen core algebraic theorems were additionally machine-checked in Lean 4 (branch directory `proofs/lean`). Grey tags such as (ledger C3-T1) refer to entries of the claim ledger kept with the proofs.

Results at a glance Table 1 lists the main statements; Fig. 2 shows how they depend on each other. Section 2 builds the algebra, Section 3 proves the grouping principle and its stability, Section 4 treats the interaction term, Section 5 asks when a solution determines its equation, Section 6 moves to nonlinear equations, and Section 26 collects the limits. Proofs are in Appendix A; Appendix E is a primer.

2 Derivatives become polynomials

In brief. Before asking what a solution knows, we turn operators into polynomials and solutions into objects that carry an ideal of equations; the one-variable case shows every idea in miniature.

Table 1: Main results in the order in which they are proved. "Proved" means a complete proof that survived an independent review.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P) \iff (P/P_i)(\partial)u \notin \mathrm{EP}$ for every irreducible $P_i \mid P$ (Theorem 5.1)	[proved]
Data limits	no finite certificate; black box impossible; decidable with a bound or a holonomic model (Proposition 5.5)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, overlap of dimension $\delta(P)$ (Theorem 3.1)	[proved]
Terms vs. factors	a split per additive term is false in general (Proposition 3.3)	[proved]
Stability	the split is stable iff $\gamma_N > 0$; for two factors, positive Friedrichs angle (Theorem 3.4)	[proved]
Interaction term	$E_{\mathrm{int}} = \prod(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum \Delta_n$; $O(N)$ value, $\Theta(2^N)$ table (Theorem 4.1)	[proved]
Instability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m (Proposition 4.2)	[proved]
Nonlinear	Burgers: closed form before the shock, sharp behaviour across it (Section 6)	[proved]/ [proved mod. Kružkov]

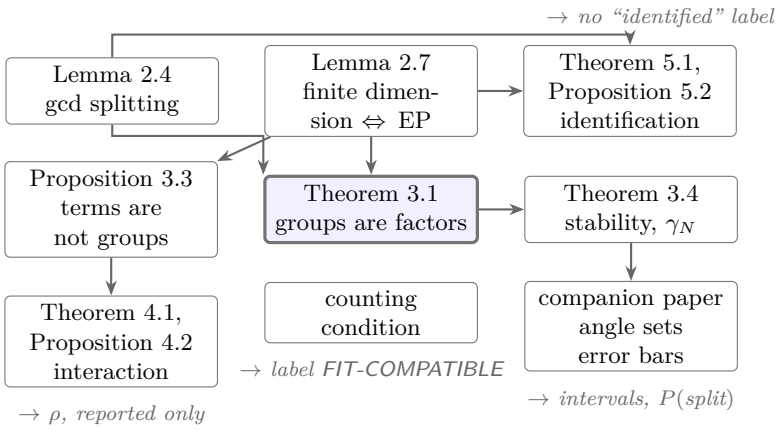


Figure 2: How the results of Section 2 depend on each other (arrows) and where the algorithm uses them (italic). The shaded result is the grouping principle; everything else either prepares it or measures how well it can be carried out on finite data.

2.1 Derivatives become polynomials

Throughout, $\Omega \subset \mathbb{R}^2$ is a domain in the variables (t, x) and every operator has constant coefficients,

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials $\psi_{\tau,\xi}(t, x) = e^{\tau t + \xi x}$, $(\tau, \xi) \in \mathbb{C}^2$, are common eigenfunctions of all such operators: $\partial_t \psi_{\tau,\xi} = \tau \psi_{\tau,\xi}$, $\partial_x \psi_{\tau,\xi} = \xi \psi_{\tau,\xi}$ and therefore

$$P(\partial)\, \psi_{\tau,\xi} = P(\tau, \xi)\, \psi_{\tau,\xi}. \tag{2}$$

Differentiation is thus replaced by multiplication, and operators are replaced by polynomials.

Definition 2.1 (Operator ring, symbol, characteristic variety)**.** The constant-coefficient operators form the commutative ring $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \cong \mathbb{C}[\tau, \xi]$. The polynomial $P(\tau, \xi)$ is the *symbol* of $P(\partial)$, and its zero set $V(P) = \{(\tau, \xi) \in \mathbb{C}^2 : P(\tau, \xi) = 0\}$ is the *characteristic variety*.

By (2), $\psi_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when $(\tau, \xi) \in V(P)$. A Fourier mode in space is the case $\xi = i\kappa$ with wavenumber $\kappa \in \mathbb{R}$; then $P(\tau, i\kappa) = 0$ is the *dispersion relation*, and the roots $\tau = \lambda(\kappa)$ are the temporal rates of the mode $e^{\lambda(\kappa)t + i\kappa x}$.

Convention We keep the algebraic variable ξ in statements about the ring and write $\xi = i\kappa$ only when we evaluate on Fourier modes. With this convention a symbol has real coefficients exactly when the equation is real, and no factor i ever appears inside $\mathcal{R}$.

Example 2.2 (Transport plus diffusion: the running example). The operator $P = (\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$, with speed $c > 0$ and diffusivity $\nu > 0$, has symbol

$$P(\tau, \xi) = Q_1 Q_2, \qquad Q_1 = \tau + c\,\xi, \qquad Q_2 = \tau - \nu\,\xi^2.$$

On Fourier modes the two factors give two dispersion branches, $\lambda_1(\kappa) = -ic\kappa$ (a wave that travels without decay) and $\lambda_2(\kappa) = -\nu\kappa^2$ (a standing mode that decays). A solution is any superposition of modes on the two branches. The equation is one object; its *two factors* are two groups of modes. Recovering those groups from data is the subject of this paper. We use $c = 1$, $\nu = 0.3$ in all numerical illustrations.

2.2 What a solution knows: the annihilator

A single function satisfies many equations: e^x is annihilated by $\partial_x - 1$, by $\partial_x^2 - 1$ and by every multiple of $\partial_x - 1$.

Definition 2.3 (Annihilator)**.** For a distribution u on Ω, $\mathrm{Ann}(u) = \{Q \in \mathcal{R} :\ Q(\partial)u = 0\}$.

$\mathrm{Ann}(u)$ is an ideal: it is closed under sums and under multiplication by any operator, because $(SQ)(\partial)u = S(\partial)\big(Q(\partial)u\big) = 0$. It collects every constant-coefficient equation that u satisfies. We use three classical facts about $\mathbb{C}[\tau, \xi]$ [2]: unique factorization into irreducible polynomials; Bézout's theorem (two polynomials without a common factor have finitely many common zeros); and Hilbert's Nullstellensatz (if q vanishes on the zero set $V(I)$ of an ideal I, then $q^k \in I$ for some k). All three require the complex field, which is why symbols are always considered over $\mathbb{C}$.

Lemma 2.4 (Splitting off the greatest common divisor)**.** *Every nonzero ideal $I \subset \mathcal{R}$ factors as $I = g \cdot J$, where $g = \gcd(I)$ and J is an ideal with finite zero set $V(J)$.*

The proof (Appendix A) is elementary: divide by g, then use Bézout on two elements of J without a common factor. Applied to $I = \operatorname{Ann}(u)$ the lemma gives the decomposition

$$\operatorname{Ann}(u) = G_u\, J_u, \qquad G_u = \gcd \operatorname{Ann}(u), \qquad \Sigma_{\mathrm{p}}(u) = V(J_u) \text{ finite.} \tag{3}$$

The polynomial G_u is the part of the equation that u fills out along whole curves of $\mathbb{C}^2$; the finite set $\Sigma_{\mathrm{p}}(u)$, the *point spectrum* of u, lists the isolated exponentials that u happens to contain. We call G_u the *intrinsic operator* of u: it depends on u alone, whereas an annihilator P supplied from outside may contain spurious factors.

Reading the two pieces The two objects in (3) answer different questions. The polynomial G_u is a differential equation: its zero set $V(G_u)$ is a union of curves in $\mathbb{C}^2$, and each curve is a continuous family of exponential modes $e^{\tau t+\xi x}$, all solving $G_u(\partial)v = 0$. The point spectrum $\Sigma_{\mathrm{p}}(u)$ is a list: the finitely many individual exponential modes that are present in this particular function u and are not accounted for by G_u. In short, G_u identifies an equation that describes a whole family of modes; $\Sigma_{\mathrm{p}}(u)$ identifies the isolated modes that u happens to contain. Example 2.5 shows the two extreme cases.

Example 2.5 (Two extremes). For $u = e^x$, $\operatorname{Ann}(u) = (\tau,\ \xi - 1)$ is the ideal of the single point $(0, 1)$; hence $G_u = 1$ and $\Sigma_{\mathrm{p}}(u) = \{(0, 1)\}$: the function knows one exponential and no equation. For $u = f(x - ct)$ with a profile f that is not a finite sum of exponentials times polynomials, $\operatorname{Ann}(u) = (\tau + c\xi)$; hence $G_u = \tau + c\xi$ and $\Sigma_{\mathrm{p}}(u) = \emptyset$: the function knows its equation completely (Theorem 5.1).

2.3 Exponential-polynomials: the solutions that remember too little

Definition 2.6 (Exponential-polynomials)**.** EP is the set of finite sums $\sum_j p_j(t, x)\, e^{\tau_j t+\xi_j x}$ with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. $\mathrm{EP}(\Omega)$ denotes their restrictions to Ω.

For a function w let $\mathcal{R}w = \operatorname{span}\{\partial_t^j \partial_x^k w :\ j, k \geq 0\}$ be the space of all its derivatives. The map $Q \mapsto Q(\partial)w$ is onto $\mathcal{R}w$ with kernel $\operatorname{Ann}(w)$, so $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$ as vector spaces.

Lemma 2.7 (Finite-dimension criterion)**.** *Let Ω be connected. Then $w \in \mathrm{EP}(\Omega)$ if and only if $\dim_{\mathbb{C}} \mathcal{R}w < \infty$. In that case $V(\operatorname{Ann} w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.*

Proof idea. We show that w is an exponential-polynomial if and only if its derivatives span a finite-dimensional space. ($\Rightarrow$) For one term $w = p(t, x)\, e^{at+bx}$,

$$\partial_t w = (p_t + a\, p)\, e^{at+bx}, \qquad \partial_x w = (p_x + b\, p)\, e^{at+bx}.$$

Differentiation never raises the degree of p, so every derivative of w lies in the finite-dimensional space of polynomials of degree at most $\deg p$ times e^{at+bx}; a finite sum of such terms stays in a finite sum of such spaces. ($\Leftarrow$) If $\mathcal{R}w$ is finite-dimensional, ∂_t and ∂_x map it to itself and act on it as two commuting matrices A and B. Their minimal polynomials m_A and m_B give two constant-coefficient differential equations,

$$m_A(\partial_t)\, w = 0, \qquad m_B(\partial_x)\, w = 0,$$

one in each variable. On every rectangle inside Ω their common solutions are exponential-polynomials, and connectedness extends this representation to all of Ω. The joint eigenvalues (a, b) of A and B are the exponents that occur in w. Details are in Appendix A. □

In brief. Deciding whether $w \in \mathrm{EP}$ is a finite-dimensional linear-algebra question: compute the rank of the derivatives of w. The joint eigenvalues of the two matrices are the exponents, that is, the points of $\mathbb{C}^2$ at which w lives. This is the computational core of everything that follows: the matrix pencil of the companion algorithm paper is exactly this eigenvalue computation, done for ∂_t one wavenumber at a time.

Example 2.8 (Rich and poor solutions of the heat equation). A single heat mode $e^{-\nu\kappa^2 t + i\kappa x}$ and any finite sum of such modes belong to EP. The heat kernel $t^{-1/2}e^{-x^2/(4\nu t)}$ does not: its x-derivatives are polynomials of increasing degree times the same Gaussian, so they span an infinite-dimensional space.

Starting from: one variable x, an ODE $p(D)u = 0$ with constant coefficients, $D = d/dx$, p known. *Why:* in one variable every object of the later sections (minimal annihilator, spectral projectors, companion matrix) appears with explicit formulas and no geometry of curves. *Role:* prototype for Section 5.1 (identification) and Section 3 (the Lagrange projectors reappear as Eq. (11)).

This section introduces, in the simplest setting, objects that reappear in Sections 5.1 and 3: the minimal annihilator, the companion matrix and the spectral projectors.

Remark 2.9 (Setting and symbols).
- We consider an ODE $p(D)u = 0$ in one variable x, with $D = d/dx$ (in this section only; elsewhere D is a diffusion constant).
- The polynomial $p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j}$ has distinct roots $\lambda_j \in \mathbb{C}$ with multiplicities $m_j \geq 1$, so that $m = \deg p = \sum_j m_j$ is the *order* of the equation (not to be confused with the Sobolev index m of Section 3).
- The set of solutions is the *kernel* $\ker p(D)$, and

$$\ker p(D) = \Big\{ \sum_j q_j(x) e^{\lambda_j x} : \ \deg q_j < m_j \Big\},$$

where each q_j is a polynomial of degree $< m_j$.

Remark 2.10 (Why this form). For one root, $(D - \lambda)\big(q\, e^{\lambda x}\big) = q' e^{\lambda x}$: applying $D - \lambda$ lowers the degree of q by one. Hence $(D - \lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $q^{(m_j)} = 0$, i.e. $\deg q < m_j$.

Each root therefore contributes the m_j functions $x^k e^{\lambda_j x}$ $(0 \leq k < m_j)$: these are the natural basis functions of the equation, finitely many, and together they form a basis of $\ker p(D)$, a space of dimension m.

Remark 2.11 (Example). $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$ has $\lambda_1 = 1$, $m_1 = 2$, $\lambda_2 = -2$, $m_2 = 3$, order $m = 5$. A basis of the solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2 e^{-2x},$$

so every solution is

$$c_1 e^x + c_2 x e^x + c_3 e^{-2x} + c_4 x e^{-2x} + c_5 x^2 e^{-2x}.$$

2.4 Identification by the Wronskian

Proposition 2.12 ((ODE identification)). *Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of independent basis functions actually present in u; it equals p if and only if $u, u', \dots, u^{(m-1)}$ are linearly independent (nonvanishing Wronskian), i.e. iff every root of p appears in u with its full multiplicity.*

Reason. $\dim \operatorname{span}\{u, u', u'', \dots\}$ is the degree of the minimal annihilator (Lemma 2.7 in one variable); a missing root lowers it.

2.5 The companion matrix and spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ (monic) and put $X = (u, u', \dots, u^{(m-1)})^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC} X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the matrix exponential has the spectral form

$$\begin{aligned} e^{xC} &= \sum_{j=1}^{m} e^{\lambda_j x}\, \Pi_j, \\ \Pi_j &= \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \end{aligned} \tag{4}$$

where the Π_j are the Lagrange projectors onto the eigenspaces. The last row of C carries the whole equation; C is a *matrix over the base field* for an ODE, but, as soon as the coefficients depend on x, or there is a second variable, its last row has entries that are operators and the object becomes a matrix over a noncommutative or polynomial ring. The invariant statement is that the whole equation corresponds to the cyclic module $\mathcal{R}/\operatorname{Ann}(u)$ (Section 2); the companion matrix is a way to compute with it relative to one chosen direction.

Proposition 2.13 ((Projectors of the companion matrix) (ledger II-T2) [proved]). *The projectors Π_j are explicit rational functions of the coefficients, with poles exactly where roots collide; they are polynomial in the coefficients if and only if the corresponding varieties are disjoint. They are* not *orthogonal in general: the companion matrix is not normal for $m = 2$ [false] (the contrary was an early conjecture).*

Hence Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on the coefficient vector.

2.6 Tensor construction versus module construction

One may encode a PDE in a higher-order tensor of derivatives, $J = \partial_{x_{i_1}} \otimes \cdots \otimes \partial_{x_{i_n}}$, or by the companion form. The comparison, (ledger Gate 3): the *principal symbol tensor* $\sigma_m(P) \in \operatorname{Sym}^m(\mathbb{C}^2)$ only sees the highest-order part of P (for the heat operator it sees ξ^2 alone), so it is an invariant "at infinity"; the *cyclic module* $\mathcal{R}/\operatorname{Ann}(u)$ is the complete invariant. All subsequent results use the module.

Remark 2.14 (Takeaway). This section shows how an ODE's polynomial encodes its solution space. Its roots determine the exponential rates, their multiplicities determine the polynomial prefactors, the companion matrix represents differentiation, and spectral projectors isolate the different modes. The later sections generalize these ideas to multiple commuting derivatives, where the same structure survives but requires more algebraic machinery: the minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Section 5.1) and the Lagrange projectors reappear as Eq. (11) (Section 3).

3 Groups are factors, not terms

In brief. The factor decomposition, its stability measured by an angle, and the regularity of the components.

3.1 Groups are factors, not terms

Suppose the operator is known and factored as

$$P = P_1^{e_1} \cdots P_N^{e_N}, \qquad Q_j = P_j^{e_j} \text{ pairwise coprime.} \tag{5}$$

Two ways of splitting a solution suggest themselves: one piece per *factor* Q_j, or one piece per additive *term* of the operator (for $u_t = \nu u_{xx} - c u_x$, one piece for diffusion and one for transport). Only the first works.

Theorem 3.1 (Factor decomposition)**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and let P be as in* (5)*. Every $u \in \ker P(\partial)$ (in $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$) can be written*

$$u = \sum_{j=1}^{N} u_j, \qquad Q_j(\partial)\, u_j = 0. \tag{6}$$

The tuples $(u_1, \ldots, u_N)$ that sum to zero form a space of exponential-polynomials of dimension exactly

$$\delta(P) = \sum_{i<j} \dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j), \tag{7}$$

the number of intersection points of the varieties $V(Q_i)$, counted pair by pair and with multiplicity.

The number $\dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j)$ therefore has two readings at once. Geometrically it counts the points where the curves $V(Q_i)$ and $V(Q_j)$ meet, with multiplicity. Analytically it is the dimension of the space of exponential-polynomials that solve both Q_i and Q_j, one exponential for each meeting point (times polynomials at a point of higher multiplicity). For two groups these are exactly the functions that can be moved from one group to the other without changing u: each meeting point of the two curves is one shared mode. For three or more groups $\delta(P)$ still gives the dimension of the ambiguity, but the ambiguous tuples need not be sums of modes shared by pairs.

In one variable the statement is Bézout's identity: if $aQ_1 + bQ_2 = 1$, then $u = bQ_2 u + aQ_1 u$ splits u into a solution of Q_1 and a solution of Q_2. In two variables "coprime" no longer gives $1 \in (Q_1, Q_2)$; it only gives finitely many common zeros, and the finite-dimensional quotient

$\mathcal{R}/(Q_1, Q_2)$ measures exactly how much the two groups overlap. The proof combines the exact sequence $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ with the vanishing of the obstruction $\mathrm{Ext}^1(M, C^\infty(\Omega))$ for the finite-dimensional module $M = \mathcal{R}/(Q_1, Q_2)$, which reduces to $H^1_{dR}(\Omega) = 0$ (compare the fundamental principle of Ehrenpreis and Palamodov [6, 18]); see Appendix A.

Example 3.2 (Running example: two groups with a two-dimensional overlap). For $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$ the common zeros are $(0, 0)$ and $(c^2/\nu, -c/\nu)$, both simple, so $\delta(P) = 2$. The ambiguity is spanned by the constant function and by $e^{(c^2/\nu)t-(c/\nu)x}$: a constant (or that growing exponential) can be moved from one group to the other without changing u. Among Fourier modes ($\xi = i\kappa$, $\kappa \in \mathbb{R}$) only the first point is visible: the $\kappa = 0$ mode belongs to both groups.

Projectors from the eigenvalues When each factor is an evolution factor, $Q_j = \tau - \Lambda_j(\xi)$ with pairwise distinct branches Λ_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = \Lambda_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - \Lambda_k(\partial_x))$ to $u = \sum_l u_l$ removes every u_l with $l \neq j$ and multiplies u_j by $\prod_{k\neq j}(\Lambda_j - \Lambda_k)(\partial_x)$. On a spatial Fourier mode κ at which no two branches meet this factor is a nonzero number, hence

$$\hat{u}_j = \prod_{k\neq j} \frac{\partial_t - \Lambda_k(i\kappa)}{\Lambda_j(i\kappa) - \Lambda_k(i\kappa)}\, \hat{u}(t, \kappa). \tag{8}$$

This is a multiplier, rational in κ, with poles exactly where two branches meet ($\Lambda_j(i\kappa) = \Lambda_k(i\kappa)$), that is, at the visible part of the overlap counted by $\delta(P)$. Formula (8) is the Lagrange interpolation formula for the spectral projectors of a matrix with distinct eigenvalues, applied to ∂_t restricted to one wavenumber: the *eigenvalues decide the groups*. For the running example, $\hat{u}_1(t, \kappa) = (\partial_t + \nu\kappa^2)\hat{u}/(\nu\kappa^2 - ic\kappa)$, which is singular only at $\kappa = 0$.

Why additive terms do not form groups

Proposition 3.3 (Terms do not give components). *If $P = \mathcal{A}_1 + \mathcal{A}_2$ with constant coefficients, then $\ker P \cap (\ker \mathcal{A}_1 + \ker \mathcal{A}_2) \subseteq \ker P \cap \ker(\mathcal{A}_1\mathcal{A}_2)$, and the right-hand side is finite-dimensional whenever $V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$ is finite.*

Proof. If $u = a + b$ with $\mathcal{A}_1 a = 0$ and $\mathcal{A}_2 b = 0$, then $\mathcal{A}_1\mathcal{A}_2 u = \mathcal{A}_2\mathcal{A}_1 a + \mathcal{A}_1\mathcal{A}_2 b = 0$ because the operators commute. At a common zero of P and $\mathcal{A}_1\mathcal{A}_2$ either $\mathcal{A}_1 = 0$, and then $\mathcal{A}_2 = P - \mathcal{A}_1 = 0$, or $\mathcal{A}_2 = 0$ and then $\mathcal{A}_1 = 0$; so $V(P, \mathcal{A}_1\mathcal{A}_2) = V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$. When this set is finite, the ideal $(P, \mathcal{A}_1\mathcal{A}_2)$ has finite codimension, its common kernel consists of exponential-polynomials with exponents in that set (Lemma 2.7), and it is finite-dimensional. □

For the heat equation, $\mathcal{A}_1 = \partial_t$ and $\mathcal{A}_2 = -\nu\partial_x^2$: every sum $f(x) + a(t) + b(t)x$ of a solution of $\mathcal{A}_1$ and one of $\mathcal{A}_2$ is killed by $\mathcal{A}_1\mathcal{A}_2 \sim \partial_t\partial_x^2$, while a heat mode $e^{-\nu\kappa^2 t}\cos\kappa x$ is not. A generic heat solution is *not* "a diffusion part plus a stationary part". The leftover of any term-wise split is the interaction term of Section 4.1.

Where the decomposition fails The hypotheses of Theorem 3.1 are not technical. On a disconnected domain different pieces can satisfy different equations. On a domain with a hole the decomposition can fail: the Laplacian factors as $(\partial_x + i\partial_y)(\partial_x - i\partial_y)$, yet $\log|z|^2$ is

harmonic on an annulus and is not a sum of a holomorphic and an antiholomorphic single-valued function. The obstruction is the first de Rham cohomology of Ω, which is why the algorithm works on rectangles.

3.2 How well separated are the groups? The Friedrichs angle

Theorem 3.1 is algebraic: it says that the split exists, not that it is stable. Stability is a geometric property of the component spaces $M_j = \ker_{H^s} Q_j$, the solutions of each factor in the Sobolev space $H^s(\Omega)$. Let

$$\mathcal{S}:\ \textstyle\bigoplus_{j=1}^{N} M_j \to H^s(\Omega), \qquad \mathcal{S}(u_1,\dots,u_N) = \sum_j u_j,$$

be the *sum operator*, with $\|(u_1,\dots,u_N)\|^2 = \sum_j \|u_j\|_{H^s}^2$, and define the *conditioning constant*

$$\gamma_N = \inf_{0 \neq v \perp \ker \mathcal{S}} \frac{\|\mathcal{S}v\|^2}{\|v\|^2} = \inf \sigma\big(\mathcal{S}^*\mathcal{S}\big|_{(\ker \mathcal{S})^\perp}\big). \tag{9}$$

It is the worst case, over all tuples of pieces, of the size of the sum relative to the size of the pieces. Tuples in $\ker \mathcal{S}$ (the overlap of Theorem 3.1) are excluded. By the triangle and Cauchy–Schwarz inequalities, $\|\sum_j u_j\|^2 \le \big(\sum_j \|u_j\|\big)^2 \le N \sum_j \|u_j\|^2$, hence

$$0 \le \gamma_N \le N.$$

If the spaces are mutually orthogonal, $\|\sum_j u_j\|^2 = \sum_j \|u_j\|^2$ and $\gamma_N = 1$. At the other end, if M_2 approaches M_1 one can take $u_2 \approx -u_1$: the pieces stay of unit size while their sum becomes arbitrarily small, so $\gamma_N \to 0$. A small γ_N is the signature of groups that can almost cancel each other.

For two groups, $\gamma_2 = 1 - c_F$, where the *Friedrichs cosine*

$$c_F = \sup\big\{|\langle a, b\rangle| :\ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\big\} = \cos\theta_F$$

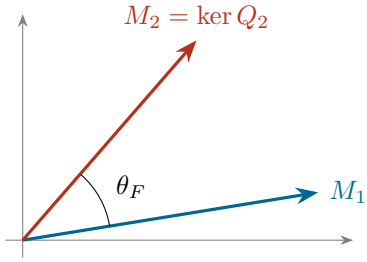
is the cosine of the smallest nonzero angle θ_F between the two spaces (Fig. 3) [3]. In regression terms, M_1 and M_2 are two blocks of regressors and c_F is the largest correlation between a combination of the first block and one of the second: the closer the blocks are to collinear, the less the data can attribute to each.

Theorem 3.4 (Stability of the split)**.** *Every $u \in \ker_{H^s} P$ has components $u_j \in H^s$ if and only if $\sum_j M_j$ is closed and dense in $\ker_{H^s} P$. Closedness is equivalent to $\gamma_N > 0$, and then*

$$\Big(\sum_j \|u_j\|_{H^s}^2\Big)^{1/2} \le \gamma_N^{-1/2}\, \|u\|_{H^s} \tag{10}$$

for the minimum-norm components. If the component spaces are finite-dimensional, as for the sampled kernels used by the algorithm, the sum is always closed and density holds whenever the sampled kernel of P is spanned by those of its factors, so stability reduces to $\gamma_N > 0$.

Density in H^s is a genuine condition on infinite-dimensional spaces. On convex domains solutions of $P(\partial)u = 0$ are limits of exponential solutions, uniformly on compact subsets [15, 11], and every exponential solution lies in $\sum_j M_j$; whether this extends to the H^s norm up to the boundary for general P we do not claim.



$c_F = \cos\theta_F, \quad \gamma_2 = 1 - c_F$
error amplification $\gamma_2^{-1/2}$
running example: $\theta_F \approx 0.60$, $\gamma_2 \approx 0.17$

Figure 3: The Friedrichs angle θ_F between the solution spaces of two factors. A small angle means that large pieces can almost cancel, so recovering the pieces from their sum amplifies errors by up to $\gamma_2^{-1/2} = (1-\cos\theta_F)^{-1/2}$. The values quoted are those measured on the running example in the companion algorithm paper.

Two warnings come with the theorem. First, $\gamma_N > 0$ can fail on infinite-dimensional component spaces: the pieces of a smooth u may then be one derivative rougher than u itself. Second, for $N \geq 3$ groups the pairwise cosines do not determine γ_N: two pairs can be well separated while the three spaces are nearly dependent, so the full Gram operator of $\mathcal{S}$ must be used.

3.3 An explicit projector for evolution factors

For an evolution operator with $P = \prod_{j=1}^{N}(\tau - a_j(\xi))$ and distinct a_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = a_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$ kills every u_l with $l \neq j$ (it contains the factor $\partial_t - a_l$) and multiplies u_j by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Hence

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\, u, \tag{11}$$

the Lagrange formula of Eq. (4) with C replaced by ∂_t. The result is a Fourier multiplier in x that is rational in ξ, with poles where $a_j(\xi) = a_k(\xi)$; this matches the characterization "projectors rational in ξ iff the factors are coprime" (ledger II-T2/T3).

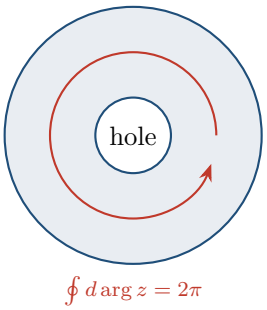
3.4 Where decomposition fails

Proposition 3.5 ((Holes) (ledger C12-P10) [proved]). *On a domain with a hole, the decomposition fails for some P.*

Figure 4 shows the standard obstruction. In the theorem the hypothesis "simply connected" means $H^1_{dR}(\Omega) = 0$, exactly what removes this obstruction; a cohomological criterion for non-simply connected Ω is also available (ledger OP-FUN, C12-Q6).

3.5 Regularity: when are the components in H^m?

The decomposition is an algebraic statement; analysis enters when we want components in a Sobolev space. For two components the governing condition is geometric: the two component spaces M_1 and M_2 must be separated by a positive angle (Figure 5); the next theorem makes this precise for any number of components. Two difficulties appear along the way; they are collected in the box after the theorem.



$u = \log|z-\alpha|^2$ solves $\Delta u = 0$ on the annulus,
but $u \neq h(z) + \overline{g(z)}$ with h, g
single-valued holomorphic: the logarithm
has monodromy (winding number $\neq 0$).

Figure 4: The Laplacian $\partial_x^2 + \partial_y^2 = (\partial_x + i\partial_y)(\partial_x - i\partial_y)$ factors, but the factor decomposition fails on a punctured disc: the obstruction is topological ($H^1_{dR} \neq 0$).

Proposition 3.6 ((Loss of a derivative) (ledger C12-C1) [false])**.** *"If $u \in H^m(\Omega)$ then all $u_j \in H^m$" is false: one derivative can be lost. In other words, the pieces of a smooth function may be rougher than the function itself.*

Theorem 3.7 ((Regularity criterion) (ledger C12-T3, C12-P7) [proved])**.** *All components lie in H^m for every $u \in \ker_{H^m} P$ iff the sum $\sum_j \ker_{H^m} Q_j$ is* closed *and* dense *in $\ker_{H^m} P$. Closedness is equivalent to $\gamma_N > 0$, where*

$$\gamma_N = \inf \sigma\big(S^*S|_{(\ker S)^\perp}\big),$$
$$S : \bigoplus_{j=1}^{N} \ker Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

For $N = 2$ this is the condition $c_F < 1$ on the Friedrichs cosine. Density holds on every bounded convex Ω, for every P and m (ledger OPF5B-T1) [proved mod. Hörmander [10], Stein].

Remark 3.8 (Reading γ_N). • The space $\bigoplus_j \ker Q_j$ consists of the N-tuples $v = (u_1, \dots, u_N)$ with norm $\|v\|^2 = \sum_j \|u_j\|^2_{H^m}$;
- $Sv = \sum_j u_j$ is their sum;
- S^* is the adjoint of S;
- $(\ker S)^\perp$ removes the tuples that sum to zero (the ambiguity of Theorem 3.1);
- $\inf \sigma$ is the bottom of the spectrum.

Since S^*S restricted to $(\ker S)^\perp$ is self-adjoint and nonnegative, the bottom of its spectrum is the minimum of the Rayleigh quotient

$$\frac{\langle S^*Sv, v\rangle}{\|v\|^2} = \frac{\|Sv\|^2}{\|v\|^2}.$$

Hence the formula of the theorem is equivalent to the more intuitive

$$\boxed{\gamma_N = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|^2_{H^m}}{\|v\|^2}}$$

Remark 3.9 (In words). γ_N is the worst case, over all tuples of pieces, of the ratio between the size of the sum and the size of the pieces.

- If $\gamma_N > 0$ the pieces cannot nearly cancel each other, and

$$\|v\| \le \gamma_N^{-1/2} \|Sv\| :$$

  the pieces are controlled by the sum, so recovering them from u is stable.
- If $\gamma_N \to 0$ there are large pieces whose sum is tiny, and the recovery is unstable.

The case $N = 2$. One finds $\gamma_2 = 1 - c_F$, where

$$c_F = \sup\{|\langle a, b\rangle| : \ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\}$$

is the Friedrichs cosine: the part of v lying in $M_1 \cap M_2$ contributes the ratio 2 and does not affect the infimum, and on the rest

$$\|a + b\|^2 \ge (1 - c_F)(\|a\|^2 + \|b\|^2)$$

with near-equality attained. Hence $c_F < 1 \iff \gamma_2 > 0$, as stated.

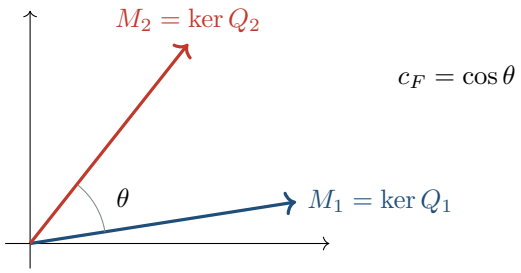


Figure 5: Friedrichs angle θ between two component spaces $M_1 = \ker Q_1$ and $M_2 = \ker Q_2$. A larger angle means a better-conditioned recovery of the pieces from their sum.

For $N = 2$, Figure 5 reads as follows: $c_F = \cos\theta$; if $\theta > 0$ the sum $M_1 + M_2$ is closed and the pieces depend continuously on u; as $\theta \to 0$ the split becomes unstable. For $N \ge 3$ the pairwise angles do not suffice; γ_N, the smallest nonzero eigenvalue of S^*S, is the right conditioning number: it is the conditioning of the linear map "components $\mapsto$ sum", exactly like the smallest nonzero singular value in a least-squares problem.

Remark 3.10 (Warning). [title=Two problems with regularity] **Problem 1: stability [false].** Algebraically the split always exists (Theorem 3.1), but the pieces may be less regular than u. When the angle between the component spaces tends to 0 (that is, $\gamma_N \to 0$) the map "$u \mapsto$ pieces" becomes unstable, and one derivative can be lost (Proposition 3.6). For the algorithm: check $\gamma_N > 0$ before using the components in H^m (Step 5).

Problem 2: three or more components [open]. For $N \ge 3$ the pairwise cosines $c_F(i, j)$ do not decide whether the sum of kernels is closed: an abstract Hilbert-space counterexample exists (ledger S-CHECK-4), and whether it can occur for differential operators is open (ledger C12-Q2).

Remark 3.11 (Why it matters). Algebraically the split always exists (Theorem 3.1); analytically, the components are well behaved exactly when the component spaces are separated by a positive angle (Theorem 3.7). The next section stops asking for a split into factors and asks what survives when one splits by additive terms instead.

4 Inside one group: the interaction term

In brief. What remains when a factor is built from several terms: an exact formula, its cost, its instability, and what happens when the terms do not commute.

4.1 Inside one group: the interaction term and its energy

A single factor may have several additive terms: $Q_2 = \tau - \nu\xi^2$ has one, but the advection–diffusion factor $\tau + c\xi - \nu\xi^2$ has two. Proposition 3.3 says that such a factor cannot be split further into one solution per term. We now measure exactly what is left over.

Consider an evolution $u_t = \mathcal{A}u$ with $\mathcal{A} = \mathcal{A}_1 + \cdots + \mathcal{A}_K$ (the terms), the flow $\Phi(t) = e^{t\mathcal{A}}$ and the flows of the single terms $\Phi_n(t) = e^{t\mathcal{A}_n}$. If term n acted alone it would change the datum by $\Delta_n u_0 = (\Phi_n(t) - \mathrm{Id})u_0$. The *interaction term* is the part of the evolution that is not the sum of these single-term changes,

$$E_{\mathrm{int}}(t) = u(t) - u_0 - \sum_{n=1}^{K} \big(\Phi_n(t)u_0 - u_0\big). \tag{12}$$

Theorem 4.1 (Interaction term)**.** *If the terms commute (in particular, for constant coefficients),*

$$E_{\mathrm{int}}(t) = \Big[\prod_{n=1}^{K}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{K}\Delta_n\Big]u_0 = \sum_{\mathcal{E}\subseteq\{1,\dots,K\},\ |\mathcal{E}|\geq 2} \Big(\prod_{n\in\mathcal{E}}\Delta_n\Big)u_0.$$

For two terms $E_{\mathrm{int}} = \Delta_1\Delta_2 u_0$. On Fourier modes, with symbols $\hat{\mathcal{A}}_n(\kappa)$ of the terms,

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \Big[\prod_{n} e^{t\hat{\mathcal{A}}_n(\kappa)} - 1 - \sum_{n}\big(e^{t\hat{\mathcal{A}}_n(\kappa)} - 1\big)\Big]\hat{u}_0(\kappa). \tag{13}$$

Proof. Commuting terms give $\Phi = \prod_n \Phi_n = \prod_n(\mathrm{Id}+\Delta_n)$. Expanding the product over subsets $\mathcal{E} \subseteq \{1,\dots,K\}$ and removing the subsets with zero or one element gives the formula; on Fourier modes every operator is multiplication by a number. □

Formula (13) costs $O(K)$ operations per wavenumber, although writing out all cross terms would need $2^K - K - 1$ of them. For advection–diffusion, $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t}-1)(e^{-\nu\kappa^2 t}-1)| \leq 2$: the interaction is of the same size as the solution, so it is not a small correction that a regularizer could remove.

Proposition 4.2 (No stable removal)**.** *Let $Q = \tau + c\xi - \nu\xi^2$ (advection–diffusion, terms $-c\partial_x$ and $\nu\partial_x^2$) on a bounded rectangle $\Omega = (0,T)\times(a,b)$. For every $s \geq 0$,*

$$\sup\Big\{\|E_{\mathrm{int}}\|_{L^2(\Omega)}/\|u\|_{H^s(\Omega)} :\ u \in \mathrm{EP},\ Q(\partial)u = 0,\ u \neq 0\Big\} = \infty,$$

so $u \mapsto E_{\mathrm{int}}$ has no continuous extension to $\ker_{H^s} Q$. Moreover, a solution of $Q(\partial)u = 0$ is a sum $v + w$ of solutions of the single-term evolutions $(\partial_t + c\partial_x)v = 0$ and $(\partial_t - \nu\partial_x^2)w = 0$ only if it lies in a finite-dimensional space of exponential-polynomials.

The test solutions are $e^{\lambda_\zeta t+\zeta x}$ with real $\zeta > c/\nu$ and $\lambda_\zeta = \nu\zeta^2 - c\zeta$, legitimate data on a bounded interval: diffusion alone amplifies them by $e^{\nu\zeta^2 t}$, the solution grows only like $e^{\lambda_\zeta t}$, and the interaction must cancel a difference that grows faster than any power of ζ (Appendix A). On a periodic domain with bounded wavenumbers the Fourier multiplier above is bounded by 2; the instability comes from the non-periodic window, which is the situation of real data. The second statement means that a model which writes the data as a sum of single-term pieces, with any penalty on the pieces, leaves a remainder on generic data, and that remainder is the interaction.

Energy We measure the size of a group and of the interaction by L^2 energies on the observation window. The *energy share* of group j and the *interaction ratio* are

$$w_j = \frac{\|u_j\|}{\sum_k \|u_k\|}, \qquad \rho = \frac{\|E_{\text{int}}(T)\|}{\|u(T) - u_0\|}, \tag{14}$$

where T is the length of the time window. The shares w_j say how much of the data each group explains; ρ says how far the evolution inside a multi-term factor is from the superposition of its terms ($\rho = 0$ for a factor with a single term). Interactions occur only *inside* a factor: different factors superpose exactly, which is another way of saying that factors, not terms, are the right groups.

4.2 Definition via the flows of the single terms

Let $S(t) = e^{tA}$ and $S_n(t) = e^{tA_n}$. The contribution that term n *alone* would make to the solution is $S_n(t)u_0 - u_0$. The remainder is the interaction term

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{15}$$

(The superscript "add" records that this is the *additive* definition; below we write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.)

4.3 Exact formula in the commuting case

Remark 4.3 (Why constant coefficients commute, and what Δ_n is). Constant-coefficient operators are polynomials in ∂_t, ∂_x (Section 2) and the ring $\mathcal{R}$ is commutative, so any two terms commute: $A_1A_2 = A_2A_1$. For commuting operators

$$e^{t(A_1+A_2)} = e^{tA_1}e^{tA_2}, \qquad \text{hence} \qquad S = \prod_n S_n;$$

without commutativity this fails (Section 4.6).

We call

$$\Delta_n = S_n(t) - \mathrm{Id}$$

the *increment* produced by term n alone, where Id is the identity operator and t is fixed; thus $S_n = \mathrm{Id} + \Delta_n$, and $\Delta_n u_0 = S_n(t)u_0 - u_0$ is exactly the contribution of term n in (15). Products of operators are compositions.

Remark 4.4 (Fourier variables). For constant coefficients it is convenient to pass to Fourier variables. With

$$\hat{u}(\kappa) = \int u(x) e^{-i\kappa x}\, \mathrm{d}x,$$

where $\kappa \in \mathbb{R}$ is the frequency:

- the derivative ∂_x becomes multiplication by $i\kappa$;
- each term A_n becomes multiplication by its *symbol* $a_n(\kappa)$ (a polynomial in $i\kappa$);
- e^{tA_n} becomes multiplication by the number $e^{ta_n(\kappa)}$.

Operators turn into numbers and compositions into products of numbers, so for a fixed κ every computation below is scalar.

Theorem 4.5 ((Interaction term) (ledger P1-T2, P1-T3) [proved])**.** *Assume the A_n commute (e.g. constant coefficients). Write $\Delta_n = S_n(t) - \mathrm{Id}$. Then*

$$E_{\mathrm{int}}^{\mathrm{add}} = \Big[\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n \Big] u_0. \tag{16}$$

For $N = 2$ this is simply $E_{\mathrm{int}} = \Delta_1 \Delta_2 u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})u_0$. In Fourier variables, with $\hat{A}_n = a_n(\kappa)$,

$$\widehat{E_{\mathrm{int}}}(t, \kappa) = \Big[\prod_{n} \big(e^{ta_n(\kappa)}\big) - 1 - \sum_{n} \big(e^{ta_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa).$$

Proof. Commutation gives $S = e^{t\sum A_n} = \prod_n S_n$. Substitute $S_n = \mathrm{Id} + \Delta_n$ in (15). For $N = 2$, expand the product:

$$\begin{aligned} S_1 S_2 &= (\mathrm{Id} + \Delta_1)(\mathrm{Id} + \Delta_2) \\ &= \mathrm{Id} + \Delta_1 + \Delta_2 + \Delta_1 \Delta_2, \end{aligned}$$

and $\sum_n (S_n - \mathrm{Id}) = \Delta_1 + \Delta_2$, so the remainder in (15) is $\Delta_1 \Delta_2 u_0$.

For general N, the Δ_n commute, so

$$\prod_{n} (\mathrm{Id} + \Delta_n) = \sum_{S \subseteq \{1, \dots, N\}} \prod_{n \in S} \Delta_n;$$

removing the subsets with $|S| = 0$ and $|S| = 1$ leaves

$$E_{\mathrm{int}} = \sum_{|S| \geq 2} \Delta_S\, u_0, \qquad \Delta_S = \prod_{n \in S} \Delta_n.$$

In words: the interaction is the sum of products of at least two increments, and appears only when at least two terms act together. □

A concrete case. Take advection–diffusion, $u_t = -cu_x + Du_{xx}$, with Gaussian datum $u_0 = g(x; \sigma^2)$, where $g(x; v) = e^{-x^2/2v}/\sqrt{2\pi v}$. The two flows act as a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x - ct; \sigma^2), \qquad S_2 u_0 = g(x; s^2), \qquad u = g(x - ct; s^2), \qquad s^2 = \sigma^2 + 2Dt.$$

Hence

$$E_{\mathrm{int}} = g(x - ct; s^2) - g(x - ct; \sigma^2) - g(x; s^2) + g(x; \sigma^2).$$

Figure 6(a) plots it: it is comparable in size to u itself, so it cannot be discarded as a small correction. In Fourier variables the ratio $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t} - 1)(e^{-D\kappa^2 t} - 1)|$ is at most 2; it vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$ (the translation returns to the identity), see Figure 6(b).

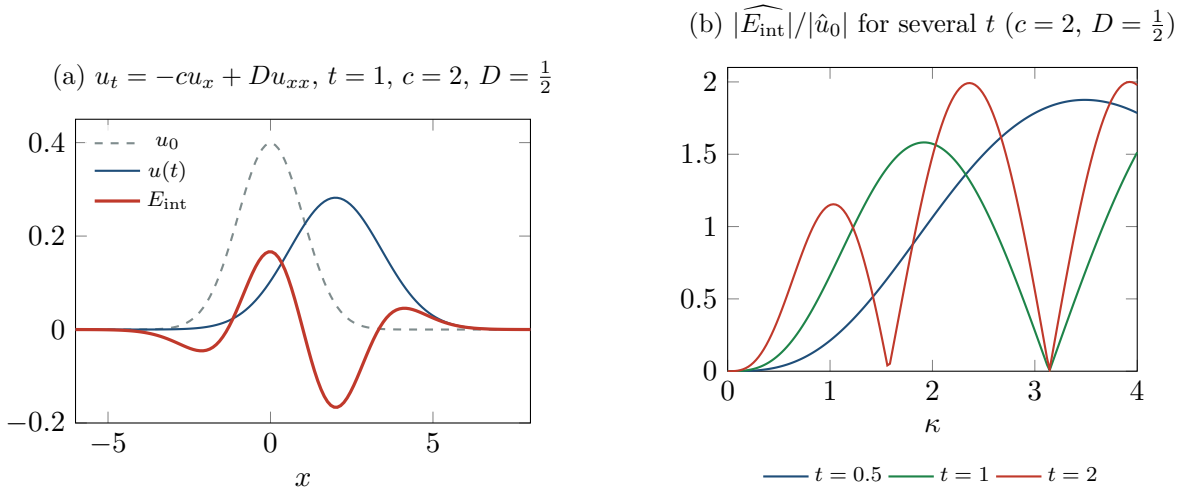


Figure 6: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$, $c = 2$, $D = \frac{1}{2}$). (a) u_0, $u(t)$ and E_{int} at $t = 1$. (b) The Fourier ratio $|\widehat{E_{\text{int}}}|/|\hat{u}_0|$ for three times.

4.4 Cost: $O(N)$ versus 2^N

Expanding (16) in the Δ_n produces one monomial per subset of $\{1, \dots, N\}$ with at least two elements: $2^N - N - 1$ cross terms (Figure 7: the red nodes are the cross terms). Writing the *complete table* of cross terms is therefore exponential. Evaluating a *value* is cheap, because we never expand the product. The algorithm uses exactly the product (Fourier) form of Theorem 4.5,

$$\widehat{E_{\text{int}}}(t, \kappa) = \Big[\prod_{n=1}^{N} e^{t a_n(\kappa)} - 1 - \sum_{n=1}^{N} \big(e^{t a_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa),$$

which for a fixed κ needs the N numbers $e^{t a_n(\kappa)}$, $N - 1$ products and N subtractions: $O(N)$ operations once $\hat{u}_0(\kappa)$ is known.

Remark 4.6 (Two remarks on cost). **What "value" means.** Here "value" means one Fourier value; recovering $E_{\text{int}}(t, x)$ at a point of space needs an inverse Fourier transform, whose cost is not counted in the $O(N)$.

Where 2^N comes from. The count $2^N - N - 1$ refers only to writing out all cross terms one by one; the product form never does so (Step 6 of the algorithm).

Proposition 4.7 ((Complexity) (ledger P1-T3) [proved]). *In the commuting exponential-type class, a value of E_{int} costs $O(M(N + \sum d))$ for M evaluation points; aggregated by order $O(MN^2)$; the full table $\Theta(M 2^N (L+1)^2)$.*

4.5 Time-dependent perturbation theory

Remark 4.8 (Why this subsection). The product formula (16) needs commuting terms. When terms do not commute, or when boundary conditions are imposed, there is no closed formula, but E_{int} is still defined by (15). Perturbation theory expands the flow in powers of the interaction. From this expansion one reads

(i) that E_{int} is well defined for all times (convergence),

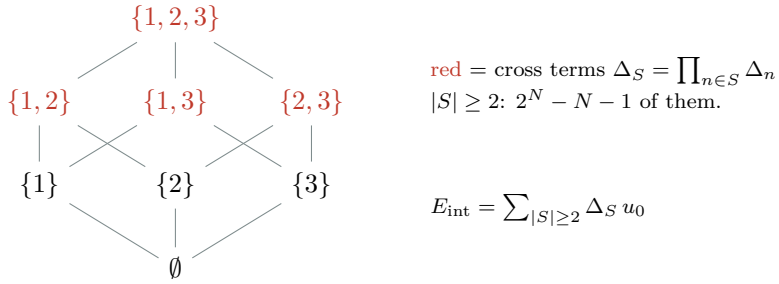


Figure 7: Subset lattice for $N = 3$ terms: one node per subset S of terms, one monomial Δ_S per node with $|S| \geq 2$. The count $2^N - N - 1$ is what makes the full table exponential.

(ii) approximations obtained by truncating the series order by order, and
(iii) the size of E_{int} for small t.

Split $A = A_1 + B$, where A_1 is the principal term (for instance diffusion) and B, the remaining terms, is treated as a perturbation of A_1. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k\geq 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s, \end{aligned} \tag{17}$$

the same expansion that gives the interaction picture of quantum mechanics [4].

Remark 4.9 (Reading the series). • $S^{(0)} = e^{tA_1}$ is the flow of A_1 alone;
• $S^{(k)}$ collects the contributions in which B acts k times, at intermediate times $s \in (0, t)$, with the flow of A_1 in between.

The terms with $k \geq 1$ are therefore the part of the flow in which B acts on the evolution generated by A_1, that is, the interaction. (When A_1 and B commute, the first-order term is

$$\int_0^t e^{(t-s)A_1} B e^{sA_1}\, \mathrm{d}s = t\, B e^{tA_1},$$

consistent with Section 4.)

What it gives. The series converges for all t when A_1 generates an even-order dissipative semigroup (e.g. diffusion) and B is a lower-order L^∞ perturbation, also with Dirichlet conditions (ledger P3-T1) [proved mod. Miyadera–Voigt [7]]: so E_{int} is well defined and can be approximated by truncation even without a closed formula. Each $S^{(k)}$ is the order-k part of the interaction. For small times, for boundary-induced interaction, with k the order of contact of the datum with the boundary, $\|E_{\text{int}}\| \sim t^{k/2+3/4}$ (ledger P3-T3, OP-F5A) [proved]: the smoother the datum is against the boundary (larger k), the faster the interaction becomes negligible as $t \to 0$.

4.6 Non-commuting terms

If $[A_1, A_2] \neq 0$, then $e^{t(A_1+A_2)} \neq e^{tA_1} e^{tA_2}$; by Baker–Campbell–Hausdorff,

$$e^{tA_1} e^{tA_2} = \exp\Big(t(A_1 + A_2) + \tfrac{t^2}{2}[A_1, A_2] + O(t^3)\Big).$$

Proposition 4.10 ((Lie-algebra dichotomy) (ledger P1-T4, P1-T5) [proved]). *If the Lie algebra generated by the terms is solvable, an exact formula with finitely many scalar functions exists (Wei–Norman). Otherwise only a local expansion exists, and* ***no finite product of the flows of the individual terms*** *reproduces* $e^{t(X+\varepsilon Y)}$.

Remark 4.11 (Why it matters). For commuting terms the interaction is explicit and cheap to evaluate, but it cannot be made to vanish nor be computed stably in H^m; for non-commuting terms even the explicit form is lost in general. These are the linear-case limits against which the nonlinear extension of Section 6 must be read.

5 When does a solution determine its equation?

In brief. The identification criterion, uniqueness, and what finite data can and cannot certify. This explains why the algorithm of the companion paper never reports "identified".

5.1 When does a solution determine its equation?

Let Ω be open and connected (on a disconnected set the pieces may obey different equations), let $u \in \mathcal{D}'(\Omega)$ satisfy $P(\partial)u = 0$, and let $P = \prod_i P_i^{e_i}$ (up to a nonzero constant factor) be the factorization of P into pairwise non-associate irreducible polynomials P_i with multiplicities e_i. The question is whether $\mathrm{Ann}(u) = (P)$: then every constant-coefficient equation satisfied by u is a consequence of P, and P is determined by u.

Theorem 5.1 (Identification criterion). *Put* $v_i = (P/P_i)(\partial)\, u$ *and* $v_i' = (P/P_i^{e_i})(\partial)\, u$.

(a) $\mathrm{Ann}(u) = (P)$ *if and only if* $v_i \notin \mathrm{EP}(\Omega)$ *for every* i.

(b) $V(\mathrm{Ann}\, u) = V(P)$ *if and only if* $v_i' \notin \mathrm{EP}(\Omega)$ *for every* i.

(c) If P *is square-free, (a) and (b) coincide. If* P *is monic in* τ *of degree* m, *then* $\mathrm{Ann}(u) = (P)$ *if and only if the only* $(\varphi_0, \dots, \varphi_{m-1}) \in \mathbb{C}[\xi]^m$ *with* $\sum_{j<m} \varphi_j(\partial_x)\, \partial_t^j u = 0$ *is zero.*

The idea is to strip one irreducible factor and look at what is left. The function v_i solves $P_i(\partial)v_i = 0$. If v_i is a generic solution of that factor, no smaller operator kills u. If v_i is an exponential-polynomial, some h coprime to P_i kills it, and then $h \cdot P/P_i$ is an annihilator of u that is not a multiple of P. The converse uses Lemma 2.4; the full proof is in Appendix A.

Proposition 5.2 (Uniqueness and the intrinsic operator). *(a) If* $\mathrm{Ann}(u) = (P)$, *then* P *is determined by* u *up to a nonzero constant and is the unique annihilator of minimal degree.*

(b) $\mathrm{Ann}(u) = (P)$ *if and only if* $\Sigma_{\mathrm{p}}(u) = \emptyset$ *and* $G_u \sim P$ *(equality up to a constant).*

(c) For any annihilator P, *the multiplicity of* P_i *in* G_u *equals* $e_i - \max\{k \le e_i : (P/P_i^k)(\partial)u \in \mathrm{EP}\}$. *Hence* G_u *is computed from any annihilator by at most* $\sum_i e_i$ *membership tests "*$\in \mathrm{EP}$*".*

Example 5.3 (The heat equation, continued). For $P = \tau - \nu\xi^2$ (irreducible) the criterion reads: $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}$. One mode $e^{-\nu\kappa^2 t + i\kappa x}$ has $\mathrm{Ann}(u) = (\tau + \nu\kappa^2,\ \xi - i\kappa)$, a point of $V(P)$; it cannot distinguish the heat equation from the pure decay $\partial_t + \nu\kappa^2$, or from any other equation whose variety passes through the point $(-\nu\kappa^2, i\kappa)$. The heat kernel is not in EP, so every constant-coefficient equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$.

Remark 5.4 (Curves and points). The two cases of Example 5.3 sit at the two ends of the decomposition (3). A finite sum of modes $u \in \mathrm{EP}$ has an annihilator with finitely many zeros, so

$$u \in \mathrm{EP}: \qquad G_u = 1, \qquad \Sigma_{\mathrm{p}}(u) = \{(\tau_j, \xi_j)\} \text{ finite (its exponents)},$$

whereas the zero set of a nonconstant polynomial in two variables is a whole algebraic curve. For the heat kernel the situation is reversed:

$$u = t^{-1/2} e^{-x^2/(4\nu t)}: \qquad G_u = \tau - \nu\xi^2, \quad V(G_u) \text{ a curve}, \qquad \Sigma_{\mathrm{p}}(u) = \emptyset.$$

By Proposition 5.2(b) it is exactly this pair, an empty point spectrum together with $G_u \sim P$, that gives $\mathrm{Ann}(u) = (P)$: the solution fills the whole curve of its equation and contains no isolated mode besides.

5.2 What finite data can and cannot certify

Theorem 5.1 speaks about u, not about finitely many numbers. Two facts frame what an algorithm can promise.

Recall that $\mathcal{R}u = \mathrm{span}\{\partial_t^j \partial_x^k u : \ j, k \geq 0\}$ is the vector space generated by the derivatives of u (Section 2.3). The *jet* of order M of u at a point $z_0 = (t_0, x_0)$ is the list of its derivatives of order at most M at that point; for example

$$j^2_{z_0}(u) = (u, \ u_t, \ u_x, \ u_{tt}, \ u_{tx}, \ u_{xx})\big|_{(t,x)=z_0}.$$

Proposition 5.5 (Finite data)**.** *Let Ω be connected and $z_0 \in \Omega$.*

(i) If $\dim_{\mathbb{C}} \mathcal{R}u \leq n$, the jet of order $2n-1$ of u at z_0 determines $\mathrm{Ann}(u)$ among all functions u' with $\dim_{\mathbb{C}} \mathcal{R}u' \leq n$.

(ii) For every u and every M there is a polynomial, hence an exponential-polynomial, with the same jet of order M at z_0; and for finitely many sample points there is an exponential-polynomial with the same values. Consequently, without a bound on $\dim \mathcal{R}u$, no finite set of jets or samples proves $\mathrm{Ann}(u) = (P)$.

Proof. (ii) The Taylor polynomial of order M at z_0 has the same jet, and polynomial interpolation in two variables matches any finite set of values. Both are exponential-polynomials, and an exponential-polynomial never has $\mathrm{Ann} = (P)$ when P has an irreducible factor (Theorem 5.1: all v_i are then exponential-polynomials). (i) The space $\mathcal{R}u$ is invariant under differentiation and, by Lemma 2.7, consists of real-analytic exponential-polynomials. Let K_M be the subspace of elements of $\mathcal{R}u$ whose jet of order M vanishes at z_0. Then $K_{M+1} = \{f \in K_0 : \ \partial_t f, \partial_x f \in K_M\}$, so once $K_{M+1} = K_M$ the chain is constant, and a constant K_M is a derivative-invariant space of functions vanishing at z_0 with all their derivatives, hence $\{0\}$. Since $\dim K_0 \leq n-1$, the chain reaches $\{0\}$ by $M = n-1$: the jet of order $n-1$ at z_0 is injective on $\mathcal{R}u$, and likewise on $\mathcal{R}u'$. For Q of degree at most n, $Q \in \mathrm{Ann}(u)$ iff the jet of order $n-1$ of $Q(\partial)u$ vanishes at z_0, a condition on the jet of order $2n-1$ of u; the same holds for u'. Finally $\mathrm{Ann}(u)$ has codimension at most n, so a Gröbner basis for a degree-compatible order consists of polynomials of degree at most n; the same holds for $\mathrm{Ann}(u')$, and two ideals generated in degree at most n that contain the same polynomials of degree at most n are equal. □

The consequence for learning is a design rule: without an a priori bound on $\dim \mathcal{R}u$, which data never supply, an algorithm can refute a candidate equation, or report that it is compatible with the data at the resolution available, but it cannot certify it. Its output must carry a label that says which of these happened.

6 Nonlinear equations

In brief. The same two questions, asked of Burgers-type equations: the interaction term for the flows of single terms, behaviour before, at and after a shock, and identification relative to a dictionary.

Starting from: a scalar equation in one space variable, $u_t = \sum_n T_n(u)$, each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; Ω bounded and simply connected; the exact solution is known. *Why:* to test which of the notions of Sections 5.1–4 survive when the terms are nonlinear. *Role:* extension of the chain. *Result:* weaker, relative statements (exact expansion of E_{int}, sharp asymptotics across a shock for Burgers, identification relative to a dictionary). Benchmarks: Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS only as controls.

We consider $u_t = \sum_n T_n(u)$ with each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$. Benchmarks: Burgers $u_t = -uu_x + \nu u_{xx}$, KdV–Burgers, Fisher–KPP; integrable cases (KdV, NLS) serve as controls.

6.1 Interaction term for flows of single terms

With S_n the (nonlinear) flow of $u_t = T_n(u)$ alone, (15) still defines $E_{\text{int}}^{\text{add}}$. Its Taylor expansion in t:

Theorem 6.1 ((Jet expansion) (ledger S4T-T1) [proved]). $E_{\text{int}}^{\text{add}} = \dfrac{t^2}{2} \displaystyle\sum_{n \neq m} T_n'[\, T_m(u_0)\,] + O(t^3)$,

where $T_n'[v]$ is the Fréchet derivative of T_n at u_0 in the direction v.

Example 6.2 (Burgers). $T_1 = -uu_x$, $T_2 = \nu u_{xx}$. Then $T_1'[v] = -(vu_x + uv_x)$ and $T_2'[v] = \nu v_{xx}$, so

$$T_1'[T_2 u_0] = -\nu\,(u_0'' u_0' + u_0 u_0'''),$$
$$T_2'[T_1 u_0] = -\nu\,(3u_0' u_0'' + u_0 u_0'''),$$

and half the sum times t^2 gives

$$\begin{aligned} E_{\text{int}} &= \tfrac{1}{2}(T_1'[T_2 u_0] + T_2'[T_1 u_0])t^2 + O(t^3) \\ &= -\nu(u_0 u_0''' + 2u_0' u_0'')t^2 + O(t^3). \end{aligned}$$

6.2 Before, at, and after the shock

The interaction term of Burgers changes qualitatively with time, and each regime calls for different tools:

Before shock formation smoothness permits perturbative expansions and derivative-based estimates.

At breaking the inviscid gradient blows up, invalidating estimates that require smoothness.

After shock formation pointwise behavior can be discontinuous, so L^1 estimates become more informative than pointwise bounds.

Figure 8 recalls why a shock appears at all in the flow of the single term $-uu_x$: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t \sin \xi$$

(the parameter s in the figure plays the role of the time t); since

$$\frac{\partial x}{\partial \xi} = 1 + t\cos\xi$$

vanishes first at $\xi = \pi$, the breaking time is $t^* = 1$.

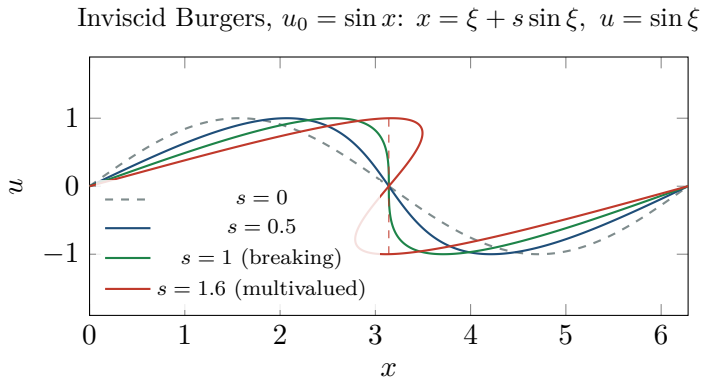


Figure 8: Characteristic solution of the inviscid flow of $-uu_x$. At $s = 1$ the profile has a vertical tangent: the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$ (equal-area rule), shown dashed. This breaking is what makes E_{int} discontinuous after the shock.

Before the shock ($t < t^*$): smooth regime. Smoothness permits perturbative expansions and derivative-based estimates.

- E_{int} has a closed form at first order in ν, with remainder $O(\nu^2\varepsilon^2 t^3)$ (ledger S4T-T2) [proved]; it blows up at the breaking time t^* (ledger S4T-P3). (This is the same family of expansions as Theorem 6.1 and Example 6.2.)
- The Cole–Hopf substitution $u = -2\nu\varphi_x/\varphi$ turns Burgers into the heat equation $\varphi_t = \nu\varphi_{xx}$; the inverse scattering transform linearizes KdV and NLS. These are used as the "exact linearization" of the nonlinear solution [9].

At breaking ($t = t^*$): estimates that need smoothness fail. The inviscid gradient u_x blows up (vertical tangent, $s = 1$ in Figure 8), so every statement that needs bounds on $u_x, u_{xx}, \ldots$ stops being usable at t^*: the closed form of E_{int} blows up, and estimates that require smoothness, such as those behind the jet expansion of Theorem 6.1, are no longer valid.

After the shock ($t > t^*$): discontinuous regime. The flow of $-uu_x$ alone forms a shock, so E_{int} jumps by amounts independent of ν; this is forced by the definition. A function with a jump is not in H^1 in one dimension, so the Sobolev norms of Sections 3 and 4 are no longer the natural ones: pointwise values are unreliable, while the *measure* of the discrepancy is well

behaved. The meaningful quantities are therefore L^1 sizes, collected below:

$$\|E_{\mathrm{int}}\|_{L^1} = \Theta(\nu), \tag{18}$$

$$\sup_{t\le T} \|E_{\mathrm{int}}\|_{L^1} = \nu|\log\nu| + O(\nu), \tag{19}$$

$$\|S - S_1S_2\|_{L^1} = \Theta(\sqrt{\nu}). \tag{20}$$

The first line holds at fixed t, each shock contributing $4\nu\log(2\cosh(\rho/2))$; the second has exact leading coefficient 1. Sources: the first line is (ledger OPA-T1) [proved mod. Kružkov uniqueness [13]]; the second is (ledger OPF5A-T1/T2); the third is (ledger OPF5D-T6) [proved]. The second line holds for data with a single birth of shock such as $\sin x$, and is conditional on a localization lemma in general [conditional].

Regime	What happens	Tools and results
Before the shock, $t < t^*$	the profile is smooth, characteristics do not cross	jet expansion (Thm. 6.1), first-order closed form of E_{int}, Cole–Hopf
At breaking, $t = t^*$	the inviscid gradient blows up (vertical tangent)	smoothness-based estimates fail; E_{int} blows up
After the shock, $t > t^*$	the single-term flow is discontinuous (equal-area rule)	L^1 estimates: $\Theta(\nu)$, $\nu\|\log\nu\|$, $\Theta(\sqrt{\nu})$

Table 2: The three regimes of the Burgers interaction term and the tools that apply in each.

6.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$ of Section 2; instead one fixes a finite candidate list. Choose a dictionary $\mathcal{D}$ of monomials of degree $\le d$ in the jet, of order $\le k$, and ask which polynomial relations $F(u, u_x, \dots) = 0$ the data satisfy. Notation: V is the span of $\mathcal{D}$, $I(u)$ the ideal of all differential relations satisfied by u, $[F]$ the differential ideal generated by F, and $\mathcal{J}_u$ the jet-level obstruction (a different object from the cofactor J_u of Section 2, hence the calligraphic letter).

Theorem 6.3 ((Dictionary criterion) (ledger S4ID-T1) [proved]). *For smooth u on any Ω there is an exact sequence $0 \to [F]\cap V \to I(u)\cap V \to \mathcal{J}_u \cap \psi(V) \to 0$. In particular*

$$I(u)\cap V = [F]\cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*"; the literal "$I(u)\cap V = \langle F\rangle$" is false in complete dictionaries (they contain $\partial_x F$, etc.) (ledger S4ID-P1). Concrete criteria: for Burgers with N shocks, u identifies the equation in the evolutionary dictionary iff $d \le N$ (ledger S4ID-T2, OP-T7); for N-soliton KdV the identifying polynomial has total degree $(N+2)2^{N-1}$ (generic $N = 2$: 8) (ledger OP-K15) [proved mod. Novikov 1974 [16]]. Finally, "closed form $\Rightarrow$ does not identify" is itself a theorem (ledger S4ID-T4).

6.4 Linear versus nonlinear: the same idea, two structures

Sections 2–4 and the present one answer the same question: *what does a single solution u remember about its equation?* In both settings we collect all relations satisfied by u and ask

	Linear approach (Sections 2–4)	Nonlinear approach (this section)
Space of candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$ of constant-coefficient operators	a finite *dictionary* $\mathcal{D}$ of jet monomials of degree $\leq d$ and order $\leq k$; $V = \operatorname{span} \mathcal{D}$
Relations satisfied by u	$\operatorname{Ann}(u)$, an ideal of $\mathcal{R}$	$I(u)$, the ideal of all differential relations
What the equation generates	the principal ideal (P)	the differential ideal $[F]$
Criterion for identification	$\operatorname{Ann}(u) = (P)$ (Theorem 5.1)	$I(u) \cap V = [F] \cap V$, i.e. F *modulo its differential consequences* (Theorem 6.3)
What escapes the equation	the cofactor J_u: the finitely many exponentials in u ($\sigma_p(u) = V(J_u)$)	the jet-level obstruction $\mathcal{J}_u \cap \psi(V)$ (exact sequence of Theorem 6.3)
Solutions that do not identify	$u \in \mathrm{EP}$	closed-form solutions ((ledger S4ID-T4))
Concrete criteria	finite linear algebra (Lemma 2.7)	Burgers with N shocks identifies iff $d \leq N$; N-soliton KdV: degree $(N+2)2^{N-1}$

Table 3: The linear and nonlinear identification problems side by side.

when these relations are generated by the equation we were given. What changes is the algebraic structure in which the relations live.

Remark 6.4 (Where the parallel breaks). **Linear case.** The solutions form a vector space and the annihilators an ideal: one can add, multiply by operators and factor, which is why the Nullstellensatz applies and the test is a finite-dimensional computation.

Nonlinear case. There is no linear structure on the solutions: one must fix a finite space V of candidates in advance, and the identification is *relative to* V, never absolute. The literal statement

$$I(u) \cap V = \langle F \rangle$$

is false in complete dictionaries, which contain $\partial_x F$, so the correct statement is "modulo differential consequences".

Remark 6.5 (Takeaway). Both approaches ask what one solution determines about its equation. The linear one answers with the annihilator ideal $\operatorname{Ann}(u)$ and its generator G_u; the nonlinear one answers with a dictionary and the differential ideal of relations, and only up to differential consequences. In the algorithm of the companion paper Steps 1–6 use the structure of $\operatorname{Ann}(u)$, while Step 7 switches to the dictionary and returns a relative identification.

Part I-B: the full development of Phases 0–5

The sections above give the main line of Phases 0–5 in the order of the grouping principle. The following sections are the full-length chapters of the complete edition (abstract and reading guide, overview, algebra primer, the algebraic language, identification, finite data and decidability, decomposition, regularity, the interaction term, nonlinear equations, the algorithm, limits, and the open problems and directions of Phases 0–5), with every theorem, example and

warning; labels that coincide with those above are prefixed by `f:` in the source. Proofs are cited as "Dim. An/Bn" in the companion proofs file.

Abstract of the complete edition

We are given one exact solution $u(t, x)$ of a differential equation that we do not know. What does u remember about its equation? This edition answers the question for the results of Phases 0–10 of the project. The answer has four parts for constant coefficients (Part I) and four more for what lies beyond (Part II).

Identification. For constant coefficients, u determines its operator P exactly when no partial quotient

$$w_i = \Big(\frac{P}{P_i}\Big)(\partial)\, u$$

is an exponential-polynomial. No finite amount of data can certify this, and a black-box test is impossible. One extra integer, a bound on the exponential-polynomial part of u, makes it a finite computation.

Decomposition. On a simply connected region, u splits into one component per irreducible *factor* of the symbol. The ambiguity has the exact finite dimension $\delta(P)$. The split per additive *term* of the operator is false. The components keep the Sobolev regularity of u exactly when a conditioning number γ_N is positive.

Interaction. For an evolution equation whose N terms commute, the part of the solution due to the interplay of the terms is

$$E_{\mathrm{int}} = \Big[\prod_{n=1}^{N}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N}\Delta_n\Big]u_0.$$

One value costs $O(N)$ operations, while the full table of cross terms costs $\Theta(2^N)$. The map $u_0 \mapsto E_{\mathrm{int}}$ is not continuous in Sobolev norms and cannot be regularised.

Nonlinear equations. For Burgers, KdV–Burgers and KdV the same questions have sharp answers. The interaction has a t^2 jet and a closed form before the shock, and size $\Theta(\nu)$ and $\nu|\log\nu|$ after it. Identification holds relative to a dictionary, modulo the differential consequences of the equation. For the N-soliton of KdV the identifying polynomial has degree $(N+2)2^{N-1}$.

Beyond constant coefficients (Part II). *Systems:* a vector solution determines a matrix operator iff every socle of its module has dimension at most one, which holds for every non-singular square system in two or more variables; gauge can never be separated from physics by one field. *Variable coefficients:* a potential $V(x)$ is read from any non-zero solution, a potential $V(t, x)$ can hide in silent time strips, and a huge coefficient class identifies nothing. *Convergence and noise:* identification holds in the limit without a certificate, no learner decides whether a law exists, and noise is handled only at a finite level of the dictionary. *The encoder and the dictionary:* an explicit Volterra transformation removes a known potential; a universal dictionary exists for approximation but never for exact identification, and a misfit has a finite certificate while a fit never does. Phase 10 corrected several statements of Phases 1–3: a continuity no-go that was false as stated, and the failure of density on regions with holes.

How to read this edition

Layout. Every chapter opens with a box with three entries. *Starting from* lists the assumptions in force. *Why* says what problem the chapter solves. *Role* says where its output is used. Inside a chapter the order is always the same: motivation, then intuition, then statement, then a worked example, then consequences.

Boxes. The colours are fixed:

- blue boxes are theorems, propositions and lemmas;
- green boxes are definitions;
- violet bars are worked examples;
- amber boxes give the intuition;
- red boxes are warnings and counterexamples.

Proofs. Proofs are not in this volume. Each statement names its proof in the companion *Catalogue of proofs*, for example Dim. A3. Parts I and II of the catalogue contain the load-bearing proofs of Phases 0–5 (A1–A31) and of Phases 6–10 (A32–A53); Parts III and IV contain all the others.

Trust labels. Every statement carries one label. [proved] means a complete proof that passed an independent review. [proved mod. X] means complete except for a cited published theorem X whose source was checked. [proof written, not audited] means a full proof exists but no independent audit has been done. [from cited source] means a published theorem we rely on. [conditional] means the result depends on an unverified lemma. [checked on examples] means it was only checked on examples. [open] means unsolved. [false] means a counterexample is proved. The reviewers were independent AI agents, not human referees. Grey tags such as (ledger C3-T1) are the identifiers in the project ledger `claims/LEDGER.md`.

Prerequisites. Linear algebra (eigenvalues, Jordan form), calculus in two variables, the Fourier transform, and linear ODEs with constant coefficients. Chapter 8 introduces every algebraic notion from scratch: ideals, annihilators, exponential-polynomials and the Chinese remainder theorem.

Scope. This edition covers Phases 0–10; a separate pair of volumes covers Phases 0–5 alone. In case of a difference the present edition prevails.

7 Overview of the full development

Starting from: one function $u(t, x)$, known exactly (no noise) on an open region $\Omega \subset \mathbb{R}^2$, which solves a linear equation $P(\partial_t, \partial_x)u = 0$ with constant coefficients. The operator P is *not* given. *Why:* to state precisely what we want to learn from u and to give a map of the answers before any detail. *Role:* frames the three questions Q1–Q3. Sections 10–11 answer Q1, Sections 12–13 answer Q2, Section 14 answers Q3. Section 15 extends all three to nonlinear equations and Section 16 assembles them into an algorithm. Part II (Sections 19–17) goes beyond constant coefficients and states the limits of the whole project.

7.1 The question

Suppose we observe a physical field $u(t, x)$: a temperature, a wave, a concentration. We know u exactly on some region Ω of the (t, x) plane, and we know that it obeys some linear law. We do not know which law. Three questions arise naturally.

Q1. Identification. Can the operator P be reconstructed from u? When is the answer unique?

Q2. Decomposition. If P is built from pieces, can u be split into the parts that each piece generates?

Q3. Interaction. If the pieces are the *terms* of the operator (diffusion, advection, dispersion, ...), which part of u is due to no single term but to their interplay? Can it be computed exactly and cheaply?

The setting is deliberately analytic. There is no statistical model, no learned weights and no noise. The aim is an *exact* procedure, together with a precise statement of what is impossible.

Example 7.1 (Three solutions of the heat equation). Take the heat equation $u_t = Du_{xx}$ with $D > 0$. Consider three of its solutions.

- A single Fourier mode:
$$u_1 = e^{-D\kappa^2 t}\cos(\kappa x).$$
- A sum of two modes:
$$u_2 = e^{-Dt}\cos x + e^{-4Dt}\cos 2x.$$
- The heat kernel, for $t > 0$:
$$u_3 = \frac{1}{\sqrt{t}}\, e^{-x^2/(4Dt)}.$$

All three solve the same equation. We will see that only u_3 determines it. The function u_1 also solves $u_{xx} = -\kappa^2 u$ and $u_t = -D\kappa^2 u$, and from u_1 alone we cannot tell which of these laws is "the" law. The difference is that u_1 and u_2 are finite sums of exponentials, while u_3 is not (Section 10, Example 10.5).

7.2 Overview of the answers

Figure 9 shows the logical chain, read from left to right. Each box is a task. Each arrow is a theorem that allows the next task to be carried out, under its own hypotheses.

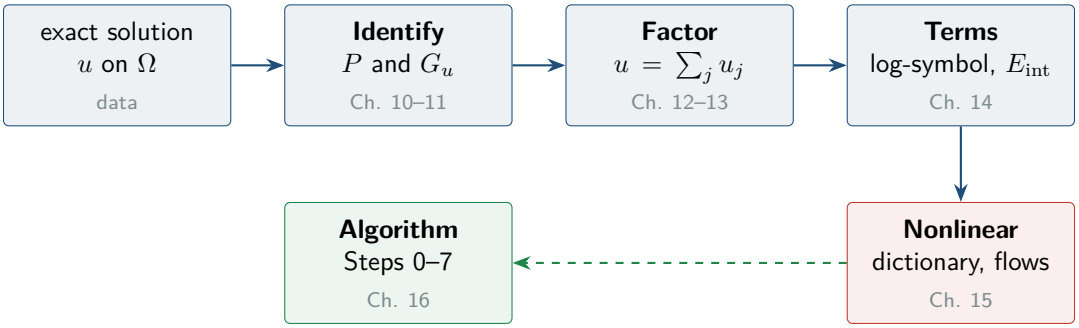


Figure 9: Logical structure of the method. The three linear stages (blue) are followed by the nonlinear extension (red). The algorithm (green) chains all of them. A step can only be trusted under the hypotheses of the theorem that precedes it.

Table 4 lists the main statements with their status. Every row points to the chapter where the statement is made precise.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P)$ iff no $w_i = (P/P_i)(\partial)u$ is an exponential-polynomial (Ch. 10)	[proved]
Spectral form	$\mathrm{Ann}(u) = (P)$ iff the point spectrum is empty and $G_u \sim P$ (Ch. 10)	[proved]
Finite data	a jet of order $2d-1$ suffices for exponential-polynomials; no finite certificate otherwise; black box impossible; decidable with a bound (Ch. 11)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, ambiguity $\delta(P)$ (Ch. 12)	[proved]
Terms	splitting per additive term is false (Ch. 12)	[proved]
Regularity	components in H^m iff the sum of kernels is closed ($\gamma_N > 0$) and dense (Ch. 13)	[proved]
Interaction	closed product formula; $O(N)$ per value, $\Theta(2^N)$ per table (Ch. 14)	[proved]
Stability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m; no stable regularisation (Ch. 14)	[proved]
Nonlinear	Burgers: jet, closed form before the shock, $\Theta(\nu)$ and $\nu\lvert\log\nu\rvert$ after (Ch. 15)	[proved] / [proved mod. Kružkov]
Dictionary	identification modulo differential consequences; KdV degree $(N+2)2^{N-1}$ (Ch. 15)	[proved] / [proved mod. Novikov]

Table 4: Main results, in the order in which they appear.

7.3 Notation at a glance

Table 5 collects the symbols used throughout, with the place where each one is defined. Some letters have different meanings in different chapters; the caption lists them.

Symbol	Meaning	Defined in
(t, x), Ω	time and space; the open region where u is known	Ch. 7
$\partial = (\partial_t, \partial_x)$	partial derivatives	Ch. 9
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	ring of constant-coefficient operators, $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$	Def. 9.1
P, P_i, e_i	the operator (its symbol), its irreducible factors, their multiplicities	Ch. 9
$Q_j = P_j^{m_j}$	pairwise coprime factors used for decomposition	Ch. 12
$V(P)$	characteristic variety: zeros of the symbol in $\mathbb{C}^2$	Def. 9.1
$\mathrm{Ann}(u)$	the ideal of all operators that annihilate u	Def. 9.2
G_u, J_u	$G_u = \gcd \mathrm{Ann}(u)$ and the cofactor ideal: $\mathrm{Ann}(u) = G_u\, J_u$	Lemma 9.4
EP	exponential-polynomials	Def. 9.7
$\sigma_p(u)$	point spectrum $V(J_u)$	Ch. 10
$\delta(P)$	dimension of the ambiguity of the factor decomposition	Ch. 12
c_F, γ_N	Friedrichs cosine; conditioning number of the sum map	Ch. 13
A_n, $S_n(t)$, Δ_n	terms of an evolution, their flows, increments $S_n - \mathrm{Id}$	Ch. 14
E_{int}	interaction term	Eq. (24)
T_n, V, $I(u)$	nonlinear terms; dictionary span; ideal of relations of u	Ch. 15

Table 5: Notation guide. Overloaded letters: S is a flow in Section 14 and the sum map in Section 13; D is d/dx in Section 9.4 and a diffusion constant elsewhere; J denotes ideals.

7.4 Roadmap

The chapters build on each other in the following order.

- **Section 8** is a primer. It explains ideals, annihilators, exponential-polynomials, the Chinese remainder theorem and the companion matrix, starting from integers and one-variable ODEs. A reader who already knows these notions can skip it.
- **Section 9** fixes the algebraic language: symbols, the annihilator ideal, its gcd splitting and the finite-dimension test for exponential-polynomials. It ends with the one-variable prototype.
- **Sections 10 and 11** answer Q1. The first gives the exact criterion and the intrinsic operator G_u. The second asks what can be decided from finite data.
- **Sections 12 and 13** answer Q2. The first proves the factor decomposition and shows why splitting by terms fails. The second asks when the pieces keep the regularity of u.
- **Section 14** answers Q3: the exact interaction term, its cost, and its instability.
- **Section 15** extends the three questions to nonlinear equations.
- **Section 16** turns the theorems into an algorithm with two fully worked examples.
- **Sections 19–23** (Part II) go beyond constant coefficients: systems and gauge, variable coefficients, convergence and noise, the analytic encoder, and the universal dictionary.
- **Section 24** records what Phase 10 changed in the statements of the earlier phases.
- **Section 25** extends the algorithm to all of the above, with a contract for every step.
- **Section 17** states what is impossible, what is open, and how reliable each result is, for the whole project.

8 A primer for readers new to the algebra

Starting from: only linear algebra, polynomials and linear ODEs with constant coefficients. *Why:* the whole theory speaks the language of rings and ideals. Every notion used later is introduced here from a familiar starting point: integers first, then ODEs in one variable, then two variables. *Role:* nothing later depends on reading this chapter, but every later chapter uses its vocabulary. It covers ideals and annihilators (§8.1), exponential-polynomials (§8.2), the Chinese remainder theorem (§8.3) and the companion matrix (§8.4).

8.1 Ideals and annihilators, starting from integers

8.1.1 Integers

The multiples of 6 form the set

$$(6) = \{\dots, -12, -6, 0, 6, 12, \dots\}.$$

This set has two properties. The sum of two multiples of 6 is a multiple of 6. A multiple of 6 times any integer is again a multiple of 6. A subset of a ring with these two properties is called an *ideal*.

Now take the ideal generated by 4 and 6, that is, all combinations $4a + 6b$ with integers a, b:

$$(4, 6) = \{4a + 6b : \ a, b \in \mathbb{Z}\}.$$

Since $2 = 6 - 4$ lies in it, and every element is even, we get $(4, 6) = (2)$. The ideal "remembers" only the greatest common divisor of its generators. In $\mathbb{Z}$ every ideal is generated by one number.

8.1.2 Operators in one variable

Replace integers by differential operators with constant coefficients in one variable x, and replace "is a multiple of 6" by "kills the function u".

Definition 8.1 (Annihilator (one variable))**.** For a function $u(x)$, the annihilator of u is the set of all polynomials Q such that the operator $Q(d/dx)$ kills u:

$$\mathrm{Ann}(u) = \Big\{ Q \in \mathbb{C}[\xi] : \; Q\Big(\frac{d}{dx}\Big)u = 0 \Big\}.$$

It is an ideal. If Q kills u and S is any polynomial, then SQ kills u too, because $S(d/dx)$ applied to 0 is 0. Sums of annihilators are annihilators.

Example 8.2 (Annihilators of familiar functions). Write $\partial = d/dx$. The operator $\partial - 1$ kills e^x, and so does every multiple of it. The basic cases are:

$$\begin{aligned} \mathrm{Ann}(e^x) &= (\partial - 1), \\ \mathrm{Ann}(\sin x) &= (\partial^2 + 1), \\ \mathrm{Ann}(xe^x) &= \big((\partial - 1)^2\big), \\ \mathrm{Ann}(e^x + e^{2x}) &= \big((\partial - 1)(\partial - 2)\big). \end{aligned}$$

The function e^{x^2} has annihilator $\{0\}$: no nonzero polynomial in ∂ kills it, because each derivative $\partial^k e^{x^2}$ equals a polynomial of degree exactly k times e^{x^2}.

In one variable every ideal has a single generator, exactly as in $\mathbb{Z}$. The generator of $\mathrm{Ann}(u)$ is the *minimal annihilator* of u: the ODE of lowest order that u satisfies.

8.1.3 Two variables: the new feature

With two variables t and x we have two derivatives. An operator is a polynomial in both. Here an ideal may need *several* generators.

Example 8.3 (An ideal with two generators). Let $u = e^{2t+3x}$. Then

$$(\partial_t - 2)u = 0 \qquad \text{and} \qquad (\partial_x - 3)u = 0.$$

Every operator of the form $A\,(\partial_t - 2) + B\,(\partial_x - 3)$ kills u, and these are all of them. So

$$\mathrm{Ann}(u) = (\partial_t - 2,\ \partial_x - 3).$$

No single polynomial generates this ideal: a polynomial in the ideal vanishes at the point $(2, 3)$, and a single polynomial vanishes on a whole curve, not at a single point.

This is the source of every subtlety in the theory. A solution of a PDE has an annihilator that may contain a "curve part" (one polynomial) and a "point part" (finitely many points). Lemma 9.4 of the next chapter makes this precise.

8.2 Exponential-polynomials

An *exponential-polynomial* is a finite sum of polynomials times exponentials:

$$w(t,x) = \sum_{j=1}^{r} p_j(t,x)\, e^{\tau_j t + \xi_j x}.$$

These are exactly the solutions of ODEs with constant coefficients, in each variable. They are the "trivial" solutions of the theory: they live on finitely many points of the characteristic variety.

The key property is a finite-dimension test. Differentiate w as often as you like: you always stay inside a finite-dimensional space.

Example 8.4 (Which functions are exponential-polynomials?).

Function	In EP?	Span of all derivatives
xe^x	yes	span$\{xe^x,\ e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite: $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$
e^{-x^2}	no	$\partial_x^k e^{-x^2} = (\text{degree } k)\cdot e^{-x^2}$: infinite dimension
$1/(1+x^2)$	no	denominators $(1+x^2)^{k+1}$ grow: infinite dimension
$f(x+t)$, f a Gaussian	no	same as the Gaussian

The test "is the span of the derivatives finite-dimensional?" is Lemma 9.8.

8.3 The Chinese remainder theorem

8.3.1 Integers

Solve the system

$$x \equiv 1 \pmod 2, \qquad x \equiv 2 \pmod 3.$$

The unique solution modulo 6 is $x \equiv 5$. In general, if a and b are coprime, a remainder modulo a and a remainder modulo b determine a unique remainder modulo ab. In the language of rings:

$$\mathbb{Z}/(ab) \;\cong\; \mathbb{Z}/(a) \times \mathbb{Z}/(b).$$

8.3.2 ODEs in one variable

Let $p(\partial) = (\partial - 1)(\partial - 2)$. Its solutions are $u = ae^x + be^{2x}$. The two pieces can be extracted by *projector operators*:

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = a\, e^x,$$
$$u_2 = \frac{\partial - 1}{2 - 1}\, u = b\, e^{2x}.$$

Check the first line: $(\partial - 2)(ae^x + be^{2x}) = -ae^x$, and dividing by $1 - 2 = -1$ gives ae^x. The sum is direct:

$$\ker p(\partial) = \ker(\partial - 1) \oplus \ker(\partial - 2).$$

This works because $\partial - 1$ and $\partial - 2$ are coprime: there are polynomials a, b with $a(\partial - 1) + b(\partial - 2) = 1$ (here $a = 1$, $b = -1$).

8.3.3 Two variables: what changes

For the wave operator $P = \tau^2 - \xi^2 = (\tau - \xi)(\tau + \xi)$ the two factors are coprime, but no combination of them equals 1. Indeed

$$(\tau - \xi,\ \tau + \xi) = (\tau,\ \xi),$$

the ideal of the single point $(0, 0)$ where the two lines meet. The quotient

$$\mathcal{R}/(\tau, \xi) \cong \mathbb{C}$$

has dimension 1. This number is the ambiguity of the splitting: the d'Alembert components $f(x + t)$ and $g(x - t)$ are defined only up to a constant moved from one to the other. Theorem 12.6 says that this finite overlap is the *only* failure of the integer picture.

8.4 The companion matrix

A linear ODE of order m becomes a first-order system for the vector of derivatives. For

$$u'' - 3u' + 2u = 0$$

put $X = (u, u')^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}.$$

The characteristic polynomial of C is $\lambda^2 - 3\lambda + 2$, the symbol of the ODE, with roots 1 and 2. The Lagrange projectors are

$$\Pi_1 = \frac{C - 2\,\mathrm{Id}}{1 - 2}, \qquad \Pi_2 = \frac{C - \mathrm{Id}}{2 - 1},$$

and the matrix exponential splits into modes:

$$e^{xC} = e^x\,\Pi_1 + e^{2x}\,\Pi_2.$$

This is the same formula as the projectors of the previous section, written for matrices. It reappears as the Lagrange formula (23) for evolution factors.

Remark 8.5 (Takeaway). Four ideas carry the theory. An *ideal* records all equations a function satisfies. Its *gcd* separates a curve part from finitely many points. *Exponential-polynomials* are the functions with finitely many derivatives up to linear dependence. The *Chinese remainder theorem* splits a solution into pieces, up to a finite overlap.

9 The algebraic language

Starting from: constant coefficients; functions or distributions on $\Omega \subset \mathbb{R}^2$ with complex values. The vocabulary of Section 8. *Why:* three notions carry the whole theory. The *symbol* turns derivatives into polynomials. The *annihilator* collects all equations a function satisfies. *Exponential-polynomials* are the trivial solutions. *Role:* foundation for every later chapter. It proves the two lemmas that every identification argument uses (Lemmas 9.4 and 9.8) and ends with the one-variable prototype (Section 9.4).

9.1 Derivatives as polynomials

A linear operator with constant coefficients is a polynomial in the two partial derivatives:

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j\, \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials are eigenfunctions of every derivative. Write

$$e_{\tau,\xi}(t,x) = e^{\tau t + \xi x}.$$

Then

$$\partial_t\, e_{\tau,\xi} = \tau\, e_{\tau,\xi}, \qquad \partial_x\, e_{\tau,\xi} = \xi\, e_{\tau,\xi},$$

and therefore, for every operator P,

$$P(\partial)\, e_{\tau,\xi} = P(\tau,\xi)\, e_{\tau,\xi}. \tag{21}$$

So applying the operator to an exponential is the same as multiplying by the value of a polynomial. This is the reason why all the algebra below is about polynomials in two variables.

Definition 9.1 (Operator ring, symbol, characteristic variety)**.** The ring of constant-coefficient operators is

$$\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \;\cong\; \mathbb{C}[\tau, \xi].$$

The polynomial $P(\tau,\xi)$ is the *symbol* of $P(\partial)$. The *characteristic variety* of P is its zero set

$$V(P) = \{(\tau,\xi) \in \mathbb{C}^2 :\; P(\tau,\xi) = 0\}.$$

By (21), the exponential $e_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when (τ,ξ) lies on $V(P)$. Fourier modes are the special case $\xi = i\kappa$ with real frequency κ.



Figure 10: Real slices of two characteristic varieties. A single exponential solution is a single point of the variety. A rich, non-exponential solution "fills" the whole curve.

9.2 Annihilators and what a solution knows

A single function is usually killed by many operators. For example, e^x is killed by $\partial_x - 1$, by $\partial_x^2 - 1$, by ∂_t, and by every combination of these.

Definition 9.2 (Annihilator)**.** The annihilator of u is the set of all operators that kill u:

$$\mathrm{Ann}(u) = \{Q \in \mathcal{R} : \ Q(\partial)u = 0\}.$$

$\mathrm{Ann}(u)$ is an ideal of $\mathcal{R}$: if $Q(\partial)u = 0$ and $S \in \mathcal{R}$, then

$$(SQ)(\partial)u = S(\partial)(Q(\partial)u) = 0,$$

and the sum of two annihilators is an annihilator. Three classical facts about $\mathbb{C}[\tau, \xi]$ are used throughout.

Remark 9.3 (Three facts about polynomials in two variables). **(F1) Unique factorisation.** Every nonzero polynomial factors as

$$P = c \prod_i P_i^{e_i},$$

with P_i irreducible and pairwise non-proportional (the "primes"), $e_i \geq 1$ their multiplicities, and c a constant. The factorisation is unique up to constants, so gcd is defined as for integers.

(F2) Bézout. Two polynomials without a common factor have only finitely many common zeros, at most the product of their degrees. Geometrically: two curves without a common component meet in isolated points.

(F3) Nullstellensatz (Hilbert). If a polynomial q vanishes on the common zero set $V(I)$ of an ideal I, then $q^k \in I$ for some $k \geq 1$. In particular, an ideal without common zeros is the whole ring.

All three are classical [1, 8]. They hold over $\mathbb{C}$; the Nullstellensatz fails over $\mathbb{R}$.

The next lemma says that every ideal splits into a "curve part" (one polynomial) and a residue that vanishes only at finitely many points.

Lemma 9.4 (Splitting off the gcd (ledger C3-L2) [proved])**.** *Let $I \subset \mathcal{R}$ be a nonzero ideal and $g = \gcd(I)$. Then*

$$I = g \cdot J, \qquad J = \{f/g : \ f \in I\},$$

where J is an ideal whose zero set $V(J)$ is finite.

[proof: Dim. C3-L2]

Remark 9.5 (Intuition). Divide every element of I by the common factor g. What is left has no common factor, so by Bézout any two "generic" elements of J meet only at finitely many points.

For the annihilator of a function we use a fixed notation:

$$G_u = \gcd \mathrm{Ann}(u), \qquad \mathrm{Ann}(u) = G_u \cdot J_u.$$

G_u is the "genuinely infinite-dimensional" part of the equation of u. J_u records the finitely many exponentials that happen to be present in u.

Example 9.6 (Two extreme cases). **A single exponential.** For $u = e^x$, seen as a function of (t, x):

$$\mathrm{Ann}(u) = (\tau,\ \xi - 1), \qquad G_u = 1, \qquad J_u = \mathrm{Ann}(u).$$

The ideal is that of the single point $(0, 1)$. The function has no infinite part.

A travelling profile. For $u = f(x + t)$ with f a Gaussian:

$$\mathrm{Ann}(u) = (\tau - \xi), \qquad G_u = \tau - \xi, \qquad J_u = \mathcal{R}.$$

The whole ideal is the curve part; there are no isolated points. This is proved in Section 10.

9.3 Exponential-polynomials

Definition 9.7 (Exponential-polynomials)**.** EP is the set of finite sums

$$w(t, x) = \sum_{j=1}^{r} p_j(t, x)\, e^{\tau_j t + \xi_j x},$$

with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. We write $\mathrm{EP}(\Omega)$ for their restrictions to Ω.

For a function w, let $\mathcal{R}w$ be the space of all its derivatives and their linear combinations:

$$\mathcal{R}w = \mathrm{span}\{\partial_t^j \partial_x^k w :\ j, k \geq 0\} = \{Q(\partial)w :\ Q \in \mathcal{R}\}.$$

The map $Q \mapsto Q(\partial)w$ goes from $\mathcal{R}$ onto $\mathcal{R}w$ and its kernel is $\mathrm{Ann}(w)$. Hence

$$\mathcal{R}w \ \cong\ \mathcal{R}/\,\mathrm{Ann}(w).$$

Lemma 9.8 (Finite-dimension criterion (ledger C3-L3) [proved])**.** *For a function or distribution w on a connected Ω:*

$$w \in \mathrm{EP}(\Omega) \iff \dim_{\mathbb{C}}(\mathcal{R}w) < \infty.$$

In that case $V(\mathrm{Ann}\, w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.

[proof: Dim. C3-L3]

Remark 9.9 (Intuition). ($\Rightarrow$) Derivatives of $p\, e^{\tau t + \xi x}$ are polynomials of degree at most $\deg p$ times the same exponential, so they stay in a finite-dimensional space.

($\Leftarrow$) On the finite-dimensional space $\mathcal{R}w$, the derivatives ∂_t and ∂_x act as two commuting matrices. Commuting matrices can be triangularised together. Their joint eigenvalues are the exponents, and their nilpotent parts produce the polynomial prefactors.

Example 9.10 (The test on two functions). For $w = xe^x$ the derivatives are $xe^x + e^x$, $xe^x + 2e^x$, and so on: all lie in $\mathrm{span}\{xe^x, e^x\}$, of dimension 2. So $w \in \mathrm{EP}$.

For $w = e^{-x^2}$, the k-th derivative is a polynomial of degree exactly k times e^{-x^2}. These are linearly independent for all k, so $\dim \mathcal{R}w = \infty$ and $w \notin \mathrm{EP}$.

Remark 9.11 (Why it matters). Lemma 9.8 is the computational heart of the method. The question "is w an exponential-polynomial?" is a question of finite-dimensional linear algebra. It is the test used in the identification criterion (Theorem 10.2) and in Step 3 of the algorithm.

9.4 One variable: the ODE prototype

In one variable every object of the later chapters appears with explicit formulas and without the geometry of curves. We consider an ODE $p(D)u = 0$ in the variable x, with $D = d/dx$ in this section only.

Remark 9.12 (Setting). The symbol factors over $\mathbb{C}$ as

$$p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j},$$

with distinct roots λ_j and multiplicities $m_j \geq 1$. The order is $m = \sum_j m_j$. The solution space is

$$\ker p(D) = \Big\{ \sum_j q_j(x)\, e^{\lambda_j x} : \ \deg q_j < m_j \Big\}.$$

The form of the solutions comes from one computation. For one root,

$$(D - \lambda)(q\, e^{\lambda x}) = q'\, e^{\lambda x} :$$

applying $D - \lambda$ lowers the degree of q by one. So $(D-\lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $\deg q < m_j$.

Example 9.13 (Reading the solution space from the roots). Let $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$. The roots are 1 (multiplicity 2) and -2 (multiplicity 3), and the order is 5. A basis of solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2e^{-2x}.$$

9.4.1 Identification by the Wronskian

Proposition 9.14 (ODE identification (ledger R4-T1) [proved]). *Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of basis functions actually present in u. It equals p if and only if*

$$u,\ u',\ \ldots,\ u^{(m-1)}$$

are linearly independent (nonvanishing Wronskian), that is, iff every root of p appears in u with its full multiplicity.

[proof: Dim. R4-T1]

Example 9.15 (A missing root). Let $p(\lambda) = (\lambda - 1)(\lambda - 2)$ and $u = 3e^x$. Then $u' = u$, so u and u' are dependent and the minimal annihilator is $\lambda - 1$, of degree 1. The root 2 is missing from u, and u cannot tell p from $\lambda - 1$.

9.4.2 The companion matrix and the spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ and put $X = (u, u', \ldots, u^{(m-1)})^\top$. Then $X' = CX$ with the companion matrix

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC} X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the exponential has the spectral form

$$e^{xC} = \sum_{j=1}^{m} e^{\lambda_j x}\, \Pi_j, \qquad \Pi_j = \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \tag{22}$$

where the Π_j are the Lagrange projectors onto the eigenspaces.

Proposition 9.16 (Projectors of the companion matrix (ledger II-T2, II-T3) [proved])**.** *For an operator monic in τ with pairwise coprime factors, the projectors onto the factor components are explicit rational functions of ξ. Their poles lie exactly where two factor varieties meet. They are polynomial if and only if the varieties are disjoint, that is, iff $(Q_i, Q_j) = \mathcal{R}$ for all $i \neq j$.*

[proof: Dim. II-T2]

Remark 9.17 (Warning). [title=The companion matrix is not normal in general] An early conjecture said that the companion matrix is normal exactly when $p = s^m - c$ with $|c| = 1$. For $m = 2$ this is [false]: the polynomial $s^2 + s - 1$ is a counterexample. The correct normality criterion for $m = 2$ is (ledger II-P5) [proved], and it is formalised in Lean. Consequence: Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on coefficient vectors.

9.4.3 The jet is a cyclic vector

Proposition 9.18 (Cyclic vector (ledger II-C1) [proved])**.** *Let P be monic in τ of degree k. Then* $\mathrm{Ann}(u) = (P)$ *if and only if the t-jet*

$$u,\ \partial_t u,\ \ldots,\ \partial_t^{k-1} u$$

is linearly independent over the ring $\mathbb{C}[\partial_x]$.

[proof: Dim. II-C1]

This is the two-variable Wronskian: it replaces "the derivatives are independent" by "the time derivatives are independent with coefficients that are operators in x".

Remark 9.19 (Takeaway). In one variable the polynomial encodes the solution space. Its roots give the exponential rates, the multiplicities give the polynomial prefactors, the companion matrix represents differentiation, and the Lagrange projectors isolate the modes. In two variables the same structure survives. The minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Section 10) and the Lagrange projectors become Eq. (23) (Section 12).

10 Identification: what one solution determines

Starting from: $\Omega \subset \mathbb{R}^2$ open and *connected*; $u \in \mathcal{D}'(\Omega)$; constant coefficients; an operator P with $P(\partial)u = 0$ is available (any annihilator, not necessarily the "true" one), with irreducible factorisation $P = c\prod_i P_i^{e_i}$. *Why:* before splitting a solution into pieces one must know its equation. We ask exactly when u alone pins down P, and what the intrinsic operator of u is when it does not. *Role:* first link of the chain in Figure 9. Its output is the pair (G_u, J_u), which every later chapter uses, and the test "$\in$ EP", which Step 3 of the algorithm implements.

10.1 The question in algebraic form

Suppose $P(\partial)u = 0$. Then every multiple SP also kills u, so the ideal (P) is contained in $\mathrm{Ann}(u)$:

$$(P) \subseteq \mathrm{Ann}(u).$$

We say that u *identifies* P when the two coincide. In that case every equation satisfied by u is a consequence of P, and P is determined by u up to a constant factor.

Why can the inclusion be strict? Because u may satisfy extra equations that P knows nothing about. The single Fourier mode of the heat equation is the typical case: it also satisfies $u_{xx} = -\kappa^2 u$.

Connectedness of Ω is necessary. On a disconnected region the two pieces can satisfy different equations, and no criterion on u as a whole can see that.

10.2 The criterion

Remark 10.1 (Terminology). • P is *reduced* if all $e_i = 1$ (no repeated factor).

- P is *monic in* τ if its leading coefficient as a polynomial in τ is a nonzero constant. Then $k = \deg_\tau P$ is its order in time, and the *t-jet* of u is the vector $(u, \partial_t u, \dots, \partial_t^{k-1} u)$.
- For each irreducible factor we form the partial quotients

$$w_i = \Big(\frac{P}{P_i}\Big)(\partial)\, u, \qquad w_i' = \Big(\frac{P}{P_i^{e_i}}\Big)(\partial)\, u.$$

In w_i we remove one copy of P_i; in w_i' we remove the whole power.

Theorem 10.2 (Identification criterion (ledger C3-T1) [proved])**.** *Let Ω be connected, $u \in \mathcal{D}'(\Omega)$ and $P(\partial)u = 0$.*
(a) $\mathrm{Ann}(u) = (P)$ if and only if $w_i \notin \mathrm{EP}(\Omega)$ for every irreducible factor P_i.
(b) $V(\mathrm{Ann}\, u) = V(P)$ if and only if $w_i' \notin \mathrm{EP}(\Omega)$ for every i.
(c) If P is reduced, (a) and (b) coincide.
(d) If P is monic in τ, the test can be carried out on the t-jet of u (Proposition 9.18).
[proof: Dim. C3-T1]

Remark 10.3 (Intuition). Strip one irreducible factor off P and look at what is left: w_i lies in the kernel of P_i.

If w_i is a "generic" member of that kernel, not a finite sum of exponentials, then no smaller operator can kill u.

If w_i is an exponential-polynomial, then u contains a finite-dimensional piece. That piece is also killed by an operator of lower order, and u cannot tell the large equation from the small one.

Proposition 10.4 (Uniqueness of the identified operator)**.** *If $\mathrm{Ann}(u) = (P)$, then every operator Q with $Q(\partial)u = 0$ is a multiple $Q = SP$ with $S \in \mathcal{R}$. In particular P is, up to a nonzero constant, the only annihilator of minimal degree.*
[proof: Dim. C3-T1]

The proof is one line: $Q \in (P)$ means $Q = SP$, and degrees add, so $\deg Q \geq \deg P$ with equality only when S is a constant.



Figure 11: The heat curve $\tau = \xi^2$. Left: a solution that is not an exponential-polynomial (the heat kernel) uses the whole curve, and $\mathrm{Ann}(u) = (P)$. Right: a finite sum of exponentials lives on finitely many points. Then $\mathrm{Ann}(u)$ contains polynomials that vanish at those points but are not multiples of P.

Example 10.5 (The heat equation, three cases). Take $P = \tau - D\xi^2$ with $D > 0$. It is irreducible, so $P/P_1 = 1$ and the criterion reads

$$\mathrm{Ann}(u) = (P) \iff u \notin \mathrm{EP}.$$

One Fourier mode. Let $u = e^{-D\kappa^2 t + i\kappa x}$. Then $u_t = -D\kappa^2 u$ and $u_{xx} = -\kappa^2 u$. So u is also killed by $\partial_x - i\kappa$ and by $\partial_t + D\kappa^2$:

$$\mathrm{Ann}(u) = (\tau + D\kappa^2,\ \xi - i\kappa).$$

This is the ideal of one point of $V(P)$, strictly larger than (P). Indeed P belongs to it:

$$P = (\tau + D\kappa^2) - D\,(\xi - i\kappa)(\xi + i\kappa).$$

A finite superposition. For $u = \sum_{j=1}^{r} c_j e^{-D\kappa_j^2 t + i\kappa_j x}$ with distinct κ_j, $u \in \mathrm{EP}$. Its annihilator is the ideal of the r points $(-D\kappa_j^2,\ i\kappa_j)$. It contains many operators that are not multiples of P.

The heat kernel. Let $u = t^{-1/2} e^{-x^2/(4Dt)}$ for $t > 0$. Its x-derivatives are polynomials of growing degree times the same Gaussian. They span an infinite-dimensional space, so $u \notin \mathrm{EP}$ (Lemma 9.8). Hence $\mathrm{Ann}(u) = (P)$: every constant-coefficient equation satisfied by the heat kernel is a multiple of $\tau - D\xi^2$.

Message. A "poor" solution remembers only finitely many exponentials and cannot tell the heat equation from many others. A "rich" solution remembers the whole equation.

Example 10.6 (The wave equation and d'Alembert (ledger C3-T5) [proved]). Let $P = (\tau - c\xi)(\tau + c\xi)$ with real $c \neq 0$, and $u = f(x - ct) + g(x + ct)$. The two partial quotients are

$$w_1 = (\partial_t + c\partial_x)\,u = 2c\,g'(x + ct),$$
$$w_2 = (\partial_t - c\partial_x)\,u = -2c\,f'(x - ct).$$

So the criterion says:

$$\mathrm{Ann}(u) = (P) \iff f \notin \mathrm{EP} \ \text{ and } \ g \notin \mathrm{EP}.$$

A single travelling wave ($g = 0$) never identifies the wave equation: it also satisfies the first-order transport equation. The same theorem covers Klein–Gordon $\tau^2-\xi^2+\mu^2$ and the telegraph operator $\tau^2 + \alpha\tau - c^2\xi^2$: both are irreducible conics, so they are identified iff $u \notin \mathrm{EP}$. For the beam $\tau^2 + \xi^4 = (\tau - i\xi^2)(\tau + i\xi^2)$ both Schrödinger parts must be outside EP.

[proof: Dim. C3-T5]

Remark 10.7 (Why it matters). Theorem 10.2 turns "does u identify P?" into finitely many membership tests "$\in \mathrm{EP}$". Each test is a finite-dimensional computation (Lemma 9.8). Whether the tests can be carried out from the data one actually has is the subject of Section 11.

10.3 The intrinsic operator and the spectral form

Theorem 10.2 says when (P) is the whole annihilator. When it is not, what does u determine? The answer uses the two pieces of $\mathrm{Ann}(u)$ given by Lemma 9.4.

Remark 10.8 (Point spectrum). Recall $\mathrm{Ann}(u) = G_u\, J_u$ with $G_u = \gcd \mathrm{Ann}(u)$. The *point spectrum* of u is the finite set

$$\sigma_p(u) = V(J_u) \subset \mathbb{C}^2.$$

Its points are the exponents (τ_0, ξ_0) of the exponentials $e^{\tau_0 t+\xi_0 x}$ that are "present" in u. The word spectrum is meant in the algebraic sense, not as the spectrum of an operator on a Hilbert space. We write $G_u \sim P$ when the two agree up to a nonzero constant.

Theorem 10.9 (Spectral characterisation (ledger P4-T2) [proved]). *Let P be an annihilator of u. Then*

$$\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset \quad \textit{and} \quad G_u \sim P.$$

Moreover the exponential-polynomial part of the derivative space is the finite quotient

$$\mathcal{R}u \cap \mathrm{EP} \;\cong\; \mathcal{R}/J_u.$$

[proof: Dim. P4-T2]

Example 10.10 (Reading the spectrum). **One exponential.** For $u = e^{\tau_0 t+\xi_0 x}$: $\mathrm{Ann}(u) = (\tau - \tau_0,\ \xi - \xi_0)$, so $G_u = 1$ and $\sigma_p(u) = \{(\tau_0, \xi_0)\}$. A single exponential has no infinite part and a one-point spectrum.

Two exponentials. For $u = e^{\tau_1 t+\xi_1 x} + e^{\tau_2 t+\xi_2 x}$ with distinct exponents: $G_u = 1$ and $\sigma_p(u)$ has two points.

A rich solution. If $\mathrm{Ann}(u) = (P)$ then $\sigma_p(u) = \emptyset$ and $G_u \sim P$.

The operator G_u is the correct "operator intrinsic to u". It depends only on u, while P depends on which annihilator was chosen. The next theorem computes it from any annihilator.

Theorem 10.11 (Intrinsic operator (ledger C3-T6) [proved]). *Let $P = c\prod_i P_i^{e_i}$ be any annihilator of u. For each i let*

$$k_i = \max\Big\{k \le e_i :\ \Big(\frac{P}{P_i^k}\Big)(\partial)u \in \mathrm{EP}\Big\}$$

(with $k_i = 0$ if no such $k \ge 1$ exists), and $f_i = e_i - k_i$. Then

$$G_u \sim \prod_i P_i^{f_i},$$

and $\mathrm{Ann}(u)$ *is principal if and only if* $G_u(\partial)u = 0$. *Computing* G_u *takes at most* $\sum_i e_i$ *tests "*$\in$ EP*".*

[proof: Dim. C3-T6]

Example 10.12 (A repeated factor). Let G be the heat kernel ($D = 1$), $P = (\tau - \xi^2)^3$ and $u = t\,G$. Then:

$$\Big(\frac{P}{P_1}\Big)(\partial)u = (\partial_t - \partial_x^2)^2(tG) = 0 \in \mathrm{EP},$$
$$\Big(\frac{P}{P_1^2}\Big)(\partial)u = (\partial_t - \partial_x^2)(tG) = G \notin \mathrm{EP}.$$

So $k_1 = 1$, $f_1 = 2$, and

$$G_u = (\tau - \xi^2)^2, \qquad \mathrm{Ann}(u) = (G_u).$$

The function tG identifies the *square* of the heat operator, not its cube.

Remark 10.13 (Warning). [title=Why G_u and not the variety] A tempting stronger statement replaces G_u by the variety $V(\mathrm{Ann}\,u)$. It is correct only for reduced P, because zero sets do not see multiplicities.

Example. Let $P = \tau^2$ and $u = e^{x^2}$. Then $u_{tt} = 0$. No nonzero polynomial in ∂_x alone kills u, so

$$\mathrm{Ann}(u) = (\tau).$$

Here $\sigma_p(u) = \emptyset$, but $G_u = \tau \not\sim P$. So $\mathrm{Ann}(u) \neq (P)$, while $V(\mathrm{Ann}\,u) = \{\tau = 0\} = V(P)$. The variety alone would wrongly say that u identifies τ^2.

10.4 Refinements

10.4.1 Evolutions: identification inside a finite class

For an evolution $u_t = a(\partial_x)u$ the symbol $P = \tau - a(\xi)$ has degree one in τ, so it is irreducible. The criterion says: $\mathrm{Ann}(u) = (P)$ iff $u \notin \mathrm{EP}$. When u *is* an exponential-polynomial, we can still ask whether a is recovered among polynomials of degree at most n.

Theorem 10.14 (Evolutions (ledger C3-T4) [proved])**.** *Let* $P = \tau - a(\xi)$ *and* $P(\partial)u = 0$ *on a connected* Ω.

(a) $\mathrm{Ann}(u) = (P)$ *if and only if* $u \notin \mathrm{EP}(\Omega)$.

(b) If $u = \sum_k q_k\, e^{\tau_k t + \xi_k x}$ *with distinct exponents and nonzero* q_k, *put* $\delta_k = \deg_x q_k$ *and*

$$D = \sum_k (\delta_k + 1).$$

If $\deg a \le n$, *then among all* b *of degree at most* n, *the evolutions* $\tau - b$ *that kill* u *are exactly*

$$b \in a + \prod_k (\xi - \xi_k)^{\delta_k + 1} \cdot \mathbb{C}[\xi]_{\le n - D}.$$

So a *is identified in degree* $\le n$ *if and only if* $D \ge n + 1$.

[proof: Dim. C3-T4]

This is Hermite interpolation: a polynomial of degree n is fixed by $n + 1$ conditions, and each mode $q_k\, e^{\tau_k t + \xi_k x}$ supplies $\delta_k + 1$ of them.

Example 10.15 (Advection–diffusion from two modes). Take $u_t = -cu_x + Du_{xx}$, so $a(\xi) = -c\xi + D\xi^2$ has degree $n = 2$. Two simple modes give $D = 2 < 3$: the pair (c, D) is not identified, because one free parameter remains. Three distinct modes give $D = 3$: both coefficients are fixed.

10.4.2 A naive transfer that fails

On a periodic domain an earlier result said: if the Fourier expansion of u has infinitely many frequencies on every factor curve, then $\mathrm{Ann}(u) = (P)$ (ledger R4-T5). On a *bounded* domain this transfer is false, because the expansion is not unique.

Remark 10.16 (Counterexample: The spectrum of an expansion is not intrinsic (ledger C3-C1) [false]). Let $P = \tau + \xi$ and $\Omega = (0,1)^2$. On $(-\pi, \pi)$ the Fourier series

$$e^s = \sum_{k\in\mathbb{Z}} c_k\, e^{iks}, \qquad c_k = \frac{\sinh(\pi)(-1)^k}{\pi(1-ik)} \neq 0,$$

converges in L^2. So $u = e^{x-t} = \sum_k c_k\, e^{ik(x-t)}$ has infinitely many exponents on the line $\tau + \xi = 0$. Yet

$$\mathrm{Ann}(u) = (\tau + 1,\ \xi - 1) \supsetneq (\tau + \xi).$$

[proof: Dim. C3-C1]

The criterion must therefore be stated on u itself, as in Theorem 10.2, not on an expansion of u.

10.4.3 Symmetries and local data

Proposition 10.17 (Translation symmetry (ledger C3-P3) [proof written, not audited]). *Suppose u is invariant under translations along a direction, $\ell(\partial)u = 0$ with $\ell = \alpha\tau + \beta\xi \neq 0$, and $P(\partial)u = 0$ with P nonconstant. If $P \not\sim \ell$, then $\mathrm{Ann}(u) \neq (P)$. If moreover $\ell \nmid P$, then $u \in \mathrm{EP}$.*

[proof: Dim. C3-P3]

So a symmetry with constant characteristic destroys identification, and it is a special case of the criterion. Symmetries with variable coefficients are invisible in $\mathbb{C}[\partial]$: the heat kernel is scale invariant, $(2t\partial_t + x\partial_x + 1)G = 0$, yet $\mathrm{Ann}(G) = (\tau - \xi^2)$. Identifiability depends on the ring of coefficients.

Proposition 10.18 (Local data (ledger C3-P5) [proof written, not audited]). *Identification can be lost on a smaller region. Let $P = \tau + \xi$, $F(s) = e^s + \varphi(s)$ with φ a smooth bump supported in $(2,3)$, and $u = F(x-t)$. On $(0,1)\times(-5,5)$, $u \notin \mathrm{EP}$ and $\mathrm{Ann}(u) = (\tau + \xi)$. On the smaller square $(0,1)^2$, $u = e^{x-t}$ and the annihilator is the ideal of a point.*

For elliptic P, every solution is real-analytic, and then $\mathrm{Ann}(u|_{\Omega'}) = \mathrm{Ann}(u)$ for every connected $\Omega' \subset \Omega$.

[proof: Dim. C3-P5]

Remark 10.19 (Takeaway). A solution identifies its equation exactly when, after removing any one irreducible factor, what is left is not a finite sum of exponentials. In general, u identifies only its intrinsic operator G_u. The difference between P and G_u is a finite set of exponentials, the point spectrum $\sigma_p(u)$.

11 Finite data and decidability

Starting from: the criterion of Section 10: $\mathrm{Ann}(u) = (P)$ iff no w_i is an exponential-polynomial. It is a statement about the whole function u. *Why:* in practice we hold finitely many numbers: a jet at a point, samples, or a formula. We ask which data suffice, which never suffice, and when the test "$\in$ EP" becomes an algorithm. *Role:* tells Step 3 of the algorithm what it can and cannot promise. The answer depends on the data model: positive for exponential-polynomials, negative for a black box, positive again with one extra integer.

11.1 Three data models

The reconstruction is exact, so the input must be of an exact kind. Three models are analysed throughout (ledger C3-D1).

M1 A closed formula for u, for example a finite sum of exponentials and Gaussians.

M2 The jet of u at a point p_0: all partial derivatives up to some order, given as exact numbers. In the "black-box" version $\mathrm{M2}^+$ we may also ask a *zero-test oracle* whether $Q(\partial)u = 0$ for a given Q.

M3 u real-analytic on a connected Ω, described by finitely many linear equations with polynomial coefficients (a *holonomic* description) with algebraic numbers as coefficients.

Remark 11.1 (Words used in this chapter).

Jet of order k the list of all partial derivatives of u of order $\le k$ at p_0: its Taylor coefficients.

Zero-test oracle a procedure that says whether a given expression built from u is identically zero.

Holonomic described by linear differential equations with polynomial coefficients whose solution space is finite-dimensional.

Weyl algebra A_1 or A_2: the ring of differential operators with polynomial coefficients.

11.2 Exponential-polynomial data: a finite threshold

When u itself is an exponential-polynomial, it never identifies its equation in the whole ring: its annihilator has finitely many zeros, while $V(P)$ is a curve (ledger C3-C2). What we can ask is whether u identifies P *among operators of degree at most* n.

Theorem 11.2 (Bézout threshold (ledger C3-T2) [proved]). *Let $u = \sum_k c_k\, e^{\langle z_k,(t,x)\rangle}$ be a sum of simple modes with distinct exponents $z_k \in V(P)$, P reduced, and $n \ge \deg P$. If every factor curve carries enough exponents,*

$$\big|\{z_k\} \cap V(P_i)\big| > n \cdot \deg P_i \qquad \text{for every } i,$$

then the annihilators of degree at most n are exactly the multiples of P:

$$\mathrm{Ann}(u) \cap \mathcal{R}_{\le n} = P \cdot \mathcal{R}_{\le n-\deg P}.$$

The threshold cannot be lowered uniformly: for $P = \tau - \xi^2$ and every n there are $2n$ exponents on the parabola that do not suffice.

[proof: Dim. C3-T2]

Remark 11.3 (Intuition). An operator Q of degree n that kills u must vanish at every exponent z_k. If P_i does not divide Q, the two curves meet in at most $n \deg P_i$ points (Bézout). More points than that force P_i to divide Q.

Example 11.4 (The parabola needs $2n+1$ points). Let $P = \tau - \xi^2$, so $\deg P = 2$. Take the $2n$ exponents

$$Z_n = \{(j^2, \pm j) : \ j = 1, \dots, n\} \subset V(P).$$

The operator $Q_n = \prod_{j=1}^{n} (\tau - j^2)$ has degree n, vanishes on Z_n, and is not a multiple of $\tau - \xi^2$ because it does not depend on ξ. So $2n$ points are not enough. With $2n+1$ points the theorem applies.

The next theorem says how many derivatives at a single point are needed.

Theorem 11.5 (Jet threshold $2d-1$ (ledger C3-T3) [proved]). *Let $0 \neq u \in \mathrm{EP}(\Omega)$ on a connected Ω, and let $d = \dim \mathcal{R}u$. Then:*

(i) the operators of degree at most n that kill u are determined by the jet of u of order $n + d - 1$ at any point;

(ii) $\mathrm{Ann}(u)$ is generated in degree at most d, so it is determined by the jet of order $2d - 1$;

(iii) the order $2d - 1$ is optimal in the worst case, already in one variable: for $u = \sum_{j=1}^{d} e^{jx}$ the jet of order $2d - 2$ does not determine $\mathrm{Ann}(u)$.

An upper bound $d' \geq d$ may replace d everywhere.

[proof: Dim. C3-T3]

Example 11.6 (One exponential, $d = 1$). For $u = e^{2x}$, $d = 1$ and the threshold is $2d - 1 = 1$. The values $u(0) = 1$ and $u'(0) = 2$ give $u' = 2u$, so $\mathrm{Ann}(u)$ contains $\xi - 2$, and τ comes from the absence of t. The value $u(0)$ alone (order 0) cannot distinguish e^{2x} from e^{3x}.

11.3 What no finite data can show

For a solution outside EP, the situation reverses.

Theorem 11.7 (No finite certificate (ledger C3-P1) [proved] / [from cited source]). *Let Ω be connected, P nonconstant and $u \in \ker P$. For every order K there is an exponential-polynomial $v \in \ker P$ with the same jet of order K at p_0 as u, and $\mathrm{Ann}(v) \neq (P)$.*

On convex Ω the same holds for any finite family of continuous linear functionals (Malgrange's density theorem) [from cited source]. On a non-convex region a single functional can certify.

[proof: Dim. C3-P1]

Remark 11.8 (Intuition). Every finite jet is matched by a finite sum of exponentials taken from the curve $V(P)$. Finitely many numbers cannot distinguish u from that sum, and the sum never identifies anything. So an algorithm that sees finitely many numbers can only *refute* identification, never confirm it.

The black-box version is sharper.

Theorem 11.9 (Black-box impossibility (ledger P4-T5) [proved]). *In the model $M2^+$ (exact jets plus a zero-test oracle), for every nonconstant P there is a solution u with $\mathrm{Ann}(u) = (P)$ such that the following holds. For every jet order k_0 and every finite set of zero tests, some exponential-polynomial $v \in \ker P$ gives the same jet and the same answers, while $\mathrm{Ann}(v) \neq (P)$. Hence no algorithm decides "$\mathrm{Ann}(u) = (P)$" correctly on all inputs.*

[proof: Dim. P4-T5]

11.4 One extra integer makes it decidable

Theorem 11.10 (Decidability with a bound (ledger P4-T4) [proved]). *In the model $M2^+$:*

(i) for every finite-dimensional space of operators $\mathcal{L}$, the annihilators of u in $\mathcal{L}$ are computable;

(ii) "$\mathrm{Ann}(u) \neq (P)$" is semi-decidable: an algorithm stops with "yes" exactly when it is true;

(iii) if an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ is given, "$\mathrm{Ann}(u) = (P)$" is decidable. With

$$N = B + \deg P - 1,$$

one has $\mathrm{Ann}(u) = (P)$ iff $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq N} = (P) \cap \mathcal{R}_{\leq N}$.

The bound also makes G_u computable.

[proof: Dim. P4-T4]

Remark 11.11 (Intuition). If identification fails, some w_i is an exponential-polynomial of dimension at most B. Its annihilator has a generator of degree at most B that is not a multiple of P_i. Multiplying by P/P_i gives a witness of degree at most $B + \deg P - 1$. So it is enough to look for witnesses up to that degree, and that is a finite linear-algebra problem.

Is the bound B the least data that works? The minimal sufficient datum is "a degree D such that, if identification fails, a witness of degree $\leq D$ exists" (ledger OP-ALG-T3) [proof written, not audited]. The bound B is strictly stronger on some regions when P has at least two irreducible factors counted with multiplicity. Whether it is strictly stronger on every region is open [open] (ledger OP-ALG-Q2).

11.5 A decidable class: holonomic solutions

Theorem 11.12 (Holonomic model (ledger OP-ALG-T1, P4-T6) [proved mod. (AC)]). *In the model M3 (real-analytic u with a holonomic description over algebraic numbers), assume (AC): jets at an algebraic point are computable in a computable field. Then*

- *"$Q(\partial)u \in \mathrm{EP}$?" is decidable for every Q;*
- *"$\mathrm{Ann}(u) = (P)$?" and "$V(\mathrm{Ann}\, u) = V(P)$?" are decidable, and G_u is computable;*
- *a bound B and a finite set of generators of $\mathrm{Ann}(u)$ are computable.*

[proof: Dim. OP-ALG-T1]

Example 11.13 (The heat kernel plus an exponential). Let $u = G + e^{t+x}$, with G the heat kernel. The slice $t = 1$ is $x \mapsto G(1, x) + e^{1+x}$, a function with a holonomic description. The algorithm finds that the slice is not an exponential-polynomial, and concludes $\mathrm{Ann}(u) = (\tau - \xi^2)$. For the slice of G alone the bound is $B = 0$: the heat kernel identifies the heat equation without further computation.

11.6 Variable coefficients: the first limits

With coefficients that depend on x, the ring of operators becomes the Weyl algebra. Two of its features break the constant-coefficient picture.

Proposition 11.14 (Restriction does not identify (ledger P4-C1) [proof written, not audited]). *For every nonzero smooth u of one variable, the action of the Weyl algebra A_1 on the module $A_1 \cdot u$ is faithful. Yet for $u = e^x$ and the class $\mathcal{L} = \mathrm{span}\{1, \partial\}$, the operator $\partial - 1$ kills u. So "the*

restriction is injective" does not detect identification. The correct object is the evaluation map $L \mapsto Lu$*, whose kernel is* $\mathrm{Ann}(u) \cap \mathcal{L}$*.*

[proof: Dim. P4-C1]

Theorem 11.15 (Saturated criterion (ledger OP-ALG-T4) [proof written, not audited])**.** *Let* P *be monic in* ∂_t *of degree* m*, with coefficients in a ring* $\mathfrak{A}$ *of operators in* x*, and* $Pu = 0$*. Then* u *identifies* P *if and only if the* t*-jet* $u, \partial_t u, \dots, \partial_t^{m-1} u$ *is free over* $\mathfrak{A}$*: no nontrivial relation* $\sum_{j<m} b_j \, \partial_t^j u = 0$ *with* $b_j \in \mathfrak{A}$*.*

[proof: Dim. OP-ALG-T4]

In the commutative case this is Proposition 9.18. With variable coefficients the reduction of the test to the finitely many quotients P/P_i fails: an explicit counterexample with $\theta = x\partial_x$ exists (ledger OP-ALG-C1) [proof written, not audited]. In the full Weyl algebra A_2, a holonomic solution never identifies its equation (ledger OP-ALG-P5) [proof written, not audited].

11.7 Summary

Data	What can be done	Meaning for the algorithm	Status
$u \in$ EP	jet of order $2d-1$ determines $\mathrm{Ann}(u)$; Bézout threshold for modes	a finite computation with a guarantee	[proved]
$u \notin$ EP, finite data	no finite certificate	output "not refuted", never "identified"	[proved] / [from cited source]
black box	"$\in$ EP?" undecidable	extra hypothesis needed	[proved]
bound B given	decidable, test degree $B + \deg P - 1$	stops with a certain verdict	[proved]
holonomic, algebraic	decidable	exact symbolic route	[proved mod. (AC)]
variable coefficients	restriction never identifies; jet freeness criterion	restrict to a finite class	[proof written, not audited]

Table 6: When the identification test can be carried out, and what each case means for Step 3 of the algorithm.

Remark 11.16 (Takeaway)*.* Identification is an infinite statement about u. It cannot be verified from finitely many numbers, and no black box can decide it. One extra integer, a bound on how much of u is exponential-polynomial, turns it into a finite computation.

12 Decomposition into components

Starting from: the operator is known and factored: $P = c \prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime. For the main theorem Ω is open, connected and simply connected (more precisely $H^1_{dR}(\Omega) = 0$). *Why:* once P is known, we want to split u into the parts generated by each piece of P, so that the pieces can be studied separately. *Role:* second link of the chain. The split exists for *factors* of the symbol, with a finite and exactly known ambiguity. It fails for additive *terms*, and that failure is what Section 14 measures.

12.1 Which pieces? Terms versus factors

An operator can be broken up in two different ways. Take the heat operator

$$L = \partial_t - D\partial_x^2.$$

Its *terms* are $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$: they are added. Its symbol $\tau - D\xi^2$ is irreducible, so it has only one *factor*. For the wave operator the situation is the opposite: $\tau^2 - \xi^2$ has two terms and also two factors, $\tau - \xi$ and $\tau + \xi$. The first question is which kind of piece gives a decomposition of the solution.

12.1.1 Terms do not give components

Proposition 12.1 (Splitting by terms fails (ledger R2-T3) [proved])**.** *Let $L = L_1 + L_2$ with constant coefficients. If $u = a + b$ with $L_1 a = 0$ and $L_2 b = 0$ solves $Lu = 0$, then u is also killed by $L_1 L_2$:*

$$\ker L \cap (\ker L_1 + \ker L_2) \;\subseteq\; \ker(L,\; L_1 L_2).$$

The common zero set of L and $L_1 L_2$ is $V(L_1) \cap V(L_2)$. When this set is finite, only a finite-dimensional space of exponential-polynomial solutions can be split by terms.

[proof: Dim. R2-T3]

Remark 12.2 (Intuition). The two operators commute, so

$$L_1 L_2\, u = L_2(L_1 a) + L_1(L_2 b) = 0.$$

A solution split by terms satisfies one extra equation. A generic solution does not.

Example 12.3 (The heat equation cannot be split by terms). Take $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$. The kernels are

$$\ker L_1 = \{f(x)\}, \qquad \ker L_2 = \{a(t) + b(t)\, x\}.$$

Every element $f(x) + a(t) + b(t)x$ of the sum is killed by $\partial_t \partial_x^2$. A heat solution is not. For $u = e^{-D\kappa^2 t}\cos \kappa x$:

$$\partial_x^2 u = -\kappa^2 u, \qquad \partial_t \partial_x^2 u = D\kappa^4 u \neq 0.$$

So a generic heat solution is *not* a sum of a solution of $u_t = 0$ and a solution of $u_{xx} = 0$.

The exceptions are few. The function $u = \frac{1}{2}x^2 + Dt$ solves the heat equation and lies in $\ker L_1 + \ker L_2$, but it is a polynomial: the exceptional space is finite-dimensional.

12.1.2 Factors do give components

Example 12.4 (The wave equation and d'Alembert). Let $c > 0$ and

$$P = \tau^2 - c^2\xi^2 = (\tau - c\xi)(\tau + c\xi) = Q_1 Q_2.$$

A direct check gives

$$(\partial_t - c\partial_x)\, f(x + ct) = 0, \qquad (\partial_t + c\partial_x)\, g(x - ct) = 0.$$

So $\ker Q_1 = \{f(x + ct)\}$ (profiles moving left) and $\ker Q_2 = \{g(x - ct)\}$ (profiles moving right). D'Alembert's formula says that every solution is a sum of the two:

$$\ker P = \ker Q_1 + \ker Q_2.$$

The pieces are unique up to a constant: $f \mapsto f + k$, $g \mapsto g - k$. This ambiguity has dimension 1, and it equals $\dim \mathcal{R}/(Q_1, Q_2)$:

$$Q_1 + Q_2 = 2\tau, \qquad Q_2 - Q_1 = 2c\xi \quad \Longrightarrow \quad (Q_1, Q_2) = (\tau, \xi), \qquad \mathcal{R}/(\tau, \xi) \cong \mathbb{C}.$$

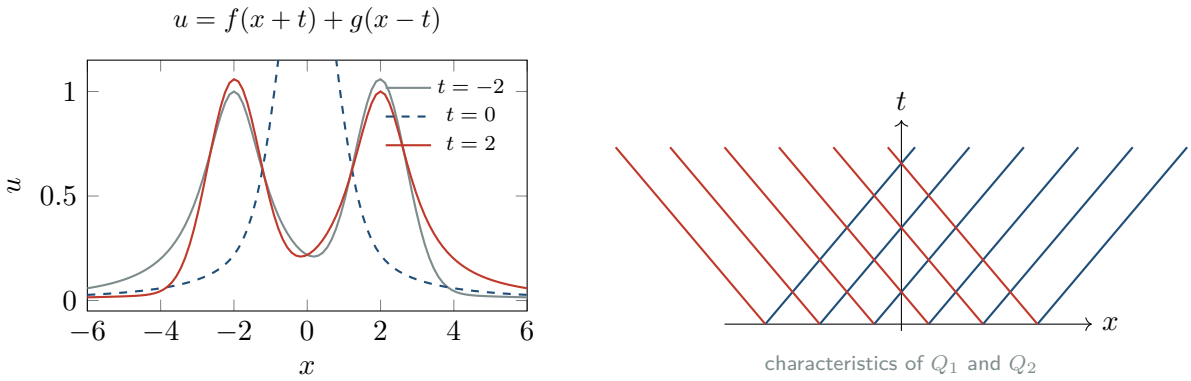


Figure 12: Left: the d'Alembert solution with $f(s) = 1/(1+s^2)$ moving left and $g(s) = e^{-s^2}$ moving right ($c = 1$). Right: the two families of characteristic lines, one per factor. Each factor of the symbol carries one rigidly translating component.

12.2 Why factors work: the Chinese remainder theorem

One variable. If two polynomials Q_1, Q_2 in one variable are coprime, there are polynomials a, b with

$$a\,Q_1 + b\,Q_2 = 1.$$

Then every u with $Q_1 Q_2 u = 0$ splits explicitly as

$$u = \underbrace{b\,Q_2\,u}_{\in \ker Q_1} + \underbrace{a\,Q_1\,u}_{\in \ker Q_2},$$

and the sum of the two kernels is direct.

Two variables. "Coprime" now means only that the common zeros are finitely many (Bézout), not that $(Q_1, Q_2) = \mathcal{R}$. The finite quotient $\mathcal{R}/(Q_1, Q_2)$ measures the failure of the one-variable picture.

Lemma 12.5 (Finite quotient (ledger C12-L3) [proved])**.** *Let $f, g, h \in \mathcal{R} \setminus \{0\}$.*

(a) If f, g are coprime, the ideal (f, g) contains a nonzero polynomial in ξ alone and a nonzero polynomial in τ alone. Hence $\dim \mathcal{R}/(f, g) < \infty$.

(b) If f is coprime to g and to h, then

$$\dim \mathcal{R}/(f, gh) = \dim \mathcal{R}/(f, g) + \dim \mathcal{R}/(f, h).$$

(c) $(f, g) = \mathcal{R}$ if and only if $V(f) \cap V(g) = \emptyset$.

[proof: Dim. C12-L3]

The geometric meaning of $\dim \mathcal{R}/(Q_i, Q_j)$ is the number of crossing points of the curves $V(Q_i)$ and $V(Q_j)$, counted with intersection multiplicity.

12.3 The decomposition theorem

Theorem 12.6 (Factor decomposition (ledger C12-T1, C12-T1′) [proved])**.** *Let $\Omega \subset \mathbb{R}^2$ be open, connected, with $H^1_{dR}(\Omega) = 0$ (for example simply connected: a rectangle, a convex set, a slit disc). Let $\mathcal{A}$ be $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $P = c\prod_{j=1}^{s} Q_j$ with pairwise coprime Q_j. Then:*

(i) every solution splits:

$$\ker_{\mathcal{A}} P = \ker_{\mathcal{A}} Q_1 + \cdots + \ker_{\mathcal{A}} Q_s;$$

(ii) the ambiguity (the s-tuples of components with sum 0) has dimension exactly

$$\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j),$$

and consists of exponential-polynomials;

(iii) the splitting is unique if and only if $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

[proof: Dim. C12-T1p]

Remark 12.7 (Intuition). At a crossing point z of two factor curves, the exponential $e^{\langle z,(t,x)\rangle}$ belongs to both kernels. It can be moved freely from one component to the other. The number of such free moves, counted with multiplicity, is $\delta(P)$.

Algebraically the proof uses the exact sequence

$$0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$$

and shows that applying $\operatorname{Hom}(\,\cdot\,, \mathcal{A})$ keeps it exact. The only possible obstruction lives on the finite-dimensional module $\mathcal{R}/(Q_1, Q_2)$, and it vanishes when Ω has no holes (Lemma 12.9).

Example 12.8 (The ambiguity on four operators).

Operator	Factors	$\delta(P)$	Ambiguity
wave $\tau^2 - \xi^2$	$\tau - \xi,\ \tau + \xi$	1	constants
beam $\tau^2 + \xi^4$	$\tau - i\xi^2,\ \tau + i\xi^2$	2	$\operatorname{span}\{1, x\}$
heat–transport	$\tau - \xi^2,\ \tau - \xi$	2	$\operatorname{span}\{1,\ e^{t+x}\}$
parallel transports	$\tau - \xi,\ \tau - \xi - 1$	0	none (unique)

For heat–transport the curves $\tau = \xi^2$ and $\tau = \xi$ meet at $(0, 0)$ and $(1, 1)$, so the shared exponentials are 1 and e^{t+x}. For the two parallel transports the lines $\tau = \xi$ and $\tau = \xi + 1$ never meet, and the splitting is unique.

The key analytic tool is a version of the Poincaré lemma "closed forms are exact" with an exponential weight.

Lemma 12.9 (Twisted Poincaré lemma (ledger C12-L1′, L2′) [proved])**.** *Let Ω be connected with $H^1_{dR}(\Omega) = 0$, and $z = (z_1, z_2) \in \mathbb{C}^2$. Write $D_t = \partial_t - z_1$ and $D_x = \partial_x - z_2$. If $\varphi, \psi \in \mathcal{A}$ satisfy the compatibility condition*

$$D_x\varphi = D_t\psi,$$

then there is $w \in \mathcal{A}$ with $D_t w = \varphi$ and $D_x w = \psi$. Consequently, for every finite-dimensional $\mathcal{R}$-module M, the obstruction group $\operatorname{Ext}^1_{\mathcal{R}}(M, \mathcal{A})$ vanishes.

[proof: Dim. C12-L1p]

Multiplying by $e^{-z_1 t - z_2 x}$ turns the twisted derivatives into ordinary ones, and the lemma becomes the classical statement that a closed 1-form on a simply connected region is exact.

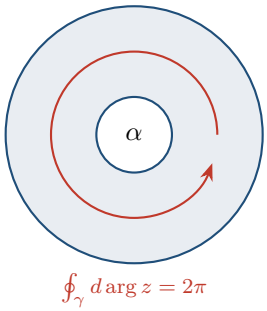
12.4 Where the decomposition fails: holes

Proposition 12.10 (Holes break it (ledger C12-P10) [proved]). *Let Ω contain a closed curve γ that winds around a point $\alpha \notin \Omega$. For the Laplacian*

$$P = \tau^2 + \xi^2 = (\tau - i\xi)(\tau + i\xi),$$

the harmonic function $u = \log|(t,x) - \alpha|^2$ is not *a sum of a solution of $\partial_t - i\partial_x$ and a solution of $\partial_t + i\partial_x$.*

[proof: Dim. C12-P10]



$u = \log|z-\alpha|^2$ is harmonic on the annulus, but $u \neq h(z) + \overline{g(z)}$ with h, g single-valued holomorphic: the logarithm has monodromy.

Figure 13: The Laplacian factors into two Cauchy–Riemann operators, but on an annulus the factor decomposition fails. The obstruction is topological: $H^1_{dR}(\Omega) \neq 0$.

For a fixed P the exact condition is cohomological.

Theorem 12.11 (Cohomological criterion (ledger OPF-T4) [proof written, not audited]). *Let Ω be connected, with first Betti number $b_1(\Omega)$. Let $\mathcal{N}$ be the space of s-tuples of global exponential-polynomial solutions with sum zero (dimension $\delta(P)$). The cokernel of the sum map*

$$\Sigma_\Omega : \bigoplus_j \ker_{\mathcal{A}(\Omega)} Q_j \to \ker_{\mathcal{A}(\Omega)} P$$

is the kernel of an explicit map $H^1(\Omega;\mathbb{C}) \otimes \mathcal{N} \to \bigoplus_j H^1(\Omega;\mathcal{K}_j)$. So the decomposition holds on Ω iff that map is injective, and

$$\dim \operatorname{coker} \Sigma_\Omega \leq \delta(P)\, b_1(\Omega).$$

[proof: Dim. OPF-T4]

Example 12.12 (Unique splittings survive holes). If $\delta(P) = 0$ (disjoint factor curves), the bound gives $\operatorname{coker} \Sigma_\Omega = 0$ on *every* connected region, with or without holes. The two parallel transports $\tau - \xi$ and $\tau - \xi - 1$ split on an annulus as well.

12.5 Explicit projectors

For evolution factors the components are given by a formula. Let

$$P = \prod_{j=1}^{N} (\tau - a_j(\xi))$$

with distinct polynomials a_j. A component is an eigenfunction of ∂_t:

$$\partial_t u_j = a_j(\partial_x)\, u_j.$$

Apply $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$. Every u_l with $l \neq j$ is killed, because the product contains its own factor. The component u_j is multiplied by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Dividing gives the Lagrange formula

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\; u. \tag{23}$$

In Fourier variables the right-hand side is a multiplier that is rational in ξ, with poles where two a_k coincide. This is Eq. (22) with the companion matrix replaced by ∂_t.

Example 12.13 (Two transports). Let $a_1(\xi) = \xi$ and $a_2(\xi) = \xi + 1$. The formula gives

$$u_1 = \frac{\partial_t - \partial_x - 1}{-1}\, u, \qquad u_2 = \frac{\partial_t - \partial_x}{1}\, u.$$

These are differential operators, with no poles, because $a_1 - a_2 = -1$ never vanishes: the two lines are disjoint and the splitting is unique.

Proposition 12.14 (Projectors and their poles (ledger II-T3, II-T4) [proved])**.** *The components are obtained by* polynomial *operators if and only if the factor curves are pairwise disjoint, $(Q_i, Q_j) = \mathcal{R}$. Only factors, not terms, admit projectors that act on one value of u at a time.*
[proof: Dim. II-T3]

Remark 12.15 (Takeaway). A "component" of u belongs to a *factor* of the symbol, not to a term of the operator. On a region without holes the factor split always exists, and the overlap between components is finite-dimensional and measured exactly by $\delta(P)$. The split by terms fails except on a finite-dimensional space.

13 Regularity of the components

Starting from: the factor decomposition of Theorem 12.6; Ω bounded; u in a Sobolev space $H^m(\Omega)$ (square-integrable derivatives up to order m). *Why:* the decomposition is algebraic. To use the components numerically we need them to be as regular as u itself, and to depend continuously on u. *Role:* completes the answer to Q2. It gives the exact condition (a conditioning number $\gamma_N > 0$), shows that it can fail, and that the geometry of Ω against the characteristics decides it. Step 5 of the algorithm checks it.

13.1 The pieces can be rougher than the whole

Remark 13.1 (Counterexample: Loss of a derivative (ledger C12-C1) [false]). The statement "if $u \in H^m(\Omega)$ then every component $u_j \in H^m(\Omega)$" is false. Let $\Omega = (0,1)^2$ and

$$f = \tau - \xi, \qquad g = \tau - \xi - 1.$$

For every m there is a smooth $u \in H^m(\Omega)$ with $f(\partial)g(\partial)u = 0$ whose unique components $u_1 \in \ker f$, $u_2 \in \ker g$ are *not* in H^m. Exactly one derivative is lost: $u_2 = f(\partial)u \in H^{m-1}$.
[proof: Dim. C12-C1]

Remark 13.2 (Intuition). In the coordinates $s = x + t$ along the characteristics, both factors are first-order ODEs along the same lines, with solutions 1 and e^t. On a long chord these two functions are clearly different. Near a corner of the square the chords become arbitrarily short, and on a short chord 1 and e^t are almost collinear. Separating a function into these two shapes then amplifies it by a factor that blows up as the chord shrinks (Figure 14).

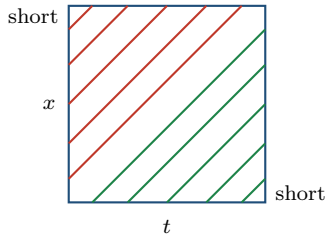


Figure 14: The unit square and the characteristic chords of $\partial_t + \partial_x$. Chords near the corners $(0,1)$ and $(1,0)$ are arbitrarily short. On a short chord the two shapes of the components are nearly collinear, and the splitting is ill-conditioned.

13.2 Two components: the chord criterion

For two transports in the same direction, the geometry decides everything.

Theorem 13.3 (Chord criterion (ledger C12-T3) [proved]). *Let Ω be bounded, open and convex, $c \in \mathbb{R}$, $d_1 \neq d_2$, and*

$$f = \partial_t + c\,\partial_x - d_1, \qquad g = \partial_t + c\,\partial_x - d_2.$$

Let $\ell(s)$ be the length of the chord of Ω along the characteristic line $x - ct = s$, and $\mathcal{H}_i = \ker_{L^2} f_i$. The following are equivalent:

(a) the chords do not shrink: $\inf_s \ell(s) > 0$;

(b) every $u \in L^2$ with $fgu = 0$ has components in L^2, with

$$\|u_1\|^2 + \|u_2\|^2 \leq C\,\|u\|^2;$$

(c) the sum $\mathcal{H}_1 + \mathcal{H}_2$ is closed in L^2;

(d) the Friedrichs cosine of the two spaces is smaller than one, $c_F(\mathcal{H}_1, \mathcal{H}_2) < 1$.

[proof: Dim. C12-T3]

Example 13.4 (Rectangles). On a rectangle $I \times J$ with $c \neq 0$, the chords in direction $(1, c)$ shrink at two corners. So the global L^2 decomposition always fails ($c_F = 1$). For $c = 0$ the chords are vertical, all of length $|I|$, and the decomposition holds.

13.3 Any number of components

For N components the right quantity is the conditioning of the sum map

$$S : \bigoplus_{j=1}^{N} \ker_{H^m} Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

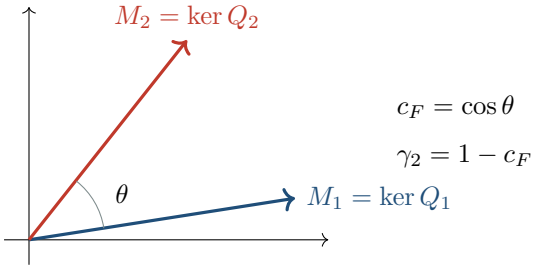


Figure 15: The Friedrichs angle θ between two component spaces. A larger angle means a better-conditioned recovery of the pieces from their sum. As $\theta \to 0$ the recovery becomes unstable.

Definition 13.5 (Conditioning number γ_N)**.** Let $(\ker S)^{\perp}$ be the tuples orthogonal to the ambiguity of Theorem 12.6. Then

$$\gamma_N = \inf_{\substack{v \in (\ker S)^{\perp} \\ v \neq 0}} \frac{\|Sv\|_{H^m}^2}{\|v\|^2}, \qquad \|v\|^2 = \sum_j \|u_j\|_{H^m}^2 .$$

Equivalently, γ_N is the bottom of the spectrum of S^*S on $(\ker S)^{\perp}$: the square of the smallest nonzero singular value of S.

Remark 13.6 (Intuition). γ_N is the worst ratio between the size of the sum and the size of the pieces. If $\gamma_N > 0$, the pieces cannot nearly cancel each other:

$$\|v\| \le \gamma_N^{-1/2} \, \|Sv\|,$$

and recovering them from u is stable. If $\gamma_N = 0$, there are large pieces with a tiny sum, and the recovery is unstable. It is exactly the smallest singular value of a least-squares problem.

Theorem 13.7 (Regularity criterion (ledger C12-P7) [proved])**.** *(a) Every $u \in \ker_{H^m} P$ has components in H^m if and only if the sum $\sum_j \ker_{H^m} Q_j$ is* closed *and* dense *in $\ker_{H^m} P$.*

(b) The sum is closed if and only if $\gamma_N > 0$. Then the minimal-energy split satisfies, optimally,

$$\sum_j \|u_j\|_{H^m}^2 \le \gamma_N^{-1} \, \|u\|_{H^m}^2 .$$

(c) For $N = 2$, $\gamma_2 = 1 - c_F$, the Friedrichs cosine of the two spaces after removing their intersection.

(d) For $N \ge 3$ the pairwise cosines do not decide. There are three closed subspaces of a Hilbert space with trivial pairwise intersections and closed pairwise sums ($c_F \le \frac{1}{2}$), whose triple sum is not closed.

[proof: Dim. C12-P7]

The condition has two parts. Closedness is the conditioning above. Density is a separate question, and on convex regions it is always true.

Theorem 13.8 (Density on convex regions (ledger OPF5B-T1) [proved mod. Hörmander, Stein])**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded and convex. For every operator P, every coprime factorisation and every $m \ge 0$, the sum $\sum_j \ker_{H^m} Q_j$ is dense in $\ker_{H^m} P$.*

[proof: Dim. OPF5B-T1]

So on a convex region only $\gamma_N > 0$ remains to be checked.

Remark 13.9 (Warning). [title=Added in the complete edition (Phase 10)] Convexity cannot simply be dropped: on a region with a hole the sum of kernels is *not* dense in general (Proposition 24.2, the Laplacian on an annulus), and the extension of this theorem to non-convex regions is false. For simply connected Lipschitz regions density was proved in Phase 10 modulo two cited facts (Section 24).

Example 13.10 (Cases where the H^m decomposition holds (ledger C12-P4) [proved]). **$P = \tau\xi$ on a rectangle.** The components are a function of t and a function of x. After removing the constants their H^m inner product vanishes, so $c_F = 0$ and $\gamma_2 = 1$.

The wave equation on a square. In characteristic coordinates $s = x + t$, $r = x - t$ the square becomes a rhombus. It can be covered by five pieces on each of which one component is controlled by u.

Heat and transport on a rectangle (ledger OPR2-T1) [proof written, not audited]. For $P = (\tau - \xi^2)(\tau + c\xi)$ the sum is dense in every H^m, so the decomposition holds exactly when $c_F < 1$.

The same operator can behave well on one region and badly on another: regularity depends on the *geometry of* Ω *against the characteristics*, not on P alone.

[proof: Dim. C12-P4]

Proposition 13.11 (Real transports, $m = 0$ (ledger OPR2-T2) [proof written, not audited])**.** *Let Ω be convex and bounded, and $Q_j = v_j \cdot \nabla - d_j$ real transports, grouped into classes of parallel directions. Assume that of any two classes at least one has chords bounded below. Then $\sum_j \ker_{L^2} Q_j$ is closed if and only if every class with at least two factors has chords bounded below. In particular, for such transports, closed pairwise sums imply a closed triple sum.*

[proof: Dim. OPR2-T2]

Remark 13.12 (Warning). [title=Three or more components: what is still open] Whether the abstract counterexample of Theorem 13.7(d) can occur for *differential* factors is open in general [open] (ledger C12-Q2). Proposition 13.11 rules it out for real transports in L^2.

Remark 13.13 (Takeaway). Algebraically the split always exists (Theorem 12.6). Analytically the components are as regular as u exactly when the component spaces are separated by a positive angle, $\gamma_N > 0$; density is automatic on convex regions. Everything delicate happens at the boundary of Ω (shrinking chords) or in its topology (holes).

14 Terms inside a factor: the interaction term

Starting from: an evolution factor $u_t = Au$ with $A = A_1 + \cdots + A_N$, the *terms* (advection, diffusion, dispersion, ...), each a polynomial $a_n(\partial_x)$; flows $S_n(t) = e^{tA_n}$; a bounded rectangle Ω and a base time t_0. The terms commute, except in Section 14.7. *Why:* Proposition 12.1 shows that splitting a solution by terms is never exact. Here we compute exactly what is left over, the interaction term E_{int}, how much it costs, and whether it can be computed stably. *Role:* third link of the chain, answering Q3. The answer is positive in a function class of exponential type (an exact product formula, cost $O(N)$ per value) and negative in Sobolev spaces (no continuity, no stable regularisation).

14.1 From the sum of operators to the interaction term

The solution of $u_t = Au$ is given by the flow (semigroup) of A:

$$u(t) = e^{tA}u_0 = S(t)\,u_0.$$

If term n acted alone, it would move the datum to $S_n(t)u_0 = e^{tA_n}u_0$, a change of $S_n(t)u_0 - u_0$.

The naive guess is that the terms act independently, so that the total change is the sum of the individual changes. It is not: the flow of a sum is not the sum of the flows. For advection ($A_1 = -c\partial_x$, a translation) and diffusion ($A_2 = D\partial_x^2$, a smoothing), the solution is translated *and* smoothed. This joint action is not "only translated" plus "only smoothed".

Definition 14.1 (Interaction term)**.** With $S_n(t) = e^{(t-t_0)A_n}$ the flow of term n alone,

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{24}$$

The multiplicative variant compares the true flow with the product of the single flows:

$$E_{\text{int}}^{\text{mult}}(t) = e^{(t-t_0)A} - \prod_{n=1}^{N} e^{(t-t_0)A_n}.$$

We write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.

14.2 The function class and the exact formula

Before computing, one must decide on which functions the flows of the single terms are defined. This is not a formality. For the heat equation there are nonzero solutions with zero initial datum (Tychonoff), so "the diffusion flow of u_0" is not unique without a class.

Definition 14.2 (The class $\mathfrak{B}$ of exponential type (ledger P1-D1))**.** For $\rho > 0$, $B_\rho(J)$ is the space of smooth f on the interval J with

$$\|f\|_{B_\rho}^2 = \sum_{j\ge 0} \rho^{-2j}\,\|f^{(j)}\|_{L^2(J)}^2 < \infty.$$

$\mathfrak{B}_\rho(\Omega)$ is the analogous space in (t,x), and $\mathfrak{B} = \bigcup_\rho \mathfrak{B}_\rho$. Every exponential-polynomial belongs to $\mathfrak{B}$, and on B_ρ the derivative is bounded: $\|\partial_x\| \le \rho$.

On $\mathfrak{B}$ every polynomial in ∂_x is a bounded operator, its exponential is an absolutely convergent series, and two such exponentials commute (ledger P1-L1, P1-L2) [proved]. So the algebra of exponentials of numbers applies to operators.

Theorem 14.3 (Closed form of the interaction (ledger P1-T2) [proved])**.** *Let $u \in \mathfrak{B}_\rho(\Omega)$ solve $(\partial_t - a(\partial_x))u = 0$ with $a = \sum_n a_n$, and put*

$$\Delta_n = e^{(t-t_0)a_n(\partial_x)} - \mathrm{Id}.$$

Then the single flows are unique in $\mathfrak{B}$, $E_{\text{int}}^{\text{mult}} \equiv 0$, and

$$E_{\text{int}}^{\text{add}}(t) = \Big[\prod_{n=1}^{N}(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_{n=1}^{N}\Delta_n\Big]u_0 = \sum_{|S|\ge 2}\ \prod_{n\in S}\Delta_n\,u_0. \tag{25}$$

For a datum $u_0 = \sum_k c_k e^{\zeta_k x}$, each mode is multiplied by the number

$$e_{\geq 2}(z) = \prod_n (1 + z_n) - 1 - \sum_n z_n, \qquad z_n = e^{(t-t_0)a_n(\zeta_k)} - 1.$$

[proof: Dim. P1-T2]

Remark 14.4 (Intuition). Expand the product $\prod_n(\mathrm{Id} + \Delta_n)$. It gives one monomial per subset S of terms. The empty subset gives Id, the singletons give the individual changes Δ_n, and everything else is interaction. So the interaction is the sum of products of *at least two* increments: it appears only when at least two terms act together.

For two terms the formula is simply

$$E_{\mathrm{int}} = \Delta_1\Delta_2\, u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})\, u_0.$$

In each Fourier mode, the interaction is the product of the two single-term responses.

Example 14.5 (Advection–diffusion with a Gaussian datum). Take $u_t = -cu_x + Du_{xx}$ and $u_0 = g(x; \sigma^2)$, where

$$g(x; v) = \frac{e^{-x^2/2v}}{\sqrt{2\pi v}}.$$

The two flows are a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x - ct; \sigma^2), \qquad S_2 u_0 = g(x; s^2), \qquad u = g(x - ct; s^2),$$

with $s^2 = \sigma^2 + 2Dt$. The interaction term is

$$E_{\mathrm{int}} = g(x - ct; s^2) - g(x - ct; \sigma^2) - g(x; s^2) + g(x; \sigma^2).$$

Figure 16(a) shows that it is comparable in size to u itself: it is not a small correction. In Fourier variables

$$\frac{|\widehat{E_{\mathrm{int}}}(\kappa)|}{|\hat{u}_0(\kappa)|} = |e^{-ic\kappa t} - 1| \cdot |e^{-D\kappa^2 t} - 1| \leq 2.$$

It vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$, where the translation returns to the identity.

Example 14.6 (A stationary state is pure interaction). For advection–diffusion, a stationary solution satisfies $-cu_x + Du_{xx} = 0$, for example $u = e^{cx/D}$. The full flow leaves it unchanged, so $u(t) - u_0 = 0$. The single flows do change it, and the interaction term exactly cancels their changes. In this mode

$$E_{\mathrm{int}} = -4\sinh^2\Big(\frac{\theta}{2}\Big)\, u, \qquad \theta = \frac{c^2(t - t_0)}{D}.$$

Equilibrium is the case where all the dynamics is interaction.

14.3 Cost: $O(N)$ versus 2^N

Expanding (25) produces one monomial per subset with at least two elements: $2^N - N - 1$ cross terms (Figure 17). Writing the *complete table* of cross terms is therefore exponential. Computing one *value* is cheap, because the product is never expanded.

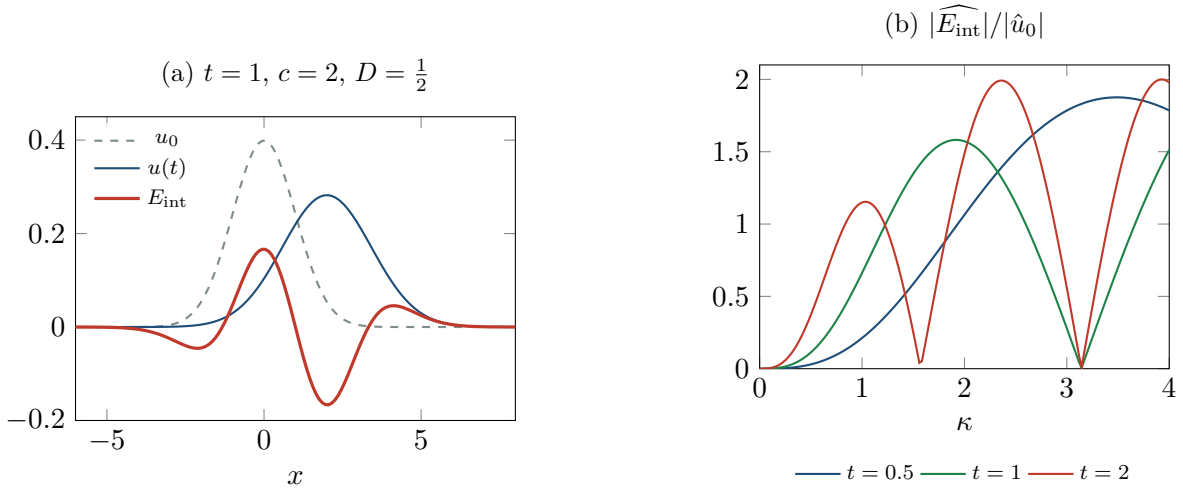


Figure 16: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$). (a) The datum, the solution and the interaction term. (b) The Fourier ratio of the interaction to the datum, for three times.

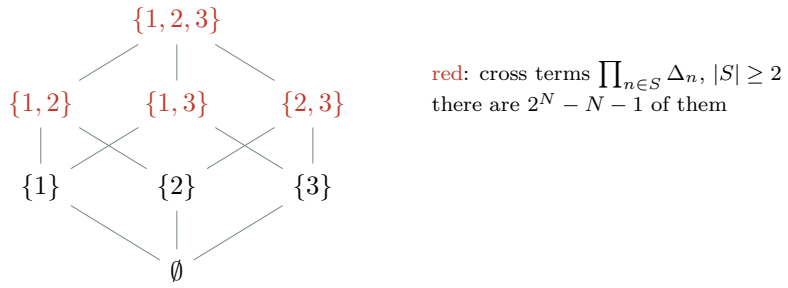


Figure 17: The subset lattice for $N = 3$ terms. Each node with at least two elements is one cross term of the interaction.

Theorem 14.7 (Cost of the interaction (ledger P1-T3) [proved])**.** *Let the datum be a sum of M exponentials with polynomial prefactors of degree at most L, and let $d_n = \deg a_n$.*

(a) One value of E_{int} costs

$$O\Big(M\big[N(L+1)^2 + (L+1)\textstyle\sum_n (d_n+1)\big]\Big);$$

for simple exponentials ($L = 0$) this is $O(M(N + \sum_n d_n))$: linear in the number of terms.

(b) All aggregates by order, $E^{(k)} = \sum_{|S|=k} \prod_{n\in S} \Delta_n u_0$ for $2 \leq k \leq N$, cost $O(MN^2(L+1)^2)$ plus the same term in d_n.

(c) The full table of cross terms has $M(2^N - N - 1)$ entries and costs $\Theta(M\,2^N(L+1)^2)$.

[proof: Dim. P1-T3]

Example 14.8 (Counting operations for one mode). For one simple mode $e^{\zeta x}$ and N terms, compute the N numbers $z_n = e^{(t-t_0)a_n(\zeta)} - 1$, then the product $\prod_n (1 + z_n)$ with $N - 1$ multiplications, then subtract $1 + \sum_n z_n$. That is about $3N$ operations. Listing the $2^N - N - 1$ products $\prod_{n\in S} z_n$ one by one costs 2^N times more for large N: already for $N = 20$, about a million.

14.4 The flows are not determined without a class

Proposition 14.9 (Non-uniqueness of single flows (ledger P1-P1(a)–(d)) [proved])**.** *On a bounded rectangle, without a function class:*

(a) for the transport term $a_n = \xi$, the flow of a datum is any smooth extension of it: infinitely many choices;

(b) for the diffusion term $a_n = \xi^2$, there are nonzero heat solutions with zero datum (Tychonoff) [from cited source];

(c) so E_{int} is not a function of u alone; inside $\mathfrak{B}$ it is unique.

[proof: Dim. P1-P1]

14.5 Stability: the interaction cannot be computed stably

Theorem 14.10 (No continuous extension to Sobolev spaces (ledger P1-P1(e)) [false])**.** *Let $Q = \tau + c\xi - D\xi^2$ with $c, D > 0$ (advection–diffusion), terms $a_1 = -c\xi$ and $a_2 = D\xi^2$, and any base time t_0. For every m,*

$$\sup\left\{\frac{\|E_{\mathrm{int}}\|_{L^2(\Omega)}}{\|u\|_{H^m(\Omega)}} : \ u \in \mathrm{EP},\ Qu = 0,\ u \neq 0\right\} = \infty.$$

So $u \mapsto E_{\mathrm{int}}$ does not extend continuously from exponential-type data to H^m, although exponential-type data are dense. The same holds for every evolution with at least two non-proportional terms; for proportional terms the interaction is continuous (ledger OPA-P3) [proved].

[proof: Dim. P1-P1e]

Remark 14.11 (Intuition). On a bounded interval, a real exponential $e^{\zeta x}$ with large ζ is a legitimate datum. The diffusion flow alone amplifies it by $e^{D\zeta^2 t}$. The true solution is only changed by the factor $e^{(D\zeta^2 - c\zeta)t}$ times the datum norm. The interaction term must cancel the gap, and the gap grows faster than any Sobolev norm of the datum.

This instability has three consequences for any algorithm that wants to be stable.

Theorem 14.12 (No regularisation removes the interaction (ledger P2-T1, P2-T2, P2-T4) [proved])**.** *(a)* ***Any penalty leaves the same residual.*** *For any regulariser (an ℓ^1 sparsity penalty, ℓ^2, total variation, ...) the residual of the best fit converges to the distance of u from the closed span of the per-term pieces.*

(b) ***Inside an evolution factor that residual is large.*** *If no single term exhausts the symbol, the part of u outside the span of the per-term kernels satisfies*

$$\|E^{\perp}(u)\|_{H^m} \ \geq\ \frac{\|g(\partial_x)u\|_{H^{m-k}}}{C}, \qquad g = \prod_n (a - a_n).$$

Only a finite-dimensional space of exponential-polynomials has a per-term representation. For advection–diffusion it is

$$\mathrm{span}\{1,\ x - ct,\ (x - ct)^2 + 2Dt\}.$$

(c) ***No uniformly Lipschitz regularisation exists.*** *Any family of Lipschitz maps converging pointwise to an unbounded map has Lipschitz constants tending to infinity. Every exact selection of components has Lipschitz constant at least $\gamma_N^{-1/2}$.*

[proof: Dim. P2-T2]

Remark 14.13 (Warning). [title=Added in the complete edition (Phase 10)] The instability above is proved for advection–diffusion. The general statement "for every operator with at least two non-proportional terms" (P1-Q6) is false as written: with constant terms, for example reaction–diffusion, the interaction is continuous from H^1. The correct dichotomy is by the number of terms of degree ≥ 1 (Theorem 24.1, Section 24).

Remark 14.14 (The one continuous split found). A *balance* decomposition of u writes each term's contribution as the time integral of that term applied to the true solution:

$$u^{(i)}(t) = \int_{t_0}^{t} a_i(\partial_x)\, u(s)\, \mathrm{d}s.$$

It is exact ($u - u_0 = \sum_i u^{(i)}$) and continuous in H^m (ledger P3-T6) [proof written, not audited]. But it needs an external principle to decide that this is "the" contribution of term i: it uses the full solution, not the flow of the term alone.

14.6 Time-dependent perturbation theory

The product formula needs commuting terms. With non-commuting terms, or with boundary conditions, there is no closed formula, but E_{int} is still defined by (24). Perturbation theory expands the flow in powers of the interaction. Split $A = A_1 + B$, where A_1 is the principal term and B the rest. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k\geq 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s. \end{aligned} \tag{26}$$

$S^{(k)}$ collects the histories in which B acts k times, at intermediate times, with the flow of A_1 in between. This is the interaction picture of quantum mechanics [4].

Theorem 14.15 (Dyson–Phillips converges for every time (ledger P3-T1) [proved]). *Let $A_1 = a(\partial_x)$ with symbol of degree q, and $B = \sum_{k\leq r} b_k(x)\partial_x^k$ with bounded coefficients and $r < q$. The symbol condition*

$$\operatorname{Re} a(i\kappa) \leq \omega - \delta|\kappa|^q \qquad (\delta > 0)$$

holds if and only if q is even and the leading coefficient has negative real part (a dissipative operator of even order). Under it, the Dyson–Phillips series converges in operator norm, uniformly on every bounded time interval, and its sum is the semigroup generated by $A_1 + B$. The same holds with Dirichlet conditions on an interval.

[proof: Dim. P3-T1]

So E_{int} is well defined and can be approximated by truncating the series even without a closed formula.

Proposition 14.16 (Interaction created by a boundary (ledger P1-P6, P3-T3, OPA-T2, OPF5A-T5) [proved]). *On an interval with Dirichlet conditions, advection and diffusion no longer commute: the boundary creates an $E_{\mathrm{int}}^{\mathrm{mult}}$ already at first order (ledger P1-P6). For small times its size*

is governed by how the datum touches the boundary. If k is the order of contact (the first nonvanishing derivative of u_0 at the endpoints), then

$$\|E_{\mathrm{int}}^{\mathrm{mult}}(t)\| = \Theta\big(t^{k/2+3/4}\big) \qquad (t \to 0).$$

For compatible data the leading constant is explicit and proportional to

$$|c|\,\frac{\tilde{N}_k}{k!}\,D^{k/2-1/4}\sqrt{u_0^{(k)}(0)^2 + u_0^{(k)}(L)^2}.$$

[proof: Dim. OPF5A-T5]

The smoother the datum is against the boundary, the faster the boundary interaction becomes negligible as $t \to 0$.

14.7 Non-commuting terms

If two terms do not commute, the flow of their sum is not the product of the flows. By the Baker–Campbell–Hausdorff formula,

$$e^{tA_1}e^{tA_2} = \exp\Big(t(A_1+A_2) + \frac{t^2}{2}[A_1,A_2] + O(t^3)\Big).$$

The commutators generate a Lie algebra, and its type decides what can be done.

Theorem 14.17 (Lie-algebra classification (ledger P1-T4, P1-P5, P1-T5) [proved])**.** *(a)* ***Finite and solvable.*** *An exact formula exists, with $d = \dim \mathfrak{g}$ entire scalar functions (Wei–Norman). Example: $\partial_x^2 + x$, and the Ornstein–Uhlenbeck operator.*

(b) ***Finite, not solvable*** *(for example $\mathfrak{sl}_2$). A closed form exists locally in time; it can fail globally.*

(c) ***Infinite adjoint orbit.*** *If X and all the iterated commutators $\mathrm{ad}_X^k Y$ are linearly independent (for example $X = x^a\partial_x$, $Y = \partial_x$ with $a \geq 3$), then no finite product of flows of the two terms, with times analytic in t and in the coupling ε and with the flows of Y switched off at $\varepsilon = 0$, reproduces $e^{t(X+\varepsilon Y)}$. Only series expansions exist.*

[proof: Dim. P1-T5]

Example 14.18 (Which potentials allow Wei–Norman). For $i\partial_t = -\partial_x^2 + V(x)$, the operators ∂_x^2 and x^2 generate the finite Lie algebra $\mathfrak{sl}_2$: the harmonic oscillator has a closed-form evolution. With $V = x^3$, repeated commutators produce polynomials of ever higher degree and the algebra is infinite-dimensional. But the hypothesis of part (c) fails there, so whether a finite product of flows is impossible for ∂_x^2 and x^k with $k \geq 3$ is still [open] (ledger P1-Q3). Part (c) applies to the transport with polynomial speed $u_t = (x^a + \varepsilon)u_x$, $a \geq 3$, which nevertheless has an explicit solution by characteristics: the theorem excludes only a factorisation into flows of the terms.

Remark 14.19 (Warning). [title=Other ways commutativity is lost]

- **Repeated factors.** For $(\tau - a)^e$ the "multiplicity term" is a nilpotent shift. It commutes with the other terms, its flow is polynomial in time (secular growth), and $E_{\mathrm{int}}^{\mathrm{mult}} = 0$ (ledger OPF-T2) [proof written, not audited].
- **Second-order evolutions** (wave, Klein–Gordon). The companion blocks do not commute, so the first-order calculus fails; a Gauss factorisation works mode by mode (ledger II-T7, P1-P4) [proved].

Remark 14.20 (Takeaway). For commuting terms the interaction is explicit and cheap to evaluate, but it can be neither removed nor computed stably in Sobolev spaces. That instability is not a defect of the formula: it is a property of the question. For non-commuting terms even the explicit form is lost, except when the terms generate a finite solvable Lie algebra.

15 Nonlinear equations

Starting from: a scalar evolution in one space variable, $u_t = \sum_n T_n(u)$, each term T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; a smooth datum u_0; the exact solution is known. Benchmarks: viscous Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS as integrable controls. *Why:* for a nonlinear equation the single-term flows do not commute and superposition fails. "Component per term" needs a new definition, and identification needs a new frame (a dictionary). We give both and compute them where everything is explicit. *Role:* fourth link of the chain. It reuses the earlier chapters through the jet expansion and the annihilator idea. Its results are weaker and relative, but sharp on the benchmark models.

15.1 The interaction term for nonlinear flows

Let S_n^t be the flow of $u_t = T_n(u)$ alone, which is now a nonlinear map. Definition (24) still makes sense:

$$E_{\text{int}}(t) = u(t) - u_0 - \sum_n \left(S_n^t u_0 - u_0\right).$$

There is no product formula, because the flows do not commute. But the Taylor expansion in time is explicit.

Theorem 15.1 (Jet of the nonlinear interaction (ledger S4T-T1) [proved])**.** *For smooth data,*

$$E_{\text{int}}(t) = \frac{t^2}{2} \sum_{n \neq m} T_n'[u_0](T_m(u_0)) + O(t^3),$$

where $T_n'[u_0](v)$ is the derivative (Fréchet derivative) of T_n at u_0 in the direction v.

[proof: Dim. S4T-T1]

Remark 15.2 (Intuition). To first order in t every flow moves u_0 by $t\,T_n(u_0)$, and these moves add exactly: there is no interaction at order t. At order t^2 the full flow sees each term react to the motion caused by the others. That reaction is the derivative of T_n in the direction of $T_m(u_0)$.

Example 15.3 (Burgers). For $u_t = -uu_x + \nu u_{xx}$ take $T_1(u) = -uu_x$ and $T_2(u) = \nu u_{xx}$. The derivatives are

$$T_1'[u](v) = -(vu_x + uv_x), \qquad T_2'[u](v) = \nu v_{xx}.$$

Substituting the other term:

$$T_1'[u_0](T_2(u_0)) = -\nu\left(u_0'' u_0' + u_0 u_0'''\right),$$
$$T_2'[u_0](T_1(u_0)) = -\nu\left(3u_0' u_0'' + u_0 u_0'''\right).$$

Half the sum, times t^2, gives

$$E_{\text{int}}(t) = -\nu\left(u_0 u_0''' + 2u_0' u_0''\right) t^2 + O(t^3).$$

15.2 Before, at and after the shock

The flow of the single term $-uu_x$ (inviscid Burgers) forms shocks: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t \sin \xi,$$

and since

$$\frac{\partial x}{\partial \xi} = 1 + t \cos \xi$$

first vanishes at $\xi = \pi$, the breaking time is $t^* = 1$ (Figure 18).

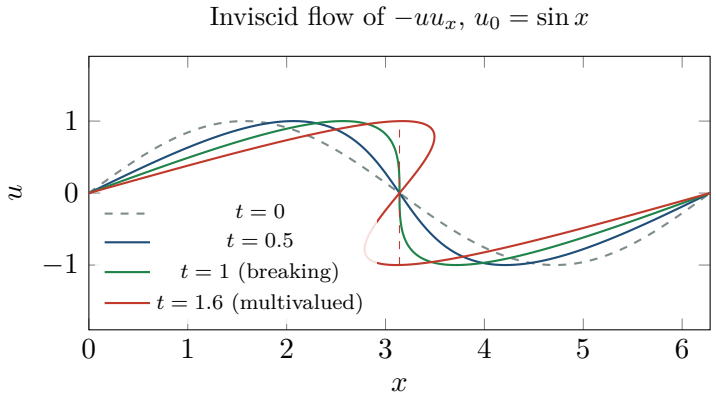


Figure 18: At $t = 1$ the profile has a vertical tangent and the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$. This breaking is what makes the interaction term discontinuous after the shock.

The interaction changes qualitatively with time, and each regime needs different tools.

15.2.1 Before the shock: a closed form

Theorem 15.4 (Closed form before the shock (ledger S4T-T2, S4T-P3) [proved])**.** *For Burgers with $t < t^*$, the interaction term has a closed form at first order in the viscosity, with remainder*

$$O(\nu^2 \varepsilon^2 t^3),$$

where ε measures the size of the datum. The closed form blows up at the breaking time t^*.

[proof: Dim. S4T-T2]

The tool is the Cole–Hopf substitution

$$u = -2\nu \frac{\varphi_x}{\varphi},$$

which turns Burgers into the heat equation $\varphi_t = \nu \varphi_{xx}$ [9]. It plays the role of an "exact linearisation". For KdV and NLS the inverse scattering transform plays the same role.

The analyticity radius of the Burgers solution with datum U_0 is at least $2/\mathrm{osc}(U_0)$, and this bound is optimal (ledger F4L-P3) [proved].

15.2.2 After the shock: sizes in L^1

The flow of $-uu_x$ alone forms a shock, so E_{int} has jumps whose size does not depend on ν (ledger S4T-T3) [proved mod. [F1]]. This is forced by the definition. A function with a jump is not in H^1, so Sobolev norms are no longer the right measure. Pointwise values are unreliable, but the *measure* of the discrepancy is well behaved. The meaningful sizes are in L^1.

Theorem 15.5 (Interaction after the shock (ledger OPA-T1) [proved mod. Kružkov])**.** *At a fixed time $t > t^*$, under a nondegeneracy condition (H_t) that holds for example for $u_0 = \sin x$,*

$$\|E_{\text{int}}(t)\|_{L^1} = \Theta(\nu) \qquad (\nu \to 0).$$

Each shock of strength ρ contributes

$$4\nu \log\left(2\cosh(\rho/2)\right)$$

to the leading term.

[proof: Dim. OPA-T1]

Theorem 15.6 (Uniform rate across the shock time (ledger OPF5A-T1, OPF5A-T2) [proved mod. Kružkov])**.** *Uniformly in time, near the shock time,*

$$\|E_{\text{int}}(t)\|_{L^1} \approx 2\nu \log \frac{1}{\max(|t - t^*|, \nu)} + O(\nu),$$

so that

$$\sup_{t \le T} \|E_{\text{int}}(t)\|_{L^1} = \nu|\log \nu| + O(\nu),$$

with leading coefficient exactly 1*. The constant at the shock time is explicit (it involves the Pearcey constant* 1.188654 …*). For data with several shock births the uniform rate is [conditional] on a localisation lemma.*

[proof: Dim. OPF5A-T2]

Proposition 15.7 (The splitting error is larger (ledger OPF5D-T6) [proved])**.** *The error of the splitting scheme "advection, then diffusion" is*

$$\|S(t)u_0 - S_2^t S_1^t u_0\|_{L^1} = \Theta(\sqrt{\nu}),$$

larger than the interaction term itself. Treating the terms one after the other costs more than ignoring their interplay.

[proof: Dim. OPF5D-T6]

15.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$: there is no ring of operators. Instead one fixes a finite list of candidates.

Remark 15.8 (Dictionary setting).
- A *dictionary* $\mathcal{D}$ is a finite set of monomials in the jet $(u, u_x, u_{xx}, \dots)$ of degree at most d and order at most k, for example $\{u,\ u_x,\ uu_x,\ u_{xx},\ u_{xxx}\}$. $V = \operatorname{span} \mathcal{D}$.
- $I(u)$ is the set of all differential polynomial relations that u satisfies.

Regime	What happens	Tools and results
before, $t < t^*$	smooth; characteristics do not cross	jet (Thm. 15.1), closed form (Thm. 15.4), Cole–Hopf
at breaking, $t = t^*$	the gradient blows up	smoothness estimates fail; the closed form blows up
after, $t > t^*$	the single-term flow has a jump	L^1: $\Theta(\nu)$, uniform $\nu\lvert\log\nu\rvert$, splitting $\Theta(\sqrt{\nu})$

Table 7: The three regimes of the Burgers interaction term.

- $[F]$ is the differential ideal generated by the equation F: F, its derivatives $\partial_x F, \partial_t F, \dots$, and their combinations.
- $\mathcal{J}_u$ is a jet-level obstruction defined from u, and ψ maps the dictionary into the jet space.

Theorem 15.9 (Dictionary criterion (ledger S4ID-T1) [proved]). *For smooth u on any Ω there is an exact sequence*

$$0 \to [F] \cap V \to I(u) \cap V \to \mathcal{J}_u \cap \psi(V) \to 0.$$

In particular

$$I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

[proof: Dim. S4ID-T1]

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*". The literal statement $I(u) \cap V = \operatorname{span}\{F\}$ is false in complete dictionaries, because they contain $\partial_x F$ (ledger S4ID-P1) [proved].

Theorem 15.10 (Burgers with N shocks (ledger S4ID-T2, OP-T7) [proved]). *For the N-shock solutions of Burgers, u identifies its equation in the evolutionary dictionary with d unknown terms if and only if $d \le N$.*

[proof: Dim. S4ID-T2]

Example 15.11 (One shock, two unknowns). The travelling front $u = -\tanh(x/(2\nu))$ is a 1-shock solution. It satisfies Burgers, but also the ODE that every travelling wave satisfies. With $d = 2$ unknown coefficients (in front of uu_x and u_{xx}) and $N = 1$, the theorem says u does not identify the pair. It fixes only their ratio.

Theorem 15.12 (KdV N-solitons (ledger S4ID-T3, OPF5C-T2) [proved mod. Novikov 1974]). *For the N-soliton solution of KdV, the annihilator in the jet variables is principal, generated by one polynomial Π_N of total degree*

$$\deg \Pi_N = (N+2)\, 2^{N-1}.$$

For generic $N = 2$ this degree is 8. The solution identifies KdV in a dictionary of degree d if and only if $d < (N+2)2^{N-1}$.

[proof: Dim. OPF5C-T2]

Proposition 15.13 (A closed form does not identify (ledger S4ID-T4) [proved]). *If u has a closed form, it lives on a smaller variety and satisfies extra relations: it does not identify its equation in a complete dictionary.*

[proof: Dim. S4ID-T4]

This is the nonlinear analogue of "exponential-polynomials never identify".

15.4 Integrable models and scattering

- **Hirota N-solitons.** The interaction is pairwise and localised: it is determined by pairs of solitons, integrates to 0, and costs $O(N^3)$ (ledger INT-P5, INT-P6) [proved].
- **Scattering identification.** For reflectionless KdV, Novikov's centraliser characterises the potential (ledger INT-T1, OPF5D) [proved]. For NLS with multiplicity, identification by the scattering data holds; without multiplicity it is [false]. The Lax and zero-curvature relations hold for all n (ledger OP-P17, OP-P18) [proved].
- **KdV–Burgers.** The jet is proved. Before the shock the dispersive correction has remainder $O((\nu + |\delta|)^2)$ (ledger OPR2-T5) [proved mod. local well-posedness].
- **Fisher–KPP.** Only a jet-level statement with an estimate that is not uniform in time (ledger S4T-P5) [proved]. No sharp analogue of the Burgers asymptotics is claimed.

Remark 15.14 (Warning). [title=Open in the nonlinear part] For analytic KdV data with nonzero reflection, is there an autonomous ODE in x? Three obstructions are proved, among them the cubic relation of Airy [open] (ledger INT-Q1). The exact minimal syzygy degrees $\delta_N(K)$ are known in nine computer-assisted cases; their general formulas are conjectures [open] (ledger OPF5C-Q1).

15.5 Linear versus nonlinear

	Linear (Sections 9–14)	Nonlinear (this chapter)
Candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$	a finite dictionary $V = \operatorname{span} \mathcal{D}$
Relations of u	the ideal $\operatorname{Ann}(u)$	the differential ideal $I(u)$
What P generates	the principal ideal (P)	the differential ideal $[F]$
Identification	$\operatorname{Ann}(u) = (P)$	$I(u) \cap V = [F] \cap V$
What escapes	the point spectrum $V(J_u)$	the obstruction $\mathcal{J}_u \cap \psi(V)$
Solutions that never identify	exponential-polynomials	closed forms
Interaction	product formula, $O(N)$	t^2 jet; $\Theta(\nu)$, $\nu\lvert\log\nu\rvert$ after shocks

Table 8: The linear and nonlinear problems side by side.

Remark 15.15 (Takeaway). Nonlinearity changes the objects (single-term flows with shocks, dictionaries instead of rings) but not the question. Where the formula is closed (before the shock, solitons) the answer is explicit. Across a shock the interaction does not vanish with the viscosity: it shrinks like $\nu\lvert\log\nu\rvert$.

16 The algorithm

Starting from: an input in one of the data models M1–M3 (Section 11) and, for each step, the hypotheses of the theorem that justifies it. *Why:* to turn the theorems into one procedure with explicit inputs, outputs and guarantees. *Role:* operational summary of Sections 10–15. Table 9 lists the steps, Figure 19 gives the control flow, and two worked examples make every step concrete.

16.1 The pipeline

Step	Task	Result relied upon	Output
0	Input	data models M1–M3	exact u
1	Annihilator	Gram matrix of a dictionary	a candidate P
2	Factor	unique factorisation (F1)	factors P_i, multiplicities e_i
3	EP test	Lemma 9.8, Thm. 10.2, Ch. 11	a verdict per factor
4	Reduce	Thms. 10.9 and 10.11	G_u, J_u, certificate
5	Decompose	Thm. 12.6, Eq. (23), Thm. 13.7	components u_j, ambiguity $\delta(P)$, check γ_N
6	Terms	Thms. 14.3, 14.7, 14.17	log-symbol, E_{int}
7	Nonlinear branch	Thms. 15.9, 15.1	relative identification, flow-based E_{int}

Table 9: The steps of the algorithm: what each step does, which result justifies it, and what it returns.

16.2 The steps in detail

Remark 16.1 (Step 1: obtain an annihilator). Let $\Phi(u) = (T_1 u, \dots, T_M u)^\top$ be the vector of M candidate terms applied to u, and

$$\mathbb{G}(u) = \int_\Omega \Phi\, \Phi^*\, \mathrm{d}\mu$$

its Gram matrix (positive semidefinite). A relation $\lambda \cdot \Phi = 0$ holds if and only if $\lambda \in \ker \mathbb{G}(u)$. The computation is exact (rational or algebraic numbers, Gaussian elimination): no learning, no statistics. If $\ker \mathbb{G}$ has dimension larger than one, u satisfies several relations, which is itself information about G_u and J_u.

Remark 16.2 (Step 2: factor). Factor $P(\tau, \xi)$ into irreducibles with multiplicities. This is a standard computer-algebra operation over the algebraic numbers.

Remark 16.3 (Step 3: exponential-polynomial test). For each i compute $w_i = (P/P_i)(\partial)u$ and decide whether $\dim \mathcal{R} w_i < \infty$. How depends on the data model:

- M1: a syntactic check (is the closed form a finite sum of polynomials times exponentials?);
- M3: a holonomic computation, decidable modulo (AC) (Theorem 11.12);
- M2: undecidable from the jet alone (Theorem 11.9); with a bound B, compare annihilators up to degree $B + \deg P - 1$ (Theorem 11.10).

Remark 16.4 (Step 4: reduce to G_u). Drop from P each factor whose test returned "EP", with the multiplicity bookkeeping of Theorem 10.11: at most $\sum_i e_i$ tests in total. The output is G_u; the dropped part is J_u, whose finite zero set lists the exponentials contained in u. The certificate reads either "$\mathrm{Ann}(u) = (P)$" or "u identifies only G_u and contains these exponentials".

Remark 16.5 (Step 5: decompose). For evolution factors $\tau - a_j(\xi)$ apply the Lagrange formula (23). For general coprime factors use the construction of Theorem 12.6; the components are determined up to the $\delta(P)$-dimensional exponential-polynomial space, fixed by a normalisation. Before using the components in H^m, check $\gamma_N > 0$ (Theorem 13.7).

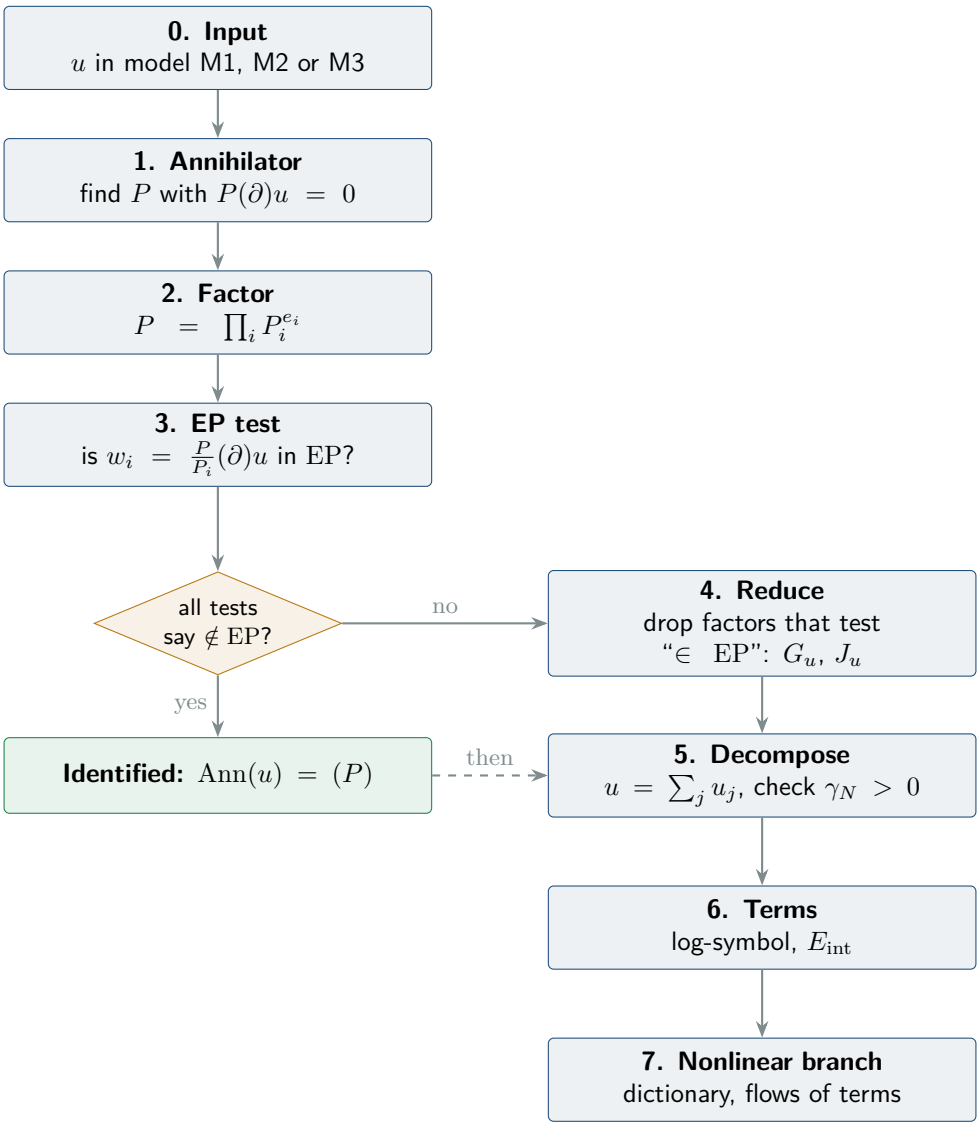


Figure 19: The algorithm as a flowchart. The left column certifies what the operator is. If every factor test says "not an exponential-polynomial", the operator is identified. Otherwise Step 4 reduces it to G_u. The right column extracts structure from u once the operator is known.

Remark 16.6 (Step 6: terms and interaction). Inside an evolution factor with symbol $a = \sum_n a_n$, the terms are read from the log-symbol, which is linear in time:

$$\log \hat{u}(t,\kappa) = \log \hat{u}_0(\kappa) + t \sum_n a_n(i\kappa).$$

Then evaluate E_{int} by the product formula (25) at cost $O(N)$ per mode. For non-commuting terms, classify the generated Lie algebra (Theorem 14.17).

Remark 16.7 (Step 7: nonlinear case). Replace Step 3 by the dictionary criterion (Theorem 15.9) and Step 6 by the flow-based interaction term and its jet (Theorem 15.1).

16.3 Worked example 1: the wave equation

We follow Steps 1–5 on the wave equation with $c = 1$ and the solution

$$u(t,x) = \underbrace{\frac{1}{1+(x+t)^2}}_{f(x+t)} + \underbrace{e^{-(x-t)^2}}_{g(x-t)}.$$

Example 16.8 (Step 1: annihilator). Take the dictionary $\{u_{tt}, u_{xx}, u_{tx}, u_t, u_x, u\}$. The kernel of the Gram matrix is the line spanned by $u_{tt} - u_{xx} = 0$. Output:

$$P = \tau^2 - \xi^2.$$

Example 16.9 (Step 2: factor).

$$P = Q_1 Q_2, \qquad Q_1 = \tau - \xi, \qquad Q_2 = \tau + \xi.$$

Two irreducible, simple factors.

Example 16.10 (Step 3: EP tests). **Remove Q_1.** We test $w_1 = (\partial_t + \partial_x)u$. The two parts behave differently:

$$(\partial_t + \partial_x) f(x+t) = 2f'(x+t), \qquad (\partial_t + \partial_x) g(x-t) = 0.$$

With $f(s) = 1/(1+s^2)$:

$$w_1 = -\frac{4(x+t)}{\left(1+(x+t)^2\right)^2}.$$

This is rational, not a finite sum of polynomials times exponentials: $w_1 \notin \mathrm{EP}$.

Remove Q_2. Symmetrically, with $g(s) = e^{-s^2}$:

$$w_2 = (\partial_t - \partial_x)u = -2g'(x-t) = 4(x-t)\, e^{-(x-t)^2}.$$

A Gaussian times a polynomial is not an exponential-polynomial: $w_2 \notin \mathrm{EP}$.

Example 16.11 (Step 4: reduce). Both tests say "not EP", so nothing is dropped. The point spectrum is empty, and the wave equation is identified:

$$\mathrm{Ann}(u) = (\tau^2 - \xi^2).$$

Example 16.12 (Step 5: decompose). $(Q_1, Q_2) = (\tau, \xi)$, so $\delta(P) = 1$. The components are $u_1 = f(x+t)$ and $u_2 = g(x-t)$, determined up to a constant moved from one to the other. Whether the components of every solution stay in H^m on a given region is decided by Theorem 13.7; on a square it holds (Example after Theorem 13.8).

16.4 Worked example 2: advection–diffusion

Now take the solution of Example 14.5,

$$u = g(x - ct;\ \sigma^2 + 2Dt), \qquad c = 2,\ D = \tfrac{1}{2},\ \sigma^2 = 1.$$

Example 16.13 (Steps 1–4). The Gram matrix of $\{u_t, u_x, u_{xx}, u\}$ has kernel spanned by $u_t + 2u_x - \frac{1}{2}u_{xx}$, so

$$P = \tau + 2\xi - \tfrac{1}{2}\xi^2.$$

It is irreducible (degree one in τ). The test is on u itself: a Gaussian is not an exponential-polynomial, so $\mathrm{Ann}(u) = (P)$ and the equation is identified.

Example 16.14 (Step 5). There is a single factor: no decomposition is needed.

Example 16.15 (Step 6: terms and interaction). The log-symbol is

$$\log \hat{u}(t,\kappa) = \log \hat{u}_0(\kappa) + t\big(-2i\kappa - \tfrac{1}{2}\kappa^2\big).$$

Its two coefficients are the two terms: advection with speed 2 and diffusion with constant $\frac{1}{2}$. The interaction term follows from (25), at cost $O(2)$ per Fourier mode:

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \big(e^{-2i\kappa t} - 1\big)\big(e^{-\kappa^2 t/2} - 1\big)\,\hat{u}_0(\kappa).$$

Figure 16 plots it.

16.5 Cost and feasibility

Component	Cost or feasibility	Source
annihilator from a dictionary	Gaussian elimination, $O(M^3)$	Step 1
factorisation	standard computer algebra	Step 2
EP test, model M1	syntactic	Step 3
EP test, model M3	holonomic, decidable mod (AC)	Thm. 11.12
EP test, black box	impossible without a bound	Thm. 11.9
EP test with bound B	linear algebra in degree $B + \deg P - 1$	Thm. 11.10
components (evolution)	one rational multiplier per factor	Eq. (23)
E_{int}: value / by order / table	$O(N)$ / $O(N^2)$ / $\Theta(2^N)$	Thm. 14.7

Table 10: Cost and feasibility of the components of the algorithm.

Remark 16.16 (Takeaway). Every step of the algorithm is a theorem with explicit hypotheses. The algorithm is exact where the theorems are positive (Steps 2, 4, 5 and 6 on exponential-type data), and it reports honestly where they are negative (Step 3 on finite data, Step 6 in Sobolev spaces).

17 Limits, open problems and status

Starting from: no new assumptions. *Why:* to state what is impossible, what is open, which statements turned out false, and how reliable each result is. *Role:* delimits every claim of this edition.

17.1 What is impossible

Each item below is a proved negative theorem, not a gap.

- a finite certificate of identification, and identification from a black box (Theorems 11.7 and 11.9);
- identification by restricting variable-coefficient operators to one solution (Proposition 11.14);
- decomposition of a solution by additive terms (Proposition 12.1);
- factor decomposition on regions with holes, for some P (Proposition 12.10);
- components in H^m in general: one derivative can be lost (Counterexample 13.1);
- a continuous or stably regularised interaction term in H^m (Theorems 14.10 and 14.12);
- a finite product of single-term flows reproducing non-commuting dynamics when the adjoint orbit is infinite (Theorem 14.17(c)).

17.2 Statements that turned out false

Statement	Counterexample or correction	Ledger
the spectrum of an expansion decides identification on a bounded domain	e^{x-t} on the unit square (Counterexample 10.16)	C3-C1
components keep the regularity of u	parallel transports on the square	C12-C1
$u \mapsto E_{\mathrm{int}}$ is continuous in H^m	real exponentials in advection–diffusion	P1-P1(e)
the companion matrix is normal iff $p = s^m - c$	$s^2 + s - 1$	R6-T3
splitting by terms lies in $\ker(L_1, L_2)$	$\frac{1}{2}x^2 + Dt$ for the heat equation	R2-T3 (corrected)
exact interaction impossible for infinite Lie algebras (as first stated)	$e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$	R5-Q5, P1-C1
NLS identification by scattering data without multiplicity	explicit counterexample	OPF5D

17.3 Open or conditional

- closedness of the sum of kernels for $N \geq 3$ differential factors [open] (ledger C12-Q2); solved for real transports in L^2;
- density of the sum of kernels on non-convex regions [open];
- minimality of the bound B on every region [open] (ledger OP-ALG-Q2);
- identification for analytic KdV data with nonzero reflection [open] (ledger INT-Q1);
- the uniform shock rate with several shock births [conditional] (ledger OPA-Q1);
- the exact values of the syzygy degrees $\delta_N(K)$ for general N [open] (ledger OPF5C-Q1);
- no finite product of flows for ∂_x^2 and x^k, $k \geq 3$ (the hypothesis of Theorem 14.17(c) fails there) [open] (ledger P1-Q3), and Theorem 14.17(c) without the assumption that the flows of Y vanish at zero coupling [open] (ledger P1-Q7).

17.4 How reliable the results are

Proofs and audits. Every statement labelled [proved] has a complete proof that passed an independent review. The reviewers were AI agents with no access to the author's reasoning, not human referees. Statements labelled [proof written, not audited] have complete proofs that were not audited; none of them is on the critical path of the main theorems.

Formal verification. Fifteen theorems of the algebraic core are formalised in Lean 4 with Mathlib, without `sorry`: the interaction identity for commuting flows, the comaximal factor decomposition with explicit projector, the normality criterion for $m = 2$, and several irreducibility results. The analytic parts (distributions, Sobolev spaces, semigroups) are not formalised, because the libraries are missing.

Citations. Many ingredients are classical: Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering. The primary sources of the load-bearing citations (Malgrange, Fulton, Kružkov, Novikov) were checked; several others only on secondary sources. A result relied upon should be traced to its primary source.

Novelty. The contribution is the package: sharp characterisations, negative theorems and counterexamples, each with a verified status. Parts of the identification theory are folklore in commutative algebra (Oberst duality); parts of the finite-data theory are known in substance (Prony, Mourrain).

17.5 What this edition does not contain

- Phases 6–11: systems and matrix coefficients, variable coefficients and the analytic encoder, the universal dictionary, convergence, the extended algorithm. They are summarised in *Punto di partenza* and will enter the complete edition after Phase 11.
- The catalogue of about 60 equation types and the tropical reading of scaling limits.
- The Lean sources.

18 Open problems and directions for improvement (Phases 0–5)

Starting from: the final report of Phases 0–5 (26 October 2026), the registry of open items, the route registry (Gates 1 and 4a) and the note on a learning pipeline. *Why:* Section 26.1 states in a few lines what is open. This chapter gives the full list for Phases 0–5, what is already proved about each item, the improvements that would strengthen the work, the routes that were scouted and not taken, and the limits of the project. *Role:* a complete record. Items that were later closed, reduced or refuted in Phase 10 are marked; for those, the complete edition (Phases 0–10) prevails.

18.1 Still open or conditional at the end of Phase 5

Item	Question	Status and obstruction	Later
INT-Q1 (analytic data)	KdV, $r \not\equiv 0$, u analytic: is there an autonomous ODE in x?	open. New partial result OPF5C-T6: relations with non-singular origin force exponential tails; order ≤ 3 gives 0 or a 1-soliton (with decay on the whole jet). Residual obstruction: singular relations at the origin, exactly where the cubic relation of Airy sits	reduced to (R) in Phase 10
OPA-Q1 general	several shock births	conditional on the localisation lemma L-loc (statement written in the audit S-CHECK-20)	closed modulo [F1]
OPA-Q2 density	density of the class Morse+(SB)	a sketch (Sard with finitely many parameters)	density proved
OPR2-Q2′	wedges with degenerate directions	partial: the tangent model is closed and direct (OPF5B-P8); localisation to the polygon is missing	still open
OPR2-Q3	repeated socle, "no finite test"	partial: proved under (I) existence of an identifying solution and (II)	characterised in Phase 10
OPF5C-Q1	$\delta_N(K)$ for general N and $N < K < 2N-2$	open; conjectures $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$	computer-assisted cases added
S4ID-P8, OPR2-P7	asymptotics with derivatives	conditional on [GT-u] (uniformity in x is not stated by Grunert–Teschl; track: Deift–Zhou 1994) and on [Hm]	[Hm] proved
INT-T1(B)	reflectionless Hirota N-solitons	conditional on [F4$^-$] (the primary source in this form was not found; verified for $N = 2$)	proved
non-convex regions	density of OPF5B-T1 outside convex sets	open	false for regions with holes

Other open questions recorded in the text.

- the closedness of the sum of kernels for $N \geq 3$ differential factors [open] (ledger C12-Q2) (solved for real transports in L^2);
- minimality of the bound B on every region [open] (ledger OP-ALG-Q2);
- no finite product of flows for ∂_x^2 and x^k, $k \geq 3$ [open] (ledger P1-Q3), and Theorem 4.10(c) without the assumption that the flows of Y vanish at zero coupling [open] (ledger P1-Q7);
- P1-Q6: no continuity of the interaction term for any operator with two non-proportional terms [open] (*false as written*, corrected in Phase 10).

Dependences that are modulo a source. [F1] Lax–Oleinik = Kružkov (the source was read; an autonomous proof is missing); [L-loc]; [GT-u]; [F4$^-$]; and Malgrange 1966 for C3-P1

with general data on convex sets (the jet case is proved without it).

18.2 Improving what is there

- **Citations whose numbers were not checked**: Krantz–Parks Thm 2.3.5 (OP-T1), Cohen GTM 138 for (AC), Hörmander SCV 1.4.4, Runge in Rudin. Of 112 sources, 34 had the theorem number checked and 75 the content (45 only in the metadata); three wrong citations were corrected (Humphreys Cor. B, Gel'fand–Dikii 1976, DLMF 9.7.10).
- **Audits**: several statements are [proof written, not audited] (not audited); TROP-5–8 are written but not audited; the Gaussian example of Section 4 and the Lagrange projector formula were derived for this edition and not literally in the ledger.
- **Algorithm**: the Gram-matrix procedure of Step 1 is one way to obtain a candidate annihilator and has to be confirmed as the intended one; a cost analysis of each step is only indicative; the nonlinear branch has no full worked example.
- **Fisher–KPP** is announced as a benchmark but only an estimate non-uniform in t is recorded (S4T-P5); a statement or a removal is needed.
- **Examples** suggested but not added: a worked example of the exact sequence of the dictionary criterion, and an example of a non-empty point spectrum with several factors.

18.3 Routes scouted and not taken (Phases 1–5)

Each candidate route was promoted, merged, parked or eliminated with a stated reason.

- **Promoted**: factor decomposition with the least common left multiple for variable coefficients (RT-1/R2b); the additive log-symbol for first-order evolutions (RT-2); the Wei–Norman and Dyson interaction analysis (RT-3); $\ker \mathbb{G} = \mathrm{Ann}(u) \cap \mathrm{span}$ and Prony $\Leftrightarrow$ Waring (RT-4); complete invariants (RT-6); the weighted Fock orthogonal basis (RT-R3a; extension to Airy open).
- **Parked**: the symbol-tensor route RT-5 (the tensor keeps only the principal symbol; kept as a geometric invariant); RT-S1e, RT-S1f (no constant-coefficient theorem; only sectorial components); RT-R1b (needs a sign hypothesis on the datum); Koopman operators ("dynamic mode decomposition is regression"; to reopen with the normal form: eigenfunctions carry the nonlinear term); Painlevé, NLSPEC, HAANTJES (triggers in the nonlinear verdicts).
- **Eliminated**: RT-S1g for the core (it gives no information for elliptic or parabolic operators), kept for the hyperbolic part.
- **Scouted in Phases 3–4**: the tropical estimate of E_{int} from the symbol alone holds only on dissipative faces (TROP-3′, TROP-8); on conservative faces (Klein–Gordon, transport) no uniform estimate exists (TROP-4).

18.4 The statistical wrapper that was proposed

A learning pipeline was drafted around the exact analytic core and *excluded* from the project by decision of the user (no machine learning). It is recorded as a direction: (1) an interpolation step returns a weighted family of smooth functions $(\hat{u}_s, \pi_s)$; (2) each is passed to the exact algorithm, which returns its factors and components; (3) the weights give a distribution over factors ("transport 0.7 ± 0.1, diffusion 0.3 ± 0.1"); (4) the distribution sets the parameters of a clustering, so that each group contains one factor; (5) repeat inside each group, with two measures of "tangle": the Friedrichs cosine between factors and E_{int} between terms. The exact

parts exist; the interpolation, aggregation and clustering are a proposal, and the note asked whether the basis functions of the first step must be chosen in advance.

18.5 What this edition does not contain

The catalogue of about 60 equation types and the tropical reading of scaling limits; the Lean sources; a human review (the reviewers were AI agents); Phases 6–10, which are in the complete edition.

Part II: beyond constant coefficients (Phases 6–10)

Sections 19–25 extend the theory of Sections 2–6 (Phases 0–5) to systems, variable coefficients, noisy and finite data, encoders, dictionaries and the extended algorithm (Phases 6–10; Phase 11 was the audit of the whole). Each result names its proof as "Dim. An" or "Dim. Bn" in the companion *Catalogue of proofs* (directory `paper/analytical_complete`).

19 Systems of equations and gauge

Starting from: a vector field $u = (u_1, \dots, u_k)$ on a connected open set $\Omega \subset \mathbb{R}^n$ that solves a system $P(\partial)u = 0$ with a *matrix* of constant coefficients $P \in R^{l\times k}$, $R = \mathbb{C}[\zeta_1, \dots, \zeta_n]$. The operator P is not given. *Why:* the equations of physics that are not scalar (Dirac, Maxwell, elasticity) are systems; Phases 1–5 treated one scalar equation. We ask the same three questions: can one solution determine the system, when can a sum of kernels be split, and what does a change of gauge do to what can be known? *Role:* extends Sections 5–3 from R to matrices over R. It supplies Step 3b of the extended algorithm (Section 25). The proofs are Dim. A32–A35 and B73–B78.

19.1 From one equation to a module

For a scalar equation the annihilator $\mathrm{Ann}(u)$ was an ideal of R. For a system the object that replaces it is a *module*. The rows of P generate a submodule $U_P \subseteq R^{1\times k}$, and the quotient

$$M_P = R^{1\times k}/U_P$$

is the module of the system: solutions of P are exactly the homomorphisms $M_P \to \mathcal{A}$ (Malgrange–Oberst duality). A vector u *identifies* P when its annihilator $\mathrm{Ann}(u) = \{r : r(\partial)u = 0\}$ equals U_P, i.e. when the map $M_P \to \mathcal{A}$, $[r] \mapsto r(\partial)u$, is injective.

Remark 19.1 (Intuition). Think of M_P as the list of everything the system says, up to consequences. A single function u must carry all of it in an injective way. The obstruction is local: at a point λ the solutions of the system that are pure exponentials $e^{\lambda\cdot x}$ form one line, but M_P may contain a plane of "independent" behaviours at that point. A plane cannot be embedded in a line.

19.2 When does an identifying solution exist?

Define the *socle* of M at a maximal ideal $\mathfrak{m}_\lambda$ as the set of elements of M killed by $\mathfrak{m}_\lambda$; it is the space of exponential behaviours $e^{\lambda\cdot x}$ the module contains.

Theorem 19.2 (Existence of an identifying solution (ledger SYS-EX) [proved]). *Let $\Omega \subseteq \mathbb{R}^n$ be non-empty, open and connected. A finitely generated module M embeds in $C^\infty(\Omega)$ (or $\mathcal{D}'(\Omega)$) if and only if $\dim_{\mathbb{C}} \operatorname{soc}_{\mathfrak{m}}(M) \le 1$ for every maximal ideal $\mathfrak{m}$. In particular a system P has an identifying solution if and only if $\dim \operatorname{Hom}_R(R/\mathfrak{m}, M_P) \le 1$ for every maximal $\mathfrak{m}$. [proof: Dim. SYS-EX]*

Example 19.3. Two ODE systems For $n = 1$ let $P = \operatorname{diag}(s-1, s-1)$ (two copies of $u' = u$). The socle at $s = 1$ has dimension 2: no function u ever identifies it. For $P = \begin{pmatrix} s-1 & 1 \\ 0 & s-2 \end{pmatrix}$ all socles have dimension ≤ 1, and $u = (ae^x - be^{2x},\, be^{2x})$ with $ab \neq 0$ identifies it. In general an ODE system of rank r with invariant factors $e_1 \mid \cdots \mid e_r$ has an identifying solution iff $e_1 = \cdots = e_{r-1} = 1$ (ledger SYS-T0) (known).

Theorem 19.4 (Square non-singular systems in two or more variables (ledger SYS-C1) [proved]). *If $n \ge 2$ and $P \in R^{k\times k}$ has $\det P \neq 0$, every socle of M_P is zero, so on every connected Ω some $u \in \operatorname{Sol}(P)$ identifies P. The associated primes of M_P are the prime factors of $\det P$. [proof: Dim. SYS-C1]*

Dirac in $3+1$ dimensions, the matrix Klein–Gordon operator and the Lamé system of elasticity are all of this type. They always have an identifying field.

Remark 19.5 (Warning). [title=Where this is not new] For $\Omega = \mathbb{R}^n$ the existence criterion is known (Oberst, Pommaret, Macaulay: the number of trajectories needed to generate a system). The new point here is only that Ω may be any connected open set, including one where $C^\infty(\Omega)$ is not an injective module.

19.3 Sums of kernels for systems

For a scalar $P = Q_1 Q_2$ we proved $\ker P = \ker Q_1 + \ker Q_2$ on a simply connected region (Theorem 3.1). For systems the analogue is stated with the *least common left multiple*: if U_1, U_2 are the row modules of two sub-systems and $N = R^{1\times k}/(U_1 + U_2)$, then $\operatorname{Sol}(Q_1) + \operatorname{Sol}(Q_2) = \operatorname{Sol}(P)$ requires $U_1 \cap U_2 = U_P$.

Theorem 19.6 (Sum of kernels (ledger SYS-S1) [proved]). *On every open Ω, $\operatorname{Sol}(Q_1) + \operatorname{Sol}(Q_2) = \operatorname{Sol}(P)$ implies $U_1 \cap U_2 = U_P$. Conversely, if $U_1 \cap U_2 = U_P$ and $\operatorname{Ext}^1(N, \mathcal{A}) = 0$ the equality holds; this is the case when Ω is convex, and when N has finite length and $H^1_{dR}(\Omega) = 0$. The intersection $\operatorname{Sol}(Q_1) \cap \operatorname{Sol}(Q_2)$ is finite-dimensional iff N has finite length, with dimension $\delta_{\mathrm{sys}} = \ell(N)$. For one scalar equation in two variables this is Theorem 3.1. [proof: Dim. SYS-S1]*

Remark 19.7 (Counterexample: Simple connectivity does not suffice for infinite-length quotients). [false] There are a simply connected open $\Omega \subset \mathbb{R}^2$ (a "U"-shaped domain) and $Q_1 = (1, 0)$, $Q_2 = (1, \partial_x)$ with $U_1 \cap U_2 = U_P$ but $\operatorname{Sol}(Q_1) + \operatorname{Sol}(Q_2) \subsetneq \operatorname{Sol}(P)$, in C^∞ and in $\mathcal{D}'$. The obstruction is the failure of P-convexity of Ω, not its topology. [proof: Dim. SYS-S1]

19.4 Dirac in Hamiltonian form

Write the free or perturbed Dirac equation as $L = I\partial_t + \sum_k \alpha^k \partial_k + B$ with arbitrary constant 4×4 matrices α^k, B (no Clifford relations are assumed).

Theorem 19.8 (Dirac coefficients are read exactly (ledger H-DIRAC) [proved]). *If $\psi \in \mathrm{Sol}(L_0)$ identifies L_0 and L is another operator of this form with $L\psi = 0$, then $L = L_0$: the matrices α^k and B are determined by ψ exactly, with no ambiguity. An identifying ψ always exists. For massive Dirac in $3+1$ dimensions $M_D \cong (R/\mathfrak{q})^2$ with $\mathfrak{q} = (p_0^2 - |p|^2 - m^2)$, and ψ identifies D iff ψ_3, ψ_4 are independent over $R/\mathfrak{q}$; in $1+1$ dimensions ψ identifies D_2 iff $\psi_2 \notin \mathrm{EP}$.* [proof: Dim. H-DIRAC]

Example 19.9. Where the Dirac coefficients come from Normalising the coefficient of ∂_t to the identity removes the ambiguity $D \sim VD$ of the covariant form $\gamma^\mu \partial_\mu$. In $1+1$ dimensions the identification test is just the one of Section 5: the second component must not be an exponential-polynomial.

19.5 Gauge, magnetic fields, and Lamé

Proposition 19.10 (Gauge covariance and what invariant data reveal (ledger G-L0, G-T2) [proved]). *Under $A \mapsto A + d\chi$, $u \mapsto e^{i\chi}u$ the covariant operators transform covariantly, so the identification tests are gauge-invariant. If the observation is invariant under the phase, a reconstruction of the potential A recovers at most its gauge class $[A]$: $A' \sim A$ iff $dA' = dA$ and the holonomies agree (Aharonov–Bohm). On a simply connected region, $A' \sim A$ iff $dA' = dA$.* [proof: Dim. G-T2]

Proposition 19.11 (Magnetic Schrödinger: one field does not identify (A, V) (ledger SYS-N3) [proved]). *For $n \geq 2$ and a nowhere-zero solution u the pairs (A', V') compatible with u form an infinite-dimensional affine space, parametrised by divergence-free fields k; their magnetic fields differ, so they are not gauge-equivalent, even in the non-stationary case. The obstruction is a count: one complex scalar equation against $n+1$ real unknowns.* [proof: Dim. SYS-N3]

Proposition 19.12 (Lamé in space (ledger LAME-3) [proved]). *For $\rho, \mu \neq 0$, $\lambda+\mu \neq 0$, $\lambda+2\mu \neq 0$, the system has $\det P = q_P q_S^2$ with Bass multiplicities 1 and 2; an identifying displacement field exists, and u identifies it iff $\operatorname{div} u$ satisfies no equation besides the wave equation of speed $\sqrt{(\lambda+2\mu)/\rho}$ and the two transverse polarisations of $\operatorname{rot} u$ satisfy no extra relation. A pure P-wave ($\operatorname{rot} u = 0$) never identifies.* [proof: Dim. LAME-3]

Proposition 19.13 (Dirac $1+1$ as an evolution: no uniform product (ledger SYS-E1, E2, E3) [proved]). *Writing $\psi_t = (X + mY)\psi$ with X the transport and Y the mass term, the operators $X, \mathrm{ad}_X^k Y$ are linearly independent, so no finite product of transport and mass flows with ξ-independent times equals $e^{t(X+mY)}$ uniformly in the mass. The Lie algebra they generate is infinite-dimensional. An explicit product (Euler angles) exists, with times that are not polynomial in ξ and with non-local Fourier multipliers.* [proof: Dim. SYS-E]

Proposition 19.14 (Systems from finite data (ledger SYS-R1) [proved]). *From exponential-polynomial data one can identify only systems with finite-dimensional solution space; for an ODE system the identifying solution exists iff $e_1 = \cdots = e_{r-1} = 1$.* [proof: Dim. SYS-R1]

Remark 19.15 (Why it matters). For a system the unknown is a module and the datum is one vector function. The only obstruction to identification is the socle, a purely algebraic and local count, and it vanishes for every non-singular square system in $n \geq 2$ variables. What a single field can never do is separate gauge from physics.

20 Variable coefficients

Starting from: a rectangle $\Omega = I \times J \subset \mathbb{R}^2$ and a solution u of $iu_t = -u_{xx} + V\,u$, i.e. $L_0 u = Vu$ with $L_0 = i\partial_t + \partial_x^2$, where the potential $V(x)$ or $V(t,x)$ is unknown. Solutions are *strong* (class SF): $u \in C^{1,2}(\Omega)$ and the equation holds almost everywhere. *Why:* in physics the coefficients are rarely constant: a Schrödinger particle feels a potential. Phases 1–5 identified constant coefficients. Here the unknown is a function, and the first question is whether a single solution can reveal it at all. *Role:* extends Section 5 from $\mathbb{C}$ to function coefficients. It supplies Step 3c of the extended algorithm (Section 25). The proofs are Dim. A36–A40 and B78.

20.1 Too much freedom: a huge class identifies nothing

Suppose we allow coefficients in any algebra $\mathcal{K}$ of functions closed under differentiation (all smooth functions, all analytic functions). Then even one solution that never vanishes carries no information beyond itself.

Theorem 20.1 (Rank-one annihilator (ledger VAR-N1, VAR-T1(a)) [proved])**.** *Let $u \in C^\infty(\Omega)$ have no zeros, with $a = u_t/u$ and $b = u_x/u$ in $\mathcal{K}$. Then the annihilator of u among differential operators with coefficients in $\mathcal{K}$ is generated by $\partial_t - a$ and $\partial_x - b$, and the module $\mathcal{K}[\partial]u$ is isomorphic to $\mathcal{K}$ itself. For an affine family $L_\theta = L_0 + \sum \theta_k T_k$ the compatible parameters are $\theta_0 + \ker \Phi_u$, so θ is identified iff $\Phi_u(\theta) = \sum \theta_k T_k u$ is injective. [proof: Dim. VAR-N1]*

Remark 20.2 (Intuition). If the coefficient class is closed under differentiation, a zero-free u can write each of its own derivatives as "a coefficient times u", so every law is a law about u itself and nothing more. To identify anything the class of coefficients must be restricted in advance.

Remark 20.3 (Warning). [title=Zeros matter] With zeros the statement fails: $u = t - x^2$ near 0 satisfies no law $\partial_t + \alpha\partial_x + \beta$ with smooth coefficients, because at the origin it would give $1 = 0$.

20.2 A potential $V(x)$ is read from any non-zero solution

For a time-independent potential the situation is the opposite: nothing can hide.

Theorem 20.4 (Identification of $V(x)$ (ledger VAR-T1(b)) [proved])**.** *Let $u \not\equiv 0$ be a strong solution with potential $V \in L^\infty_{\mathrm{loc}}(J)$, and $X_0 = \{x : u(t,x) = 0\ \forall t\}$. Then X_0 is closed and discrete; if $L_0 u = W(x)u$ for another W, then $W = V$ almost everywhere; and*

$$V(x) = \frac{iu_t + u_{xx}}{u} \qquad \text{at every } (t,x) \text{ with } u(t,x) \neq 0.$$

[proof: Dim. VAR-T1b]

Remark 20.5 (Intuition). If u vanished along infinitely many vertical lines accumulating at x_0, cutting u off to the right of x_0 would give another strong solution that vanishes on a half-strip. Schrödinger waves have infinite speed, and a Carleman estimate (a weighted energy inequality) shows that a solution vanishing on a strip vanishes everywhere. So u cannot have such lines, and the formula $V = (iu_t + u_{xx})/u$ holds on a set of full measure.

Example 20.6. The harmonic oscillator For $V = x^2$ and the Hermite functions $h_0, \dots, h_3$ the set X_0 is empty or $\{0\}$, and $(iu_t + u_{xx})/u$ returns x^2 exactly. The same formula recovers the Pöschl–Teller and Airy potentials.

20.3 A potential $V(t,x)$: silent strips

Theorem 20.7 (Time-dependent potentials (ledger VAR-T1(c)) [proved]). *For $V \in C(\Omega)$, the potentials compatible with u are $V + \{W : W = 0$ on $\Omega \setminus Z(u)\}$, where $Z(u) = \{u = 0\}$; so V is identified iff* $\operatorname{int} Z(u) = \emptyset$. *If* $\operatorname{int} Z(u) \neq \emptyset$ *there is an open interval $T' \subset I$ with $u = 0$ on $T' \times J$ (a* silent strip*). Silent strips occur: Tychonoff's series gives a solution vanishing for $t \leq 0$ and non-zero for $t > 0$. Hence "every $u \not\equiv 0$ identifies $V(t,x)$" is false. [proof: Dim. VAR-T1c]*

Remark 20.8 (Counterexample: Data that switch on from the boundary). [false] On a bounded domain without boundary conditions, a solution can be identically zero up to a time and non-zero afterwards (Tychonoff's series). While it is silent every potential supported in the silent time is compatible with it. The failure is in the time direction: unique continuation holds in x but not in t.

20.4 Stability and the phase

Theorem 20.9 (Stability of the reconstruction (ledger VAR-P3) [proved]). *On a rectangle K with $|u_1| \geq \delta > 0$: $\|V_1 - V_2\|_\infty \leq \delta^{-1}(1 + \|V_2\|_\infty)\|u_1 - u_2\|_{C^{1,2}(K)}$; the map is not continuous from C^1 to L^∞ (an explicit oscillation); and under an a priori bound M on $\|e_{tt}\|_\infty, \|e_{xxx}\|_\infty$ one has a Hölder modulus with sharp exponent $1/3$. It is a modulus of continuity, not a certificate. [proof: Dim. VAR-P3]*

Theorem 20.10 (Unobserved phase: V modulo constants (ledger VAR-P4) [proved]). *If only the class $\{e^{i\theta(t)}u\}$ is observed, the compatible potentials are exactly $\{V + c : c \in \mathbb{R}\}$. For $P(\partial_t, L_q)u = 0$ with P unknown, (P,q) is defined only modulo $q \to q + c$, $P(\tau,\ell) \to P(\tau, \ell - c)$: a redundancy of the parametrisation. [proof: Dim. VAR-P4]*

Proposition 20.11 (Dirac $1+1$ with a potential (ledger VAR-P5) [proved]). *Finite speed of propagation leaves regions that the data never visit: for $m = V = 0$ a travelling bump $\psi = f(x-t)v_+$ vanishes for $x \leq 0.5$ and $x \geq 1.7$, so every potential supported there is compatible. The matrix potential is determined at x_0 iff* $\operatorname{span}_t\{\psi(t, x_0)\} = \mathbb{C}^2$. *[proof: Dim. VAR-P5]*

20.5 Other results of this phase

- **Algebra of differential operators with polynomial coefficients** (Weyl algebra): for a D-finite u the annihilator is a principal left ideal and u identifies iff its derivatives are independent over K; Airy is irreducible and every solution identifies; the oscillator is reducible exactly for $E \in \pm(2\mathbb{N} + 1)$ (ledger VAR-T2a, VAR-T6, VAR-E1, VAR-E2).
- **Dynamical Lie algebras**: $\dim \mathfrak{g}_V < \infty$ iff $V \in \operatorname{span}\{1, x, x^2\}$; the modified Pöschl–Teller potential has an explicit evolution through a Darboux chain (ledger VAR-P8).
- **Non-autonomous splitting**: $E^{\text{mult}} = \iint [A_2(s_1), A_1(s_2)] + O(t^3)$; commuting at equal times is not enough (ledger VAR-P9). Dyson–Phillips converges for bounded time-dependent perturbations (ledger VAR-P11); Floquet logarithms are bounded but not differential (ledger VAR-P12).

These are list cards of the catalogue (Part IV); the closures made in Phase 10 are in Section 24.

Remark 20.12 (Why it matters). A function-valued coefficient is identified by one solution exactly when the solution cannot hide: for $V(x)$ it cannot (infinite speed and unique continuation); for $V(t,x)$ it can fall silent; for a huge class of coefficients, nothing is identified. Stability is of order two and the phase costs one constant.

21 Convergence, noise and the limits of learning

Starting from: one analytic function u on a connected open set $\Omega \subset \mathbb{R}^n$, observed through its Taylor jet at a point p_0 (the datum at step K is $j_K u$), or through a noisy copy $\tilde{u}$ with $\|\tilde{u} - u\| \le \delta$ in a stated norm. *Why:* Sections 5 and 5.2 proved that no finite amount of data certifies identification. A real experiment gives more and more data: does the identified operator converge, how fast, and what can a noisy measurement still say? *Role:* supplies Steps 1–4 and 7 of the extended algorithm (Section 25). The proofs are Dim. A41–A43.

21.1 Identification in the limit

At step K the rule computes, for each degree $m \le K$, the space $N_{m,K}$ of operators of degree $\le m$ whose action on u has vanishing Taylor coefficients up to order $K - m$. These spaces decrease with K and their limit is $\mathrm{Ann}(u) \cap R_{\le m}$. The *minimal-degree rule* outputs the gcd of $N_{m^*(K),K}$ at the smallest degree $m^*(K)$ where it is non-zero.

Theorem 21.1 (Identification in the limit (ledger CONV-L1, CONV-T1) [proved])**.** *If* $\mathrm{Ann}(u) \neq 0$ *with minimal degree* d*, the rule outputs* $g_d = \gcd(\mathrm{Ann}(u) \cap R_{\le d})$ *from some step on; if* $\mathrm{Ann}(u) = (P)$ *it outputs* P*, without knowing* $\deg P$*; if* $\mathrm{Ann}(u) = 0$ *the degree* $m^*(K) \to \infty$*. The sequence of outputs takes at most* $(d+1)(d+2)/2$ *values. But:*

- *the step* K^* *after which the output is correct is not bounded by any function of* d*;*
- *no function of the jet of order* K *is a correct certificate: any finite jet is shared by a polynomial, whose annihilator is a different ideal.*

[proof: Dim. CONV-T1]

Example 21.2. Convergence at an unpredictable time For $u = 1 + x^{N+2}e^{x^2}$ one has $\mathrm{Ann}(u) = (\tau)$. The rule outputs "no law of degree 1" for $K \le N+1$ and τ from $K = N+2$, for every N: nothing in the first $N+1$ derivatives announces the law.

21.2 No learner decides whether a law exists

Theorem 21.3 (Analytic Gold theorem (ledger CONV-T2, CONV-T2′) [proved])**.** *For any sequence of maps* φ_K *from jets of order* $\le K$ *to* $\{law, no\ law\}$ *(no computability or measurability assumed) there is a real entire function* u *on which* $\varphi_K(j_K u)$ *does not converge to the correct answer. The same holds restricted to the class where* $\mathrm{Ann}(u)$ *is principal or zero; hence no learner identifies* $\mathrm{Ann}(u)$ *on that class.* *[proof: Dim. CONV-T2]*

Remark 21.4 (Intuition). Build u as a limit of polynomials. Whenever the learner says "law" on the current polynomial (it must, eventually), add Taylor coefficients, chosen outside a proper algebraic set, that destroy every law of degree $\le k$. The limit function is differentially transcendental yet agrees with a polynomial where the learner said "law", infinitely often.

Remark 21.5 (Takeaway). "$\mathrm{Ann}(u) \neq 0$" is *verifiable in the limit* (a rule that is eventually right when it holds) but not *decidable in the limit.*

21.3 Noisy data

Theorem 21.6 (Noise: discontinuity, conditional stability, discrepancy rule (ledger CONV-T3, T4, T5) [proved])**.** *(T3) In spaces where polynomials are dense and* $e^{|z|^2}$ *lies, the sets* $\{\mathrm{Ann} = 0\}$

and $\{\mathrm{Ann} \neq 0\}$ *are both dense:* $u \mapsto \mathrm{Ann}(u)$ *is discontinuous everywhere. (T4) At a fixed level* n_0 *with a margin* $\rho = \sigma_2(\Phi(u)) > 0$*, the minimal singular vector of* $\Phi(\tilde{u})$ *is within* $\sin\angle \leq 2\delta/\rho$ *of the true law, an order that is optimal; under an* L^2 *error and an a priori* H^{n_0+s} *bound the rate is Hölder, and* $\sigma_1(\Phi(\tilde{u})) > \delta$ *certifies a misfit. (T5) The discrepancy rule* $m(\delta) = \min\{m : \sigma_1(\Phi_m(\tilde{u})) \leq \delta_m\}$ *returns the correct degree* d *and converges at rate* δ *once* $\delta < \delta_0 = \frac{1}{2}\min_{m<d} s_m$*; if* $\mathrm{Ann}(u) = 0$*,* $m(\delta) \to \infty$*. The threshold* δ_0 *cannot be bounded below from the data. [proof: Dim. CONV-T4]*

Example 21.7. The heat kernel from noisy data For $u = (t+1)^{-1/2}e^{-x^2/(4(t+1))}$ (minimal degree 2, $\mathrm{Ann} = (\tau - \xi^2)$), the rule returned $m(\delta) = 2$ for $\delta = 10^{-3}$ and 10^{-4} in a numerical check.

21.4 Other results of this phase

- **Locality**: when the law is a non-local operator it is invisible to $\mathrm{Ann}(u)$; an irrational fractional power or a delay gives $\mathrm{Ann} = 0$ (a delay is in the local class through an auxiliary transport) (ledger CONV-T6, CONV-P6, CONV-P7).
- **Dimension** $1 + d$: the ambiguity of the factor decomposition is infinite unless the factors are pairwise co-maximal (ledger CONV-D1); on a convex region $\ker(Q_1Q_2) = \ker Q_1 + \ker Q_2$ in C^∞ (ledger CONV-D2).
- **Boundary conditions** are invisible to $\mathrm{Ann}(u)$; stability holds for a dictionary of traces (ledger CONV-B1–B3). Splitting with a Dirichlet boundary has global order 1 (Lie) and $\Theta(h^{5/4})$ (Strang) in L^2 (ledger CONV-T8).
- **Mori–Zwanzig** for constant-coefficient 2×2 systems: exact memory form and eliminated law (ledger CONV-MZ1–3); the criterion "$E_{\mathrm{int}} = 0 \iff [\mathcal{P}, L] = 0$" is false (ledger CONV-MZ3-F).
- **A convergence contract** for chains of steps: classes (A) exact, (B) convergent with explicit error, (C) no guarantee, (Lim) in the limit; the error of a chain of Lipschitz steps is $\leq \sum_i (\prod_{j>i} L_j)\varepsilon_i$ (ledger CONV-T7).

Remark 21.8 (Why it matters). Learning an operator from growing data is possible only in the limit and without a certificate; deciding whether any law exists is impossible even in the limit; and noise can be handled only relative to a finite level of the dictionary and a margin that the data cannot supply.

22 The analytic encoder

Starting from: a Schrödinger-type equation with a variable potential $q(x)$ *known*, written $P(\partial_t, L_q)u = 0$ with $L_q = -\partial_x^2 + q$ and $L_0 = -\partial_x^2$, on a symmetric interval J. *Why:* the variable potential is a nuisance: the theory of Phases 1–5 applies to L_0. Can one change the unknown by an explicit, invertible, stable map so that the potential disappears, without machine learning? Such a map is an *encoder*. *Role:* supplies Step 5 of the extended algorithm (Section 25): the encoder is a *translation* to constant coefficients, not a source of information. The proofs are Dim. A44–A47 and B79.

22.1 What an encoder is

A *structured encoder* is a linear map W acting in x only, with (E1) finite-dimensional kernel on its own domain, (E2′) $WL_q = (L_0 + c)W$ for a constant c, and (E3) explicit. The conditions are all needed: without (E1) the zero map qualifies, and without (E3) the interaction-picture map $U_0(t)U_V(t)^{-1}$ does.

Proposition 22.1 (A constant potential has no strict differential intertwiner (ledger ENC-K0) [proved])**.** *For $q \equiv c \neq 0$ no differential operator of any order satisfies $WL_c = L_0 W$. With the relaxed requirement $WL_c = (L_0 + c)W$ the identity works.* [proof: Dim. ENC-K0]

22.2 The Volterra transformation operator

Let K solve the Goursat problem $K_{xx} - K_{ss} = q(x)K$ on the triangle $|s| \le |x| \le a$, with $K(x,x) = \frac{1}{2}\int_0^x q$ and $K(x,-x) = 0$, and put

$$Tf(x) = f(x) + \int_{-x}^{x} K(x,s) f(s)\,\mathrm{d}s, \qquad W = T^{-1}.$$

Theorem 22.2 (The symmetric Volterra encoder (ledger ENC-1, ENC-U1) [proved])**.** *For $q \in C^1$ the problem has a unique solution, and for every $f \in C^2(J)$, without any boundary condition, $L_q T f = T L_0 f$ with $(Tf)(0) = f(0)$, $(Tf)'(0) = f'(0)$. T is bijective on C, C^2 and (for $q \in C^\infty$) C^∞, with $\|K\| \le \frac{aQ}{2} e^{a\sqrt{Q}}$ and a Lipschitz dependence on q. Among all Volterra intertwiners the one preserving Cauchy data at the base point is unique; it depends on the base point already for constant q.* [proof: Dim. ENC-1]

Remark 22.3 (Intuition). In characteristic coordinates the kernel solves a Volterra integral equation by successive approximations, with the bound $(Q|\xi\eta|/4)^n/(n!)^2$. Volterra operators with a triangular kernel have no spectrum, so T is invertible, and the Goursat conditions are exactly the ones that make every boundary term in the Leibniz expansion of $L_q Tf - TL_0 f$ cancel.

Corollary 22.4. *One encoder for every equation (ledger ENC-C1) [proved] For $q \in C^\infty$ and every polynomial P: $P(\partial_t, L_q)u = 0 \iff P(\partial_t, L_0)(Wu) = 0$. The same W serves Schrödinger, heat, wave and Klein–Gordon.* [proof: Dim. ENC-C1]

Example 22.5. Constant potential For $q \equiv c$ the transformation maps the exponential $e^{ik\cdot}$ to $e^{ikx_0}[\cos\mu(x-x_0) + ik\sin\mu(x-x_0)/\mu]$ with $\mu^2 = k^2 - c$: the eigenfunction of L_c with the Cauchy data of e^{ikx} at x_0.

22.3 What no encoder can do

Theorem 22.6 (No time-independent encoder for a time-dependent potential (ledger ENC-P0) [proof written, not audited])**.** *Let W be linear with finite kernel and $W(-\partial_x^2 + V(t,\cdot))f = (-\partial_x^2 + c(t))Wf$ for all t. Then $V(t,x) - V(s,x) = c(t) - c(s)$: the potential must be autonomous up to a function of time. The reason is that otherwise infinitely many independent functions $m\varphi_k$ would lie in $\ker W$. Encoders for a genuinely time-dependent potential exist but depend on time (interaction pictures), and a pointwise representative with the identity time change exists exactly for $V = \beta(t)x + \gamma(t)$.* [proof: Dim. ENC-P0b]

Theorem 22.7 (Blind encoders and identifiability (ledger ENC-N1′) [proved])**.** *A blind encoder, one whose output does not depend on which potential produced the data, exists iff the coherence condition (C) holds for the chosen family; it is implied by identifiability and is strictly weaker (a witness: the data model that observes only a phase). The correct statement is: blind encoder $\iff$ identifiability modulo the gauge group of the data model.* [proof: Dim. ENC-N1p]

Remark 22.8 (Warning). [title=An earlier claim that turned out false] The synthesis of the first version, "the exceptions change the time", was false as written: Avron–Herbst and the whole class $V = \beta(t)x + \gamma(t)$ have a pointwise representative with the identity time change. Likewise "a blind encoder exists iff $W_V^{-1}v$ is injective" is false (ENC-N1).

22.4 Other results of this phase

- **Darboux encoders**: a differential W with $WL_q = (L_0 + c)W$ is a Darboux chain on each component of a complement of a countable closed set; for rational $V \to 0$ it exists iff V is an iterated Darboux transform of 0 (Adler–Moser) (ledger ENC-K1′, ENC-K1). They are unbounded with a non-trivial kernel, so they lose modes.
- **The interaction term with a potential**: $\|E^{\mathrm{mult}} f\| \le \frac{t^2}{2}(\|V''\|\|f\| + 2\|V'\|\|f'\|) + \frac{2t^3}{3}\|V'\|^2\|f\|$ with the optimal constant $2/3$, and $E^{\mathrm{mult}} \equiv 0$ iff V is constant (ledger ENC-A9).
- **Faddeev decomposition** on an interval and **Foldy–Wouthuysen** for the free Dirac $1+1$ are decompositions, not encoders (ledger FAD-1, FW-1).

Remark 22.9 (Why it matters). Removing a variable potential is possible by an explicit stable Volterra transformation when the potential is known, and the same map works for every equation in (∂_t, L_q). When the potential is unknown the encoder gives no information, and for a time-dependent potential no time-independent encoder exists.

23 The universal dictionary and the misfit certificate

Starting from: a filtered dictionary $\mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots$ of finite sets of operators (the *atoms*), enumerated by an explicit algorithm; a datum u *fits* a level if some non-zero combination of its atoms annihilates u. *Why:* the project was asked: can a dictionary be built that "takes everything into account"? The honest answer has three parts: in the sense of approximation, yes; as exact identification of every operator, no; and in between, a misfit can always be certified by finite data, while a fit never can. *Role:* supplies Steps 2 and 4 of the extended algorithm (Section 25). The proofs are Dim. A48–A51.

23.1 What "universal" can mean

Theorem 23.1 (Every local linear operator is a limit of dictionary elements (ledger DICT-U1) [proved])**.** *By Peetre's theorem a linear operator on $C^\infty(\Omega)$ that does not enlarge supports is a differential operator with smooth coefficients, of locally finite order. With a partition of unity and polynomial approximation of the coefficients it is a limit, in the topology of locally bounded order, of elements of the explicit dictionary $\{\chi_j(y)y^\beta\partial^\alpha\}$. It is a pointwise limit of polynomial-coefficient operators iff its order is globally finite.* [proof: Dim. DICT-U1]

The same holds for classes beyond differential operators: regular non-linear operators (differential polynomials, by a local version of a Peetre–Slovák argument), symbols with loss ε, completely monotone memory kernels, delays (as an augmented transport system) and systems, each with its own explicit atoms (ledger DICT-U2, U3, RIT1, U4).

23.2 What it cannot mean

Theorem 23.2 (No countable dictionary identifies everything (ledger DICT-N1) [proved])**.** *(i) For any countable dictionary the exactly representable operators form a meagre set (Baire). (ii) No finite data certifies the fit or the identification. (iii) If* $u \in \mathrm{EP}$*, at every high level* $\mathrm{Ann}(u) \cap R_{\le n} \supsetneq (L) \cap R_{\le n}$ *for any single law* L*: the kernel is strictly larger than the consequences of one law. (iv) A procedure with finite outputs emits only countably many operators. (v) For a filtered family of differential polynomials the set of* u *that satisfy some equation of the dictionary is meagre* and *shy (prevalence): the generic* u *satisfies none.* [proof: Dim. DICT-N1]

Remark 23.3 (Intuition). A compact set in an infinite-dimensional space has empty interior, so a countable family of finite-parameter families covers a "thin" set. For the measure statement one moves u along a direction on which the determinant of the dictionary applied to $u + sw$ is a non-zero polynomial in the single real parameter s: only finitely many s lie in the bad set.

Remark 23.4 (Warning). [title=What this theorem does not say] "Everything" holds in the sense of approximation (Theorem 23.1) plus a misfit certificate, not as exact identification. The first form of the statement for coefficients that are C^∞ functions ("no operator is identified") is false and was replaced by a rank theorem: an affine family is identified iff $\ker \Phi_u = 0$ (ledger DICT-RANGO).

23.3 Fit and misfit

Theorem 23.5 (Asymmetry of fit and misfit (ledger DICT-A, MF1–MF3) [proved])**.** *Let* $\{D_0, \dots, D_N\}$ *be independent continuous operators. The set of* u *with a* misfit *(the* $D_i u$ *independent) is open, and each such* u *has a certificate: a non-vanishing determinant* $\det\langle D_i u, \psi_j\rangle$ *for finitely many test functions, valid in a neighbourhood. The set of* u *that* fit *is closed with empty interior, and for differential atoms a fit cannot be certified even from finitely many* exact *measurements.* [proof: Dim. DICT-A]

Example 23.6. A certified misfit, an uncertifiable fit For $u = e^{-t}\sin x$ the level $\{\partial_t, \partial_x\}$ shows a misfit: the Gram matrix of (u_t, u_x) is non-singular and $\varphi = \langle \cdot, \mathrm{res}\rangle / \|\mathrm{res}\|$ certifies that no first-order law holds. The level $\{\partial_t, \partial_x^2\}$ fits ($u_t = u_{xx} = -u$), and no finite data can certify that.

Misfit certificates exist in three forms: a Hahn–Banach functional (distance to the span), a Gram quotient in a Hilbert space, and a weak form that needs no derivative of u (only integrals against test functions) (ledger MF1, MF2, MF3).

23.4 Stopping rules and the limit

Theorem 23.7 (Stopping rule and identification in the limit (ledger DICT-GR, I-LIM) [proved])**.** *If* n_0 *is the least level with an affine fit and the kernel at every higher level equals the consequences of that law, then the test stops exactly at* n_0*. With a compact class of coefficients and uniqueness, small residual means close coefficients:* $\|L_n - L\| \le \omega(C_u \varepsilon_n)$*, and with data of error* δ *one gets* $\omega(C_u \varepsilon_n + 2C_{R'}\delta)$*, a Hölder rate in the one-coefficient case.* [proof: Dim. DICT-GR]

23.5 Other results of this phase

- **Exact gain of an atom** (ledger DICT-R3); **change of frame** by the Volterra encoder, with exact intertwining and constants $\|W\|, \|W^{-1}\| \le 1 + a^2 Q\, e^{a\sqrt{Q}}$ (ledger DICT-R6).

- **Exponential families**: exact polynomialisation, kinks and the 2-kink (ledger DICT-NS).
- **Minimax rate** $\Theta(\varepsilon^{1/3})$ for the potential reconstruction under $M > \delta_0$ (ledger DM-δ-MINIMAX).

Remark 23.8 (Why it matters). A dictionary can be universal for approximation, never for exact identification. Its negative side is both topological and measure-theoretic; its positive side is a one-way certificate: misfit is witnessed by finitely many integrals, fit by none.

24 What Phase 10 changed

Starting from: the open problems and conditional statements of all earlier phases, reviewed in Phase 10 by three research sessions (phases 1–4, phase 5, phase 8), each followed by an independent audit and, where needed, a second round. *Why:* an honest edition has to say which earlier statements were refuted, which were closed, and which remain open. Phase 10 refuted five of them and closed many. *Role:* in case of a difference with Sections 4, 3.5 and 6 this section prevails. The proofs of the two corrections that change statements are Dim. A52 and A53; the other items are list cards (Part IV).

24.1 Statements that turned out false

Proposition 24.1 (The interaction no-go with constant terms (ledger S10A-T1, P1-Q6) [proved])**.** *Section 4 left open (P1-Q6) whether* $u_0 \mapsto E_{\mathrm{int}}$ *fails to be continuous in* H^m *for every operator with at least two non-proportional terms. The statement as written is* ***false****: for reaction–diffusion* $u_t = Du_{xx} + ru$ *(two non-proportional terms)* E_{int} *is continuous from* H^1*, with*

$$E_{\mathrm{int}}(u) = (e^{sr} - 1)(v(t) - v_0), \qquad s = t - t_0.$$

The correct form is: for monomial terms of distinct degrees, E_{int} *is unbounded* $H^m \to L^2$ *for every* m *and base time iff there are* at least two terms of degree ≥ 1*; with at most one the interaction is bounded for every* $m \geq 1$*. Constant terms only multiply the solution by a bounded exponential and neither create nor destroy the no-go.* [proof: Dim. S10A-T1]

Proposition 24.2 (Holes: the sum of kernels is closed but not dense (ledger S10A-P1) [proved])**.** *On the annulus* $A = \{1 < |z| < 2\}$ *the Laplacian is a product of the two coprime factors* $\partial_t \pm i\partial_x$*. The harmonic function* $\log|z|$ *is not in the closure of the sum of holomorphic and anti-holomorphic functions in any* H^m*, and for* $m \geq 1$

$$\ker{}_{H^m(A)} \Delta = (\mathcal{H}_1 + \mathcal{H}_2) \oplus \mathbb{C} \log|z| :$$

the sum is closed, of codimension $b_1 = 1$*, and not dense. Hence the density question of Section 3.5 has a negative answer on non-convex regions in general (P2-Q1 is false as a universal statement), and this is a* differential *example of "closed but not dense".* [proof: Dim. S10A-P1]

Remark 24.3 (Intuition). A hole gives a period: the flux of a harmonic function through a loop around the hole. The flux is a continuous functional that vanishes on holomorphic and anti-holomorphic pieces (their derivative has a single-valued primitive), but not on $\log|z|$.

Three more statements were refuted: the extension of the convex density theorem to non-convex regions (S10B-C1, by a hole with an interior); P2-Q1 as a universal statement (for Δ on every Lipschitz region with $b_1 \geq 1$); and, in the variable-coefficient phase, "$P11(e)$ with B merely continuous" and "the pole depends only on $A(\xi_0)$ and the factorisation".

24.2 Problems closed

- **Density of the sum of kernels** on every simply connected Lipschitz region, and for $\delta(P) = 0$ on every Lipschitz region, modulo two cited facts [F1], [F2]; the source of the collar field was replaced by a proof (ledger S10A-T3, S10COL).
- **C12-P7**: "closed but not dense" realised by the annulus.
- **C3-Q1**, the decidability question, closed as a *classification by data model* (composition of black-box impossibility, bound, analytic and jet models); no new mathematics.
- **OPF5C-T5, INT-T1(B), S10B-T2**: the multi-soliton is a Hirota solution; the reflectionless Jost computation holds.
- **Uniform shock rate with several births** (OPA-Q1): modulo the equivalence Oleinik $\Leftrightarrow$ Kružkov [F1].
- **Variable coefficients**: VAR-P11(e) by Volterra and Gronwall; a solution with $V = 0$, $|Z| > 0$ and $\operatorname{int} Z = \emptyset$ (Tychonoff and a Cantor-type set); Floquet in the quadratic class ($\operatorname{tr} M = -26/5$); a monic criterion in the Weyl algebra.
- **Systems**: Maxwell has an identifying field; at fixed real time no finite product of Dirac flows with polynomial times; the pole criterion $E_0 \cap F_0 \neq 0$.
- **Dictionary**: the L^1 rate for general completely monotone memory kernels; the minimax rate $\Theta(\varepsilon^{1/3})$.
- **Computer-assisted**: $N \leq 4$, and, after the triage, $N = 5$ and $\delta_6(12) = 7$ for the syzygy degrees (modulo a novelty hypothesis (Nov)); $\delta_6(11) = 8$ with a written proof.

24.3 Textbook examples and formalisation

Five textbook examples were checked against the whole theory: free Dirac $1+1$, the harmonic oscillator, the Pöschl–Teller potential, the fractional heat equation and Maxwell in vacuum; 80 of 80 checks pass, and three new arguments were proved. The formalisation in Lean 4 with Mathlib reached 37 entries: 27 are faithful to the ledger statement, 7 are weaker and 3 are different, none is vacuous, and all 85 `#print axioms` use only the standard axioms. Mathlib has no Weyl algebra and no C_0-semigroups, so the analytic parts are not formalised.

Remark 24.4 (Why it matters). The corrections of Phase 10 all push in one direction: the negative results of Phases 1–3 are sharper than first stated (constants do not break the no-go; holes do break density), and the closures are explicit about what they depend on.

25 The extended algorithm

Starting from: the results of Sections 19–23 and of the algorithm of the companion paper; a datum in one of the models below. *Why:* the algorithm of the companion paper turned the theorems of Phases 1–5 into a seven-step procedure for a scalar constant-coefficient equation. This section extends it to systems, variable coefficients, noise, and operators outside the catalogue, and states, for every step, what is promised and what is not. *Role:* the operational summary of the whole edition. It introduces *no new claim*: it composes rows of the ledger.

25.1 Which senses of "identify"

The word "identify" has five senses, and every step declares which one it uses.

Sense	Meaning	Unique when
ID-ann	$\mathrm{Ann}(u) = (P)$: the annihilator ideal is principal and generated by the law	always if proved; never for $u \in \mathrm{EP}$ (the point spectrum is non-empty)
ID-ker	$\ker \Phi_u = 0$ in a fixed parametric class	only if a law of the class exists (H-ex)
ID-res	coefficients determined by the residue map, e.g. $V = (iu_t + u_{xx})/u$	on $\{u \neq 0\}$, in the classes of Section 20
ID-fam	identification modulo a family: $V + c$, a gauge class, an encoder modulo $c(t)$	–
ID-lim	identification in the limit, never certified at finite time	under (H-ID)

25.2 Data models and the contract

Model	Datum	Senses allowed
DM1	a closed form that is an exponential-polynomial	only ID-ker relative to a finite class, never ID-ann
M3-hol	a closed-form holonomic non-EP function (Bessel, Airy, sech, Gaussians)	ID-ann, ID-res, ID-ker, with a symbolic proof
DM2	the jet at a point	only ID-lim: no certificate at finite time
$\mathrm{M2}^+$	jet plus a zero test, or a declared holonomic ideal	exact decidability modulo (AC)
DM3	u known exactly in $H^m(\Omega)$ or $C^\infty(\Omega)$	ID-ann, ID-res, ID-ker, ID-fam
DM-δ	$\tilde{u}$ with $\|\tilde{u} - u\|_Y \leq \delta$, norm Y declared	no identification: only the moduli and conditional stability

The contract has four classes: (A) finite and exact, (B) convergent with an explicit error ε_i, (C) no guarantee, (Lim) in the limit without a certificate. The pipeline has **two layers**.

1. *Conditioning* (Steps 1–3) fixes the model, the level and the class. Its outputs are discrete, and since $\mathrm{Ann}(\cdot)$ is discontinuous everywhere (Theorem 21.6) no Lipschitz bound applies to them. Completeness ("the right law is in the chosen level") is always (C).
2. *Conditional correctness* (Steps 4–6) has metric outputs. *At the declared level and in the declared class* the error of the chain is bounded by $\sum_i (\prod_{j>i} L_j)\varepsilon_i$, provided every step is (A) or (B) and Lipschitz in its computed inputs.

25.3 The steps

Step	Task	Result relied upon	Output
1	model	Thms. 5.5, 5.5, 21.1; Prop. 19.14; Thm. 20.9	model, ring of coefficients, senses reachable
2	level of $\mathcal{D}_\infty$ (L0–L10)	Thm. 23.2	level n and finite dictionary, or `NON-DECISO`
3	identification (scalar, system, variable)	Chs. 5, 19, 20	$\mathrm{Ann}(u) \cap R_{\leq n}$, G_u, σ_p, sense
4	misfit certificate or best approximation	Thms. 23.5, 23.7, 21.6	`MISFIT`, `FIT-COMPATIBLE` or stop
5	analytic encoder for V	Thm. 22.2, Cor. 22.4	translated datum and certificates, constant C_W
6	factors, terms and E_{int}	Chs. 3–4, 24	components, ambiguity $\delta(P)$, E_{int}
7	convergence guarantees	Thm. 21.1–21.6	double label and chain bound

Remark 25.1 (The levels L0–L10). L0–L2 scalar constant coefficients; L3 non-monic or degree ≥ 2 in τ; L4 polynomial non-linear; L5 memory and delay (augmented system); L6 constant-coefficient systems (Dirac, Lamé, Maxwell); L7 variable coefficients in x (Schrödinger, Airy); L8 coefficients in (t, x) (Stark, oscillator, Floquet); L9 non-local (fractional, multipliers); L10 general dictionary with misfit and stopping. The choice of level is a *heuristic* (completeness (C)): Theorem 21.3 excludes any a priori criterion.

Remark 25.2 (Step 3 by case). **Scalar** (L0–L5): the tests of Section 5; in DM1 never ID-ann. **Systems** (L6): the existence test (socles ≤ 1, Theorem 19.2), then, for a given u, ID-ann or, in Hamiltonian form, ID-ker with a pointwise linear system of rank 8 (Dirac $1+1$) provided at least four generic modes. **Variable** (L7–L8): the residue map ρ_u (two derivatives and a division per point) or the affine set $\theta_0 + \ker \Phi_u$.

Remark 25.3 (What the pipeline never outputs). It *never* emits "law identified" from finite numerical data. Its outputs are: `MISFIT(certificate)` at level n; `FIT-COMPATIBLE`$(n, r_n,$ `bound)`; `IDENTIFIED[sense, model]`, only with an exact symbolic proof in DM3, M3-hol or M2$^+$ (modulo (AC)), or relative to a finite class in DM1; `ID-LIM(`n`)`, a current candidate without certificate; and `NON-DECISO`, when the budget runs out.

Remark 25.4 (Warning). [title=The encoder is a translation] Step 5 is only usable when q is *given*. If q is estimated at Step 3c the Lipschitz constant of $q \mapsto W_q$ is not explicit and the transport is (C). The only proved bound is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$, which can overestimate by about four orders of magnitude at $a\sqrt{Q} = 5$; a sharper bound (a cosh or an exact Gronwall estimate) is open. A Darboux encoder is unbounded with a non-trivial kernel: it loses modes, and an identifying datum can become exponential-polynomial.

Remark 25.5 (Why it matters). The extended algorithm is two things at once: a procedure and a map of its own limits. What it can promise is correctness *at a declared level and class*; what it can never promise is that the level is the right one. That asymmetry, not a technical gap, is the content of the negative theorems of Phases 1–8.

Table 12: The kinds of claim made in the paper.

Kind	Statements
proved	groups are factors with overlap $\delta(P)$ (Theorem 3.1); terms are not groups (Propositions 3.3 and 4.2); identification criterion and its limits on finite data (Theorem 5.1 and Propositions 5.2 and 5.5); stability via γ_N (Theorem 3.4); Bayesian error bars via the angle (the companion algorithm paper); counting condition
design rule	labels instead of "identified"; distributions of ρ instead of values; decisions on the raw observations; growth of the class instead of coefficient penalties; minimum-norm split of the overlap
measured	0/20 confident wrong answers on random equations not used in development; 60/60 factor sets on grid designs with independent fields and noise; 30/30 on the development benchmark; coefficient coverage 0.94 at nominal 0.90

26 Limits, open problems, and status

Starting from: no new assumptions. *Why:* to state what is impossible, what is open, and how reliable the results are. *Role:* delimits the claims of the paper.

What is impossible (all [proved]).

- finite certificates of identification, and black-box identification (Section 5.2);
- per-term decomposition of the solution (Proposition 3.3);
- stable regularization of E_{int} (Proposition 4.2);
- identification by restricting variable-coefficient operators to one solution;
- decomposition on domains with holes (Figure 4);
- a finite product of single-term flows reproducing non-commuting dynamics.

Open or conditional.

- identification for analytic KdV data with nonzero reflection coefficient (ledger INT-Q1) [open];
- time-uniform shock estimate with several shock births (ledger OPA-Q1) [conditional];
- density of the sum of kernels on nonconvex domains [open];
- exact values $\delta_N(K)$ of an invariant for general N (ledger OPF5C-Q1) [open];
- nonlinear asymptotics with derivatives (ledger S4ID-P8) [conditional];
- closedness of the sum of kernels for $N \geq 3$ differential factors (ledger C12-Q2) [open].

Honest assessment. Many ingredients are classical (Malgrange–Ehrenpreis, Nullstellensatz, Cole–Hopf, inverse scattering). The contribution is the *package*: sharp characterizations, negative theorems, and counterexamples, each with a verified status. Independent reviewers were AI agents. Fifteen theorems of the algebraic core (the E_{int} identity for commuting flows, the comaximal factor decomposition with explicit projector, companion normality for $k = 2$, irreducibility results) are formalized in Lean 4/Mathlib without `sorry`; the analytic parts (distributions, Sobolev spaces, semigroups) are not, since Mathlib lacks the libraries. Several citations were verified only on secondary sources; a result relied upon should be traced to its primary source. Out of scope so far: a variable potential treated as noise (Schrödinger/Dirac), matrix-valued coefficients (4×4 Dirac), and any statistical preprocessing.

26.1 Limits of all eleven phases

Starting from: no new assumptions. *Why:* to state what is impossible, which statements turned out false, what remains open, and how reliable each result is, for the whole project (Phases 0–10). *Role:* delimits every claim of this edition. The analytic part of the project is *closed* by decision of the user (28 October 2026); the unresolved items are collected in the project file `PROBLEMI_APERTI.md`, none of them bearing a central result.

26.1.1 What is impossible

Each item below is a proved negative theorem, not a gap.

- a finite certificate of identification, and identification from a black box (Theorems 5.5 and 5.5); in the limit, no learner decides whether a law exists (Theorem 21.3);
- identification by restricting variable-coefficient operators to one solution (the restriction result of the complete edition); a huge class of coefficients identifies nothing (Theorem 20.1);
- decomposition of a solution by additive terms (Proposition 3.3); factor decomposition on regions with holes (Proposition 3.5);
- components in H^m in general: one derivative can be lost (Counterexample 3.6); density of the sum of kernels on regions with holes (Proposition 24.2);
- a continuous or stably regularised interaction term in H^m *when at least two terms have degree* ≥ 1 (Theorems 4.2, 4.2, 24.1);
- a finite product of single-term flows reproducing non-commuting dynamics when the adjoint orbit is infinite (Theorem 4.10(c)); a uniform finite product for Dirac $1 + 1$ with transport and mass times independent of the frequency (Proposition 19.13);
- an identifying solution when a module has a socle of dimension ≥ 2 (Theorem 19.2); one field identifying (A, V) in the magnetic Schrödinger equation (Proposition 19.11);
- a time-independent encoder for a genuinely time-dependent potential (Theorem 22.6);
- exact identification of every operator by a countable dictionary (Theorem 23.2), and any finite certificate of a fit (Theorem 23.5).

26.1.2 Statements that turned out false

Statement	Counterexample or correction	Ledger
the spectrum of an expansion decides identification on a bounded domain	e^{x-t} on the unit square	C3-C1
components keep the regularity of u	parallel transports on the square	C12-C1
$u \mapsto E_{\mathrm{int}}$ is continuous in H^m	real exponentials in advection–diffusion	P1-P1(e)
the companion matrix is normal iff $p = s^m - c$	$s^2 + s - 1$	R6-T3
splitting by terms lies in $\ker(L_1, L_2)$	$\frac{1}{2}x^2 + Dt$ for the heat equation	R2-T3 (corrected)
exact interaction impossible for infinite Lie algebras	$e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$	R5-Q5, P1-C1
NLS identification by scattering data without multiplicity	explicit counterexample	OPF5D
no continuity for *any* operator with two non-proportional terms	reaction–diffusion $u_t = Du_{xx} + ru$ (Theorem 24.1)	P1-Q6
density of the sum of kernels on every Lipschitz region	the Laplacian on the annulus (Proposition 24.2)	P2-Q1, S10B-C1
$\Phi(V, v) = W_V^{-1}v$ injective $\iff$ a blind encoder exists	the phase-only data model	ENC-N1
an eigenvalue crossing implies a pole of the projector (derogatory A)	$A = \mathrm{diag}(\xi, -\xi)$ has constant projectors	SYS-II-T2-C
every $u \not\equiv 0$ identifies $V(t, x)$	Tychonoff silent strips	VAR-T1(c)-F
the chirp–free–chirp word breaks at the lens time	explicit continuation	VAR-T10-F
Dyson–Phillips with B merely continuous	a rank-2 counterexample	VAR-Q18-CE

26.1.3 Open or conditional

These are not closed and are not declared closed. For each the ledger records what is proved.

With a proved reduction or condition.

- the Lie-flow question for algebras with (H) but without a weighted dilation [open] (ledger P1-Q7) (candidate $(x^3 + x)\partial, \partial$ proved);
- the closedness of the sum of kernels for $N \geq 3$ differential factors, reduced to triples with at least two non-compact pairs, outside real transports [open] (ledger C12-Q2);
- the graded case of the syzygy degrees [open] (ledger OPF5C-Q1); the realisability of atoms [open] (ledger OPR2-Q3); the sufficient condition (L-v) at every vertex [open] (ledger OPR2-Q2′);
- identification for analytic KdV data with nonzero reflection, equivalent to (R), modulo [GT-u$_m$] and (WP) [open] (ledger INT-Q1);
- boundedness of the Hölder constants in the Gelfand–Levitan analysis [open] (ledger [GT-u$_0$]); the justification of a differentiation in a cited analysis [open] (ledger [GT-u$_m$]).

Without a known route. Non-Lipschitz domains (P2-Q1-NL: only a model case is proved); the exact codimension for a general P (P2-Q1-COD); closedness of the sum for $(\tau - \xi^2)(\tau + c\xi)$ (C12-Q1); multiplications and a general Floquet logarithm (VAR-Q18, VAR-

P12(d)); the non-monic case in the Weyl algebra (VAR-C3Q2-open, VAR-Q13); complex-time Dirac with polynomial times (SYS-DIRAC-POL); the DM3 reading of ENC-Q4 and the real residue ENC-Q10; the optimality of the L^1 rate for general kernels (DICT-U3d-v); the C^∞ division reformulation of DICT-Q3(c).

Conditional on sources not read in full. Oberst 1993 (read only as an abstract; it concerns novelty, not a load-bearing step); the equivalence Oleinik ⇔ Kružkov [F1] (a plan exists: vanishing viscosity on Cole–Hopf, no audited proof); Malgrange 1966; the existence of global solutions in $\mathcal{S}$ (WP); Björk. The project forbids downloading files, so these items are *blocked on sources unavailable offline*.

26.1.4 How reliable the results are

Proofs and audits. Every statement labelled [proved] has a complete proof that passed an independent review. The reviewers were AI agents with no access to the author's reasoning, not human referees. Statements labelled [proof written, not audited] have complete proofs that were not audited. For Phases 6–10 the labels follow the ledger: a result is [proved] if the ledger records it as proved after the audits of Gate 8 and of Phase 10 (S-CHECK-F8, S-CHECK-F10). **The proofs in this edition, however, were rewritten from the research packages for this volume (cards A32–A53 and B73–B79) and were not themselves re-audited**; the label refers to the result, not to the rewriting. Several packages cite theorems (Peetre, Malgrange, Oberst, Marchenko) whose exact statements were verified only on abstracts or secondary sources.

Formal verification. Fifteen theorems of the algebraic core of Phases 0–5 and, in Phase 10, 37 entries (27 faithful, 7 weaker, 3 different, none vacuous, 85 `#print axioms` with standard axioms only) are formalised in Lean 4 with Mathlib. The analytic parts are not.

Citations. Many ingredients are classical. The primary sources of the load-bearing citations of Phases 0–5 were checked; for Phases 6–10 a number of citations were checked only on abstracts or secondary sources (Oberst, the transformation-operator theorem numbers, Casas for Magnus). A result relied upon should be traced to its primary source.

Novelty. The contribution is the package: sharp characterisations, negative theorems and counterexamples, each with a verified status. Parts of the identification theory are folklore in commutative algebra (Oberst duality, socle criteria, Peetre's theorem, Volterra transformation operators); parts of the finite-data theory are known in substance (Prony, Mourrain).

26.1.5 What this edition does not contain

- The proofs of every result of Phases 6–10: the catalogue contains proofs of 22 results in Part II and of 25 more in Part IV; the other 86 results appear as list cards with their trust label and the research file the companion catalogue of the catalogue onwards).
- The catalogue of about 60 equation types and the tropical reading of scaling limits.
- The Lean sources.
- Phase 11 as a research phase: this edition *is* its deliverable. The text written for it (the rewritten proofs of Parts II and IV, the corrections added to Sections 3.5 and 4, and the reorganisation of the extended algorithm in Section 25) has not had the proof-checker audit, citation audit and kill-argument review that the plan required before closing.

27 The complete list of open problems

Starting from: the registry `PROBLEMI_APERTI.md` of 28 October 2026, the closing report of Phase 10, the final report of Phases 0–5, the open-question files of every research package, and Section 11 of the extended algorithm. *Why:* Section 26.1 states what is open in a few lines. This section is the full registry: every open or conditional item of the project, with what is already proved about it and what would be needed to close it. Nothing here is a central result. *Role:* a starting point for any later research phase. The status of every item in the ledger is unchanged: a label is raised only after a proof and an audit.

27.1 Why the analytic part can be closed

None of the items below carries a central result of the project (the linear characterisations, the algebraic theorems ALG-*, the dictionary criterion S4ID-T1, the sum-of-kernels theorems C12-T3 and P2-T3, the negative theorems and the counterexamples). Each item is of one of three kinds: (i) an extension to more general classes of a result already proved in a model case; (ii) a new question born during the work; (iii) a dependence on a published source that was not available offline.

The only items from which statements of the report depend are listed with their dependence, and each is already marked "modulo" or "conditional" where it is used:

- **[F1] Lax–Oleinik = Kružkov.** The source is classical and was read (Kružkov 1970, Thm 1, §3); what is missing is an autonomous proof, for which the triage gives a plan (vanishing viscosity on Cole–Hopf). Burgers beyond the shock depends on it (S4T-T3(iii), OPA-T1, OPF5A-T1/T2/C1). This is *not* an open problem.
- **[L-loc]**: the localisation lemma on which OPA-Q1 depends for several shock births; the case of a single shock birth (for example $\sin x$) is closed modulo [F1].
- **[GT-u]**, that is [GT-u_0] and [GT-u_m]: S4ID-P8 and OPR2-P7 (asymptotics with derivatives).
- **[F4$^-$]**: INT-T1(B) (reflectionless N-solitons), verified for $N = 2$.
- **Malgrange 1966**: C3-P1 for general data on convex sets; the jet case is proved without it and the cited theorem is verified.

27.2 Research problems with no known route

Item	Question	Already obtained	What is needed
C12-Q2 = OPF-Q2	closedness of the sum of kernels for $N \geq 3$ differential factors	false in abstract Hilbert space (counterexample); $N = 2$ proved	a differential family with ≥ 2 non-compact interactions
C12-Q1	closedness for $(\tau - \xi^2)(\tau + c\xi)$	—	boundary-layer estimates (route OPR2-L1, heat × transport)

Item	Question	Already obtained	What is needed
OPF5C-Q1	syzygy degrees $\delta_N(K)$ for general N	nine exact values, computer-assisted; $\delta_6(12) = 7$ new; conjectures $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$	a structure for general N; for fixed N it is a finite computation; $\delta_6(11) = 8$ has a written proof, Hirota $N = 5$ is not concluded (a run of about 17 minutes was interrupted)
OPR2-Q3	repeated socle, "no finite test"	proved under (I) and (II)	a lifting theory of atoms
OPR2-Q2′	wedges with degenerate directions	tangent model closed and direct (OPF5B-P8); numerical check 7/7	(L-int): an explicit Hilbert–Schmidt estimate, a few hours of work; (L-v): analysis at the vertex of the cone
INT-Q1	KdV, $r \not\equiv 0$, u analytic: is there an autonomous ODE in x?	OPF5C-T6: order ≤ 3 implies 0 or a 1-soliton	singular relations at the origin and sub-dominant terms of the radiative asymptotics (upstream: [GT-u$_m$])
P1-Q7, general	Lie algebras with (H) but without a weighted dilation (Dil)	the candidate $(x^3 + x)\partial, \partial$ is proved	no route without a dilation
P2-Q1-NL	non-Lipschitz domains	Δ, $m \geq 1$, $L^{1,2} \subset L^2$ proved	general P and $m = 0$
P2-Q1-COD	exact codimension for general P	—	continuous period functionals in H^m
VAR-Q18	multiplications	a rank-2 counterexample with B merely continuous	a counterexample with rough W, v merely continuous, and control of the whole Dyson series
VAR-P12(d)	non-quadratic Floquet logarithms	partial observations (parity, dilations)	an obstruction for non-quadratic logarithms
VAR-C3Q2-open, VAR-Q13	non-monic case in the Weyl algebra	monic case (a′) proved (VAR-C3Q2-R)	a decomposition without monicity
SYS-DIRAC-POL	polynomiality for complex t, m	real t, m proved	the triage sketches a lemma E′ "by growth"; not audited
ENC-Q4	the DM3 reading	—	the class $W(t)$ must be defined first
DICT-Q3(c)	C^∞ division at variable rank	—	a general division criterion
AM-CL	closure AM	no longer needed (ledger F10-TRIAGE)	—
OPA-Q2	density of the Morse+(SB) class	a sketch (Sard with finitely many parameters)	write and audit the sketch

Item	Question	Already obtained	What is needed
non-convex regions	density of OPF5B-T1 outside convex sets	convex case proved; the extension is false on regions with holes (A53); simply connected Lipschitz modulo F1, F2	—

27.3 Items blocked on sources unavailable offline

The project forbids downloading files. Each item below is unblocked by placing the indicated text in the project folder.

Item	Source needed	After reading
[GT-u_0]	Grunert–Teschl §5, Krüger–Teschl	rewriting with explicit constants; a few days of work
[GT-u_m]	Deift–Zhou, CMP 165 (1994) 175–191; Buslaev–Sukhanov 1986	term-by-term derivation; not short
VAR-Q17(b)	Baudouin–Puel; Lasiecka–Triggiani–Zhang; Mercado–Osses–Rosier (Carleman for Schrödinger with $V(t,x)$)	an adaptation
ENC-Q10	Matveev–Salle; Bagrov–Samsonov (pairwise Darboux)	an adaptation, plus a decision on the class $\mathfrak{E}$
DICT-U3d-v	Braess; Braess–Hackbusch	for $t^{-\beta}$ probably the τ_k are not needed; the uniform limit on the class stays a hard problem
Malgrange 1966	*Ideals of differentiable functions*	only to have an autonomous proof: the theorem is already cited
(WP)	Bona–Smith; Kato 1983; Lax 1975	alternatively an autonomous proof of several days
[F4$^-$]	the primary source for reflectionless N-solitons	verified for $N = 2$
Oberst 1993	MCSS 6 (read only as an abstract)	*priority and novelty only*: SYS-EX is proved on an arbitrary Ω without it

27.4 Open questions of each research package

Systems (Phase 8).

- **Q-S8SYS-1** (closed for real times; complex times open): finite words of Dirac flows with times polynomial in ξ.
- **Q-S8SYS-3**: for fixed Ω, does (i$\Leftarrow$) of SYS-S1 hold for every pair iff $\mathrm{Ext}^1(N, C^\infty(\Omega)) = 0$ for every N that occurs, i.e. iff Ω is P-convex for all P ($C^\infty(\Omega)$ injective)?
- **Q-S8SYS-4** (derogatory A): the candidate "the projector has a pole at ξ_0 iff $A(\xi_0)$ has a non-trivial Jordan block mixing eigenvalues of coprime factors"; not proved, high cost; the pole criterion $E_0 \cap F_0 \neq 0$ was proved in Phase 10.

- **Q-S8SYS-5** (novelty of SYS-EX): read Oberst 1990 §2–4, Oberst 1995 (*Variations on the fundamental principle*) and Pillai–Shankar 1998 to decide whether SYS-EX on $\mathbb{R}^n$ is already stated.
- **Q-S8SYS-6** (Maxwell): socle of vacuum Maxwell in (E, B), an over-determined 8×6 system; closed in Phase 10 ($\mathrm{pd} \le 2 \Rightarrow \mathrm{soc} = 0$) in the symmetric class; in the general Hamiltonian class ID-ker is impossible (EX10-e-K).
- **Q-S8SYS-7** (SYS-N2): the determinantal reduction for massless Weyl needs the class group of the light cone (Knörrer/Eisenbud); not touched.

Variable coefficients (Phase 8).

- **VAR-Q10** (closed in Phase 10 by an example). **VAR-Q6′**: the Galois group of $-\partial^2+x^2-E$ over $\mathbb{C}(x)$ (SL_2 for $E \notin \pm(2\mathbb{N}+1)$, the Borel group otherwise); not needed for identification; tool: Kovacic or a direct argument.
- **VAR-Q12**: for L monic of order $m \ge 2$ in ∂_t, an intrinsic characterisation of B-independence (holonomy of $A_2 u$, characteristic dimension); is there a non-D-finite u that does not identify? **VAR-Q13**: for non-constant leading coefficient in every direction, when is $\mathrm{Ann}(u)$ equal to the saturation $A_2 \cap \mathbb{C}(t,x)\langle\partial\rangle L$? A decision procedure needs factorisation of D-modules (Grigoriev–Schwarz).
- **VAR-Q11**: for generic periodic $V(t,x)$, does $U(T,0)$ have a differential (unbounded) logarithm? Known for $V = v(t)$ and quadratic cases.
- **VAR-Q8, Q8′**: convergence of the Magnus series for unbounded operators; when $Y(t;\varepsilon)$ is normal for complex ε, as Casas 2007, Thm 3.4 requires in infinite dimension.
- **VAR-Q14**: whether Saut–Scheurer (1987) or Isakov (1993) contain VAR-L1 for $V \in L^\infty(t,x)$ in 1D (probably VAR-L2 and VAR-T1(c)(ii)); if so, cite them. **VAR-Q15**: read Schindelar–Levandovskyy–Zerz (2011) to confirm the delta of the Weyl-algebra comparison.
- **VAR-Q16**: for continuous matrix V, does "V determined on an open U" imply $\mathrm{span}_t \psi(t,x) = \mathbb{C}^2$ on a dense subset of U? **VAR-Q17**: lateral observability for mild solutions ($\psi_0 \in L^2$, $V \in L^\infty$). **VAR-Q18**: see above. **VAR-Q19**: robustness in DM-δ: construct an estimator of V with guaranteed error, or prove that the Hölder exponent $1/3$ of VAR-P3 is also optimal for the estimation problem (settled with $M > \delta_0$ in Phase 10).

Encoder (Phase 8).

- **Q1**: ENC-U1 in wider classes (kernel $K \in L^1$ for $q \in L^1$, non-triangular kernels). **Q2**: boundedness of W on $H^{2k}(J)$ with explicit constants (the induction with $(Wf)'' = W(f'') - W(qf)$ is only sketched; it is needed for rule R6 with Sobolev norms).
- **Q3** (ENC-S1): transport of the factor decomposition for coprime factors in $S = \mathbb{C}[\tau,\sigma]$ through φ and C12-T1′; to be written out, and to check whether $\mathrm{lcm}(Q, \tilde{Q})$ is the generator.
- **Q4**: a natural class of potentials with $\mathcal{S}_V \cap \mathcal{S}_{V'} \neq \{0\}$ for $V \neq V'$ and a structured encoder for each, so that separation (S) is non-empty. **Q14**: the same modulo the gauge of the data model.
- **Q5**: E^{mult} for Schrödinger on a bounded Ω with Dirichlet conditions (leading term $t^2/2\,[H_0, V]$ presumably, to be proved). **Q6**: ENC-1 in dimension $d \ge 2$: no triangular Volterra operator with local kernel (the characteristic problem is not Goursat).
- **Q7**: does the local Darboux chain extend to all of J across the zeros of the seeds, and is Z discrete (it is only shown closed and countable)? Known for rational potentials (Duistermaat–Grünbaum).

- **Q8**: a quantitative bound for the truncated Foldy–Wouthuysen series with $V \in W^{k,\infty}$ (error $O(m^{-k-1})$?). **Q10**: non-local but explicit encoders for non-quadratic V (Bluman–Shtelen classify only pointwise encoders).
- **Q11**: W commuting with ∂_t with non-constant $c(t)$ and W non-local in t (for example convolutions in t). **Q12**: lateral continuation and (E1) on solutions. **Q13**: regularity of the propagator with V merely norm-continuous (Kato asks for $t \mapsto H(t)$ strongly C^1). **Q15**: ENC-T2(c′) in the class of principal annihilators (proved for the specific family in Phase 10).
- **Q9**: theorem numbers to verify (Marchenko ch. 1, van der Put–Singer, Duistermaat–Grünbaum, Casas).

Convergence (Phase 8).

- **CONV-Q1**: without an upper bound, is $G_u = \gcd \mathrm{Ann}(u)$ identifiable in the limit when $\mathrm{Ann}(u)$ is not principal? (the rule converges to g_d, which can be a proper multiple of G_u). **CONV-Q2**: an explicit jet threshold K^* for classes wider than EP (finite growth order). **CONV-Q3**: a quantitative version of T2 in terms of the complexity of changes of mind.
- **CONV-Q4**: characterise $\mathcal{N}_{\mathcal{K}}$ when $\mathcal{K}$ is the class of augmented systems with finitely many auxiliary variables. **CONV-Q5**: the decomposition $\ker(Q_1 Q_2) = \ker Q_1 + \ker Q_2$ on non-convex domains in $1 + d$ (P-convexity, Hartogs–Ehrenpreis phenomenon). **CONV-Q17**: injectivity of $\mathcal{D}'(\Omega)$ on convex Ω (Malgrange calls it probable; to check in Palamodov 1970).
- **CONV-Q7**: T4 and T5 with stochastic noise, rates in probability. **CONV-Q8**: explicit margin ρ_d for concrete families (heat, waves) as a function of Ω. **CONV-Q14**: T4(b) and T5 for non-integer s (complex or real interpolation; withdrawn from the statement).
- **CONV-Q11**: Mori–Zwanzig with variable coefficients ("closed local law $\Longleftrightarrow$ finite elimination" proved only in the direction $\Leftarrow$ with constant coefficients); look for a Weyl-algebra / D-module version. **CONV-Q12**: the L^∞ order of Strang in the P3-T3 framework (numerically 1, an example, not proved). **CONV-Q13**: higher orders in ε for CONV-T8 and non-modal data in H^1_0 or H^2. **CONV-Q6′**: the norm of the orders 1 (Dirichlet) and 1.5 (Neumann/Robin) in Dang–Einkemmer–Ostermann 2025. **CONV-Q16**: identification of a boundary condition with positive margin through a quotient dictionary for tangential consequences (CONV-B3).

Dictionary (Phase 8).

- **Q1**: a uniform-in-t rate for general completely monotone kernels (the explicit rate is known for $t^{-\beta}$; closed in Phase 10; optimality is open). **Q2**: Hölder for $s \geq 1$ in I-LIM needs a Gagliardo–Nirenberg interpolation $\|f\|_{C^s} \leq C\|f\|_{L^2}^{\theta}\|f\|_{C^{s,1}}^{1-\theta}$, $\theta = 2/(2s+2+N)$, on Lipschitz domains with a constant (the exponent is verified by scaling; the source is missing).
- **Q3**: the intermediate case of RANGO (the pointwise residue map $A_u(y)$ is neither injective on a dense set nor onto everywhere): is identifiability decided by division at variable rank, or are there counterexamples? **Q4**: Theorem A(iii) for non-local atoms (memory, symbols, fractional): the proof uses locality.
- **Q5**: order optimality of polynomials via Kolmogorov n-widths (Pinkus 1985, source not read). **Q6**: the 2-kink has kernel 1 (only checked on examples; a proof needs exact independence of $u_{xt}, u_{xx}, u_{tt}, u$). **Q8**: prior art for N1(v)-ALG, "the generic u is differentially transcendental". **Q9**: R6 for $Y = H^{2k}$ and transport of the weak form MF3 with compactly supported test functions.

- **Q7**: theorem numbers and citations to verify: Hörmander III 18.1.13; Trefethen–Weideman (trapezoid rule); Boman 1967; Christensen 1972 and Hunt–Sauer–Yorke 1992 (prevalence); Ax 1971; Weiss–Tabor–Carnevale 1983; Whitney extension.

27.5 Open questions of the extended algorithm

The twelve questions of Section 11 of the extended algorithm:
1. robustness in DM-$\delta(Y)$ (VAR-Q19);
2. an example at level L9 with infinitely many modes and an identified non-local atom, and DICT-A(iii) for non-local atoms;
3. DICT-ILIM-Hs (closed in Phase 10 with an elementary cone lemma);
4. CONV-Q1: G_u in the limit without an upper bound;
5. VAR-T1(c)-Q10 (closed by an example in Phase 10);
6. C3-Q2 in general;
7. CONV-Q9: unbounded V;
8. the cost of rule R1 per level; the audit of C3-A1 (elimination at polynomial cost, a written proof without a ledger row); a rigorous inclusion for the misfit (A);
9. rule R6 in H^{2k} and with the weak form MF3; a bound on $\|W_q\|$ tighter than C_W (at $a\sqrt{Q} = 5$ the proved $C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2(a\sqrt{Q}) \approx 5.5 \cdot 10^3$);
10. an explicit Lipschitz constant of $q \mapsto W_q$ (ENC-1) for Step 5 with an estimated q;
11. ENC-T2(c′) in the principal class; VAR-P12(d); II-T2(b′) and Maxwell (Q6);
12. a ledger row for Trotter–Kato; the closure of the citations of the next section.

Citations to close. so that the steps "conditional on verification" become fully grounded: van der Put–Singer Prop. 1.34/Thm 1.27 (VAR-T6); DICT-B3; Whitney–Stein (DICT-ILIM, (H2)); Pazy ch. 5 (VAR-P11(e), VAR-L2-cor); Thaller ch. 5 (FW-1 with V); Lax–Oleinik = Kružkov [F1] (S4T-T3); Humphreys §4.1 Cor. B (P1-T4); Wei–Norman 1964, content (VAR-P8(b) "⇐"); Casas 2007 Thm 3.4 (VAR-T10(4)); Oberst 1993, full text; and the Phase 0–5 numbers not checked: Krantz–Parks Thm 2.3.5 (OP-T1), Cohen GTM 138 for (AC), Hörmander SCV 1.4.4, Runge in Rudin.

27.6 Statuses of the claims used by the algorithm

For completeness, the statuses recorded in the table "claims used → status" of the extended algorithm that are *not* fully proved: C3-A1 (written proof, not audited, no ledger row: every output that depends on it is "(A) conditional on C3-A1"); VAR-T6, DICT-B3, DICT-ILIM-Hs (candidate, later closed), VAR-L2-cor (not used), FW-1 with V, VAR-P8(b) "⇐", VAR-P11(e), VAR-T10(2)(4)(5), P1-T4 (Lie from a cited source), S4T-L2 and S4T-T3 (modulo [F1]), ENC-A9(b3) (cited source and example), and the five textbook examples (checked on examples only).

28 Directions for improvement and routes not taken

Starting from: the route registry `ROUTES.md` (decisions at Gates 1, 4a, 6 and 8), the research notes on the learning pipeline (`NOTE_ML`), the open questions of the extended algorithm, and the honest limits of the final report. *Why:* a record of what could make the work better: sharper constants, missing

audits, whole directions that were scouted and parked, and the statistical wrapper that was proposed and postponed. It also records which routes were eliminated and why, so that nobody repeats them. *Role:* companion to Section 27. That section lists open *questions*; this one lists *improvements and directions*.

28.1 Improving what is already there

Sharper encoder constants. The only proved bound on the Volterra encoder is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$; it overestimates by about four orders of magnitude at $a\sqrt{Q} = 5$ ($C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2 \approx 5.5 \cdot 10^3$). A cosh bound or an exact Gronwall estimate would make the transported certificates of Step 5 useful in practice. An *explicit* Lipschitz constant of $q \mapsto W_q$ would allow Step 5 with an estimated potential.

Rigorous misfit. The misfit certificate is class (A) only with a rigorous inclusion of r_n (interval arithmetic or an error bound for the quadrature); with floating-point quadrature it is class (B). The numerically stable computation is the least-squares residual, not the Gram determinant ratio.

Cost model. No general cost estimate is proved for the dictionary levels other than the linear family ($(n+1)(n+2)/2$ monomials in $1+1$); the cost of rule R1 per level, and of the Goursat solve ($O(M^2)$ evaluations of q by iterated quadrature, indicative only), should be proved.

Audits not yet run. The elimination procedure C3-A1 has a written proof, no audit and no ledger row; every output that depends on it is "(A) conditional on C3-A1". Trotter–Kato convergence has no ledger row. The final text of this edition has had no proof-checker audit, no citation audit and no kill-argument review (the Phase 11 plan required all three).

Citations. Close the list of Section 27 (van der Put–Singer, Whitney–Stein, Pazy, Thaller, Humphreys, Wei–Norman, Casas, Oberst) and read the sources that unblock the "blocked" items; read Saut–Scheurer and Isakov to decide whether the unique-continuation lemma is already known; read Schindelar–Levandovskyy–Zerz in full to confirm the stated delta.

Rates under stronger hypotheses. Explicit jet thresholds beyond exponential-polynomials; explicit margins ρ_d for the heat and wave equations as functions of the domain; stochastic noise with rates in probability; non-integer smoothness in the Hölder estimates.

Dimension. The Volterra encoder is one-dimensional: in dimension $d \geq 2$ the characteristic problem is not of Goursat type. A genuine multidimensional analogue, or a proof that none exists with a local kernel, is open.

A first-principles proof of the Burgers closure. A written proof of the equivalence Oleinik $\Leftrightarrow$ Kružkov by vanishing viscosity on Cole–Hopf would remove the "modulo [F1]" from the shock results; the triage gives the plan.

28.2 Routes scouted and parked

During the project each candidate approach (a *route*) was scouted, then promoted, merged, parked or eliminated with a stated reason. The parked routes are the natural next directions.

Route	Content / why parked	Reopen when
RT-5 (symbol tensor)	the tensor keeps only the principal symbol (R6-P1); useful as a geometric invariant; reopened for systems (RT-SYS) and parked again	a use as an invariant "at infinity"
RT-R1b	needs a sign hypothesis on the datum	such a hypothesis is acceptable
Koopman / DMD	"dynamic mode decomposition is regression", excluded by the no-learning constraint; reopen with the normal form: eigenfunctions carry the nonlinear term	the no-ML rule is lifted
Painlevé, NLSPEC, HAANTJES	triggers in the verdicts of the nonlinear scouting; Painlevé kept as a *tool* (DICT-R4)	a nonlinear exact family needs it
RT-NA (non-autonomous propagators)	parked after Phase 8	triggers VAR-Q8, Q11, Q18
RT-ENC (the encoder as a route)	parked; lemmas moved to the dictionary and variable-coefficient routes	triggers ENC-Q4, ENC-Q10
RIGID, STOKES, CONV-BORDO	not touched in Phase 8	boundary identification with margin (CONV-Q16)
Weyl algebra for PDE	scouted only for ODE and A_2	a D-module factorisation tool is available
Inverse Sturm–Liouville (spectral data)	parked	spectral data become available as a datum
U3(a) kernels, U5	universality classes not covered	a use case
Homogenisation / gamblets	approximate, a tool (S) per scale, not an encoder	oscillating or rough V treated as noise
Wave operators $\Omega^*_\pm$	outside the bounded-domain model	scattering data are the datum
Hartogs $1+d$	P-convexity on non-convex domains in several variables (CONV-Q5)	non-convex $1+d$ decomposition needed
MON, NAT, MOB	minor routes, remain parked	—
Mori–Zwanzig, RT-R2a, POLE, gamblets as *methods*	eliminated: the criteria are false or only cited; the lemmas remain (Section 21)	—

Eliminated for good. Statistical consistency as a method (excluded by the no-learning rule), SPLIT-f, free probability, ICA, quiver representations, microlocal analysis beyond the characteristic variety, and Lie/Olver symmetry methods were eliminated or merged. RT-S1g was eliminated for the core (it gives no information for elliptic or parabolic operators) and kept for the hyperbolic part.

28.3 The statistical wrapper that was proposed and postponed

At the end of Phase 5 a learning pipeline was drafted around the exact analytic core (the note *Reading the structure of a differential equation from data*). It was *excluded* from the project

by decision of the user (no machine learning of any kind, including the autoencoder/mixture idea, "Idea I", postponed to a later phase). It is recorded here as a direction.

1. **Interpolate.** A learning algorithm fits a smooth function to noisy data (t_i, x_i, y_i) and returns not one function but a weighted family $(\hat{u}_s, \pi_s)$, with $\pi_s \geq 0$, $\sum \pi_s = 1$.
2. **Analytic core.** Each plausible function is passed to the exact algorithm, which returns its factors and the component each carries.
3. **Aggregate.** Weighted by the probabilities, this gives a distribution over factors (for example "transport 0.7 ± 0.1, diffusion 0.3 ± 0.1").
4. **Group.** The distribution sets the parameters of a standard clustering (how many groups, how spread out), cutting the data into groups that ideally contain one factor each, with a small "tangle number" (the Friedrichs cosine between factors; the size of E_{int} between terms).
5. **Repeat inside each group** until all groups are clean; the factors then rebuild the equation. The dashed loop of the pipeline: each group becomes a new data set.

The exact parts are the analytic core and the two tangle measures; the interpolation, the aggregation and the clustering were left as a proposal. The note asked two questions: whether the basis functions of the first step must be chosen in advance (and therefore only discover equations whose building blocks were anticipated), and how to avoid that choice. The answer built in Phase 8 is the filtered dictionary with a misfit certificate and a stopping rule (Section 23); the statistical aggregation itself remains open.

28.4 What was never done

- the catalogue of about 60 equation types (ODE, transport, diffusion, hyperbolic, integral and memory, beams and elasticity, quantum, non-linear) used as scouting material, and the tropical reading of scaling limits (only the dissipative-face estimate TROP-3′ and its corollary for advection–diffusion are proved);
- Lean sources and a Lean statement for the analytic parts (the Mathlib libraries are missing: no Weyl algebra, no C_0-semigroups);
- an independent human review: all reviewers were AI agents;
- the decision on global installation of the project skills and on Mathpix (Step 0 of the original plan, left to the user);
- the Phase 11 audits on the final text of this edition.

28.5 Honest limits of the whole project

The overall novelty is modest: many building blocks are classical (Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering, Peetre, Volterra transformation operators, Oberst duality). The contribution is the package of characterisations, the negative theorems and the counterexamples, with verified statuses. The source policy (decision of the user, 26 October 2026) allowed results to rest on published theorems provided the source is verified: of 112 sources, 34 had the theorem number checked, 75 had the content checked (45 only in the metadata), none was not found, and three wrong citations were corrected (Humphreys Cor. B, Gel'fand–Dikii 1976, DLMF 9.7.10). Several load-bearing numbers were still unchecked at the end (Section 27). Work interrupted by API usage limits was resumed from the files with no loss recorded.

A Proofs

Throughout, $\Omega \subset \mathbb{R}^2$ is open and connected, $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acts on $\mathcal{A}$ by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. For $z = (\tau_z, \xi_z) \in \mathbb{C}^2$, $\mathfrak{m}_z = (\tau - \tau_z, \xi - \xi_z)$ is its maximal ideal and $\psi_z(t, x) = e^{\tau_z t + \xi_z x}$.

A.1 Algebraic toolkit

Proof of Lemma 2.4. Coprime polynomials meet in finitely many points. Let $f, g \neq 0$ be coprime. By Gauss's lemma they are coprime in the principal ideal domain $\mathbb{C}(\xi)[\tau]$, so $af + bg = 1$ there; clearing denominators, $a'f + b'g = r(\xi) \neq 0$ with $a', b' \in \mathcal{R}$. At a common zero $r(\xi_0) = 0$, which leaves finitely many ξ_0; exchanging the variables, finitely many τ_0. Hence $V(f, g)$ is finite.

Splitting. Put $J = \{f/g : \ f \in I\}$ with $g = \gcd(I)$; it is an ideal and $I = gJ$, and no prime divides all of J. If $J \neq \mathcal{R}$, pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_r$ and, for each a, $g_a \in J$ with $\pi_a \nmid g_a$. The sets $\{c \in \mathbb{C}^r : \pi_a \mid \sum_b c_b g_b\}$ are proper linear subspaces, so some c avoids all of them; then f and $\sum_b c_b g_b \in J$ are coprime and $V(J) \subseteq V(f, \sum_b c_b g_b)$ is finite. By the Nullstellensatz a power of $\sqrt{J}$ lies in J, and $J = \mathcal{R}$ when $V(J) = \emptyset$. □

Proof of Lemma 2.7. ($\Rightarrow$) $Q(\partial)(q\,\psi_z) = (Q(z + \partial)q)\psi_z$ for every polynomial q, so the derivatives of $q\,\psi_z$ stay in the finite-dimensional space of polynomials of degree $\leq \deg q$ times ψ_z. Moreover $\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w)$, so $V(\operatorname{Ann} w) \subseteq \{z_k\}$; conversely $Q \in \operatorname{Ann}(w)$ forces $Q(z_k + \partial)q_k = 0$ by the linear independence of distinct exponentials, whose top-degree part is $Q(z_k)$ times that of q_k, so $Q(z_k) = 0$.

($\Leftarrow$) Let $\mathcal{R}w$ be finite-dimensional and p, q the minimal polynomials of ∂_t and ∂_x restricted to $\mathcal{R}w$, so that $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On an open rectangle $I' \times J' \subset \Omega$, induction on $\deg p$ shows that every distribution with $p(\partial_t)v = 0$ is $\sum_{j,l} c_{jl}(x)\, t^l e^{\lambda_j t}$ with $c_{jl} \in \mathcal{D}'(J')$ and λ_j the roots of p (the base case uses that a distribution with $\partial_t V = 0$ on a rectangle is $c(x) \otimes 1$). Applying $q(\partial_x)$ and testing against functions dual to $t^l e^{\lambda_j t}$ gives $q(\partial_x)c_{jl} = 0$, so each c_{jl} is an exponential-polynomial of x. Thus w is an exponential-polynomial E near every point. The set where w coincides near y with a fixed E is open, and it is closed because two exponential-polynomials that agree on an open set agree everywhere; by connectedness it is Ω. □

A.2 Identification

Proof of Theorem 5.1. Let $I = \operatorname{Ann}(u) \ni P$ and write $I = G\,J$ as in Lemma 2.4. Since $P \in I$, $G \mid P$, so $G = c' \prod_i P_i^{f_i}$ with $0 \leq f_i \leq e_i$. First, $G \sim P$ iff $I = (P)$: if $G \sim P$ then $I = PJ \subseteq (P) \subseteq I$; if $I = (P)$, P divides all of I and lies in I.

(a), $\Leftarrow$ by contraposition. If $I \neq (P)$, some $f_i < e_i$, so $P/P_i = G\,H$ with $H \in \mathcal{R}$. A power $\mathfrak{n}^\ell$ of the ideal $\mathfrak{n}$ of the finite set $V(J)$ lies in J, and for $h \in \mathfrak{n}^\ell$, $h \cdot P/P_i = G(Hh) \in GJ = I$, i.e. $h(\partial)v_i = 0$. So $V(\operatorname{Ann} v_i) \subseteq V(J)$ is finite and $v_i \in \mathrm{EP}$ by Lemma 2.7.

(a), $\Rightarrow$ by contraposition. If $v_i \in \mathrm{EP}$, $V(\operatorname{Ann} v_i)$ is finite while $V(P_i)$ is infinite, so some $h \in \operatorname{Ann}(v_i)$ is not divisible by P_i. Then $Q = h \cdot P/P_i$ satisfies $Q(\partial)u = h(\partial)v_i = 0$; if $Q = PA$, cancelling P/P_i gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

(b) $V(I) = V(G) \cup V(J)$ with $V(J)$ finite. If $f_i \geq 1$, $V(P_i) \subseteq V(I)$; if $f_i = 0$, $V(P_i) \cap V(I)$ lies in the finite set $\bigcup_{f_j \geq 1} V(P_i, P_j) \cup V(J)$, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$, and $f_i = 0$ iff $v_i' \in \mathrm{EP}$ by the argument of (a) with $P/P_i^{e_i}$ in place of P/P_i.

(c) If $e_i = 1$ then $v_i = v_i'$. If P is monic of degree m in τ, divide any $Q \in \mathcal{R}$ by P in $\mathbb{C}[\xi][\tau]$: $Q = AP + \sum_{j<m} \varphi_j(\xi)\tau^j$ uniquely. Then $Q \in I$ iff $\sum_j \varphi_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $\varphi_j = 0$. □

Proof of Proposition 5.2. (a) If $I = (P)$, every annihilator is SP and $\deg SP \geq \deg P$ with equality iff S is a nonzero constant. (b) If $I = (P)$ then $G \sim P$ and $J = \mathcal{R}$, so $\Sigma_{\mathrm{p}}(u) = \emptyset$. Conversely, $V(J) = \emptyset$ gives $J = \mathcal{R}$ by the Nullstellensatz, hence $I = (G) = (P)$. (c) Let f_i be the multiplicity of P_i in G and $w = (P/P_i^k)(\partial)u$. If $k \leq e_i - f_i$, then $G \mid P/P_i^k$, say $P/P_i^k = GH$; for h in a power $\mathfrak{n}^\ell \subseteq J$ of the ideal of $V(J)$, $h \cdot P/P_i^k = G(Hh) \in I$, so $\mathrm{Ann}(w) \supseteq \mathfrak{n}^\ell$ and $w \in \mathrm{EP}$. If $k > e_i - f_i$ and $w \in \mathrm{EP}$, pick $h \in \mathrm{Ann}(w)$ with $P_i \nmid h$ (as in the proof of Theorem 5.1); then $hP/P_i^k \in I$, so $G \mid hP/P_i^k$, but P_i occurs in hP/P_i^k with multiplicity $e_i - k < f_i$, a contradiction. Hence $f_i = e_i - \max\{k : \ w \in \mathrm{EP}\}$. □

A.3 Decomposition and stability

Proof of Theorem 3.1. We prove the statement for open connected Ω with $H^1_{dR}(\Omega) = 0$, which holds for simply connected domains.

Step 1: finite modules have no obstruction. Let M be an $\mathcal{R}$-module of finite dimension over $\mathbb{C}$. Then $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0$ and $\dim \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A}) = \dim M$. By the long exact sequence it is enough to check one-dimensional modules $\mathcal{R}/\mathfrak{m}_z$, since M has a composition series with such quotients. A free resolution of $\mathcal{R}/\mathfrak{m}_z$ is the Koszul complex of $(\tau - \tau_z, \xi - \xi_z)$, so $\mathrm{Ext}^\bullet(\mathcal{R}/\mathfrak{m}_z, \mathcal{A})$ is the cohomology of $\mathcal{A} \to \mathcal{A}^2 \to \mathcal{A}$, $v \mapsto \big((\partial_t - \tau_z)v, (\partial_x - \xi_z)v\big)$, $(a, b) \mapsto (\partial_t - \tau_z)b - (\partial_x - \xi_z)a$. Multiplication by ψ_{-z} turns it into the de Rham complex of Ω with coefficients in $\mathcal{A}$, whose first cohomology is $H^1_{dR}(\Omega) = 0$ (for distributions by de Rham's theorem), and whose zeroth cohomology is the constants, of dimension one.

Step 2: two factors. For coprime Q_1, Q_2, $(Q_1) \cap (Q_2) = (Q_1 Q_2)$ by unique factorization, and $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ is exact ($1 \mapsto (1, 1)$, $(a, b) \mapsto a - b$). Applying $\mathrm{Hom}_{\mathcal{R}}(\cdot, \mathcal{A})$, and using $\mathrm{Hom}(\mathcal{R}/(f), \mathcal{A}) = \ker_{\mathcal{A}} f$, gives the exact sequence $0 \to \mathrm{Hom}(\mathcal{R}/(Q_1, Q_2), \mathcal{A}) \to \ker Q_1 \oplus \ker Q_2 \xrightarrow{(u_1,u_2)\mapsto u_1+u_2} \ker Q_1 Q_2 \to \mathrm{Ext}^1(\mathcal{R}/(Q_1, Q_2), \mathcal{A})$. The quotient $\mathcal{R}/(Q_1, Q_2)$ is finite-dimensional by Lemma 2.4, so by Step 1 the last group vanishes: the sum map is onto, and its kernel has dimension $\dim \mathcal{R}/(Q_1, Q_2)$ and consists of solutions of a zero-dimensional ideal, i.e. of exponential-polynomials.

Step 3: induction. $P' = Q_2 \cdots Q_N$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Dimensions of kernels add along the composition of the two sum maps, and $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$ because $\mathcal{R}/(Q_1, Q_2Q_3 \cdots)$ is the direct sum of its localizations at the finitely many common zeros, where the intersection multiplicities add. This gives (7). □

Proof of Theorem 3.4. The sum operator $\mathcal{S}$ is bounded from $\bigoplus_j M_j$ into $\ker_{H^s} P$, and every u in its range has components in H^s. Its range is all of $\ker_{H^s} P$ iff the range is closed and dense. By the closed range theorem the range is closed iff $\mathcal{S}$ is bounded below on $(\ker \mathcal{S})^\perp$, i.e. iff $\gamma_N > 0$, and then the minimum-norm preimage satisfies (10). For two closed subspaces, $M_1 + M_2$ is closed iff $c_F < 1$, and $\gamma_2 = 1 - c_F$ [3]. In finite dimension every subspace sum is closed, and the sum equals the span of the sampled exponentials of P, each of which lies in the kernel of one factor. □

Proof of Proposition 4.2. For real $\zeta > c/\nu$ let $u_\zeta = e^{\lambda_\zeta t + \zeta x}$, $\lambda_\zeta = \nu\zeta^2 - c\zeta > 0$; then $Q(\partial)u_\zeta = 0$ and $\|u_\zeta\|_{H^s(\Omega)} \le C_s(1+\zeta)^{2s} e^{\lambda_\zeta T}\|e^{\zeta x}\|_{L^2(a,b)}$. With base time t_0 and horizon t, Theorem 4.1 gives

$$E_{\mathrm{int}} = e^{\lambda_\zeta t_0}\bigl(e^{-c\zeta t} - 1\bigr)\bigl(e^{\nu\zeta^2 t} - 1\bigr)\, e^{\zeta x},$$

whose $L^2(\Omega)$ norm is at least a constant times $e^{\nu\zeta^2 t}\|e^{\zeta x}\|_{L^2(a,b)}$ for large ζ. With $t_0 = 0$ and $t = T$ the ratio therefore grows at least like

$$e^{\nu\zeta^2 T - \lambda_\zeta T}(1+\zeta)^{-2s} = e^{c\zeta T}(1+\zeta)^{-2s},$$

which is unbounded for every s. For the second statement, $v + w$ is killed by Q and by $R = (\tau + c\xi)(\tau - \nu\xi^2)$. At a common zero of Q and R, either $\tau = -c\xi$, and then $Q = 0$ gives $\nu\xi^2 = 0$, or $\tau = \nu\xi^2$, and then $c\xi = 0$; in both cases $(\tau, \xi) = (0, 0)$. So (Q, R) has a single zero, and its common kernel is a finite-dimensional space of exponential-polynomials (Lemma 2.7). □

B Glossary of classical tools

Each entry states a classical tool in one sentence and points to where this edition uses it.

Name	Statement in one sentence	Used in
Unique factorisation	Polynomials in two variables factor uniquely into irreducibles, up to constants.	Ch. 2
Bézout's theorem	Two curves without a common component meet in at most the product of their degrees points [8].	Ch. 2, 5.2
Hilbert's Nullstellensatz	A polynomial vanishing on the zero set of an ideal has a power in the ideal.	Ch. 5
Chinese remainder theorem	Coprime moduli split a quotient ring into a product.	Ch. 3
Malgrange–Ehrenpreis	Every nonzero constant-coefficient operator has a fundamental solution, and is onto on smooth functions on a convex set [5, 14].	Ch. 3, 5.2
Oberst duality	Solution spaces of constant-coefficient systems correspond to modules over the operator ring [17].	Ch. 5
Poincaré lemma	On a simply connected region every closed 1-form is exact.	Ch. 3
Friedrichs angle	$M + N$ is closed iff the cosine of the angle between M and N, after removing $M \cap N$, is smaller than 1.	Ch. 3.5
Miyadera–Voigt	Small relatively bounded perturbations of a generator generate a semigroup [7].	Ch. 4
Baker–Campbell–Hausdorff	$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$.	Ch. 4
Wei–Norman	Linear time-dependent systems on a solvable Lie algebra reduce to ODEs for scalar coefficients [19].	Ch. 4
Tychonoff	The heat equation has nonzero smooth solutions with zero initial datum [12].	Ch. 4
Cole–Hopf	$u = -2\nu\varphi_x/\varphi$ maps Burgers to the heat equation [9].	Ch. 6

Name	Statement in one sentence	Used in
Kružkov	Entropy solutions of scalar conservation laws are unique and L^1-stable [13].	Ch. 6
Inverse scattering	KdV and NLS are linearised by a nonlinear Fourier transform with trivial time evolution.	Ch. 6
Novikov	The stationary KdV hierarchy characterises finite-gap and soliton potentials [16].	Ch. 6

C Bibliography of the project

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D Explained: step-by-step walk-throughs

The following walk-throughs explain, in plain language and step by step, the notions used in the main text (rings, the operator ring, the annihilator, exponential-polynomials, the gcd, factors, the interaction term, the nonlinear case, the algorithm). They come from the explained edition of the first manuscript, where each was a box placed at the point named in its first line. They are self-contained and can be read in any order.

D.1 Rings, the operator ring, symbol and characteristic variety

(E1: after Figure 1 (Section 2.1)) **Step 0: what is a ring?** A *ring* is the analogue of a group in which there are two operations, addition and multiplication. A set R with $+$ and $\cdot$ is a ring if:

1. $(R,+)$ is an abelian group: addition is associative and commutative, there is an additive identity 0, and every element a has an additive inverse $-a$;
2. multiplication is associative, $(ab)c = a(bc)$;
3. multiplication distributes over addition, $a(b+c) = ab + ac$ and $(a+b)c = ac + bc$.

Unlike in a group, a ring does *not* require every nonzero element to have a multiplicative inverse, and multiplication need not be commutative. The hierarchy is:

- *group*: one operation;
- *ring*: an additive abelian group together with a multiplication compatible with the addition;
- *field*: a commutative ring with $1 \neq 0$ in which every nonzero element has a multiplicative inverse.

Examples: $(\mathbb{Z},+)$ is a group; $(\mathbb{Z},+,\cdot)$ is a ring but not a field (2 has no inverse in $\mathbb{Z}$); $(\mathbb{Q},+,\cdot)$ is a field. A ring is therefore *not* simply a group under multiplication: it is an additive abelian group on which one can also multiply, with multiplication obeying the distributive law.

Step 1: what is $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x]$? It is the ring of polynomials in the two formal variables ∂_t, ∂_x with complex coefficients. Elements of $\mathcal{R}$ are, for instance,

$$P(\partial) = \partial_t^2 + \partial_x^2, \qquad Q(\partial) = 3\partial_t\partial_x + 2, \qquad S(\partial) = i\partial_t^3 - 5\partial_x.$$

Why is it a ring? One can add operators ($P+Q$), multiply them by composing them (PQ), there is a zero operator and every operator has an additive inverse, and composition is associative

and distributes over addition. The crucial point is that these are *constant-coefficient* operators: the coefficients are complex numbers, not functions of t or x. So $2\partial_t^2 + 3\partial_x + 1$ is in $\mathcal{R}$, whereas $t\,\partial_t$ is not (its coefficient t is not constant).

Step 2: why $\mathcal{R} \cong \mathbb{C}[\tau, \xi]$? The symbol $\cong$ means *isomorphic*: the two rings have exactly the same algebraic structure. Here τ, ξ are ordinary formal polynomial variables and we simply replace

$$\partial_t \longleftrightarrow \tau, \qquad \partial_x \longleftrightarrow \xi, \qquad \text{e.g.} \qquad \underbrace{\partial_t^2 + 3\partial_t\partial_x + 2\partial_x^2}_{\text{differential operator}} \longleftrightarrow \underbrace{\tau^2 + 3\tau\xi + 2\xi^2}_{\text{polynomial}}.$$

This is legitimate because constant-coefficient partial derivatives commute, $\partial_t\partial_x = \partial_x\partial_t$, so the operator ring is commutative, exactly like the polynomial ring. At this stage τ and ξ are *not* numbers: they are formal variables that encode the operators.

Step 3: the symbol. For $P(\partial) = \partial_t^2 - \partial_x^2 + 1$ the symbol is obtained by replacing each derivative with its variable,

$$P(\tau, \xi) = \tau^2 - \xi^2 + 1.$$

The symbol translates a differential operator into an algebraic object, and polynomial algebra is often easier to study than differential operators directly. *Convention.* Other texts replace ∂ by $i\tau, i\xi$ (Fourier convention). In this paper the substitution is the *direct* one, $\partial \mapsto (\tau, \xi)$; the Fourier convention $\partial_x \mapsto i\kappa$ appears only later, in Sections 3 and 4, and is stated there.

Step 4: the characteristic variety. Given the symbol, $V(P) = \{(\tau, \xi) \in \mathbb{C}^2 : P(\tau, \xi) = 0\}$: the set of all pairs of complex numbers at which the symbol vanishes. For $P = \tau^2 - \xi^2 + 1$ it is $\tau^2 = \xi^2 - 1$, a subset of $\mathbb{C}^2$; a pair belongs to $V(P)$ exactly when it satisfies this equation. For $P = \tau - \xi$ it is the line $\tau = \xi$ of $\mathbb{C}^2$.

Example: the wave operator. $P(\partial) = \partial_t^2 - c^2\partial_x^2$ has symbol $\tau^2 - c^2\xi^2 = (\tau - c\xi)(\tau + c\xi)$, so

$$V(P) = \{\tau = c\xi\} \cup \{\tau = -c\xi\},$$

two lines (Figure 10, left). They encode the characteristic directions of wave propagation; their precise interpretation depends on the PDE and on whether the covectors are taken real or complex.

The whole passage in one picture.

$$\underbrace{P(\partial_t, \partial_x)}_{\text{operator on functions}} \xrightarrow{\partial \mapsto (\tau,\xi)} \underbrace{P(\tau, \xi)}_{\text{polynomial}} \xrightarrow{\text{zeros}} \underbrace{V(P) \subset \mathbb{C}^2}_{\text{set of points}}$$

Three related but *different* objects: an operator acting on functions, a polynomial, and a set of points of complex space.

D.2 Annihilators, ideals, (F1)–(F3) and the splitting of the gcd

(E2: after Lemma 2.5 and the paragraph on G_u, J_u (Section 2.2)) **Step 1: annihilating a function.** Take $u(x) = e^x$. Since $\partial_x e^x = e^x$, $(\partial_x - 1)e^x = 0$: the operator $\partial_x - 1$ *annihilates* e^x. It is not the only one: $(\partial_x^2 - 1)e^x = 0$, and $\partial_x^2 - 1 = (\partial_x - 1)(\partial_x + 1)$ is annihilating precisely because it contains the factor $\partial_x - 1$. More generally, if $Q(\partial_x)e^x = 0$ then every polynomial

multiple of Q annihilates e^x, for instance $(\partial_x + 3)(\partial_x - 1)e^x = 0$. This is the basic reason annihilators are ideals.

Step 2: the annihilator is the whole ideal $(\partial_x - 1)$. For $R = \mathbb{C}[\partial_x]$, any polynomial can be divided by $z - 1$: $Q(z) = (z-1)S(z) + Q(1)$. Replacing z by ∂_x and applying to e^x gives $Q(\partial_x)e^x = Q(1)e^x$, hence

$$Q(\partial_x)e^x = 0 \iff Q(1) = 0 \iff (z-1) \mid Q(z).$$

So $\mathrm{Ann}(e^x) = (\partial_x - 1) = \{S(\partial_x)(\partial_x - 1) : S \in \mathbb{C}[\partial_x]\}$, and every annihilating polynomial is a multiple of $\partial_x - 1$.

Step 3: why an ideal. An ideal is a subset closed under addition (and additive inverses) and under multiplication by any element of the ring. For annihilators: if $Qu = 0$ and $Pu = 0$ then $(Q + P)u = Qu + Pu = 0$; if $Qu = 0$ and $S \in \mathcal{R}$ then $(SQ)u = S(Qu) = S(0) = 0$. Intuition: once an operator kills a function, composing it with another constant-coefficient operator cannot bring the function back to life.

Step 4: one variable versus two variables. In $\mathbb{C}[\xi]$ every ideal is generated by a single polynomial, $I = (g)$ – like the multiples of 6 in $\mathbb{Z}$. In $\mathbb{C}[\tau, \xi]$ several generators may be needed. Consider $I = (\tau, \xi - 1) = \{\tau A + (\xi - 1)B\}$: neither generator is a multiple of the other and the ideal is not generated by one polynomial. This matters because the annihilator of a function of two variables can have several independent generators: for $u(t, x) = e^x$ we have $\partial_t u = 0$ and $(\partial_x - 1)u = 0$, and

$$\mathrm{Ann}(u) = (\partial_t,\ \partial_x - 1).$$

Both equations are needed to describe it in the two-variable ring.

Step 5: the three facts.

- **(F1) Unique factorization.** Every nonzero polynomial factors into irreducibles, uniquely up to constants and order; e.g. $\tau^2 - \xi^2 = (\tau - \xi)(\tau + \xi)$. This allows one to define the gcd of an ideal: the greatest common polynomial factor of all its elements.
- **(F2) Bézout.** Two polynomials in (τ, ξ) with no common nonconstant factor have finitely many common zeros in $\mathbb{C}^2$. Example: τ and $\xi - 1$ have no common factor and their only common zero is $(0, 1)$. Geometrically, two algebraic curves without a shared component meet in finitely many points.
- **(F3) Nullstellensatz.** If q vanishes on $V(I)$ then $q^k \in I$ for some $k \geq 1$: $q|_{V(I)} = 0 \Rightarrow q^k \in I$. It connects the geometry of a zero set to the algebra of the ideal. The power matters: q itself need not belong to I (e.g. $I = (\tau^2)$, $q = \tau$).

Step 6: the splitting lemma. Every nonzero ideal $I \subset \mathcal{R}$ can be written $I = gJ$, with $g = \gcd(I)$ and $V(J)$ finite. Here g is the polynomial factor shared by every element of I; J is the ideal of the quotients f/g; because g is the gcd, the elements of J have no common nonconstant factor, and by (F2) their common zero set is finite. So the ideal splits into a common polynomial factor and a residual ideal with finite zero set.

- *Example A (shared factor).* $I = (\tau(\xi - 1), \tau(\xi + 1))$. Both generators contain τ, so $g = \tau$ and $J = (\xi - 1, \xi + 1)$. Since $(\xi + 1) - (\xi - 1) = 2$, $J = (1) = \mathcal{R}$, whose zero set is empty (hence finite). Thus $I = (\tau)$.
- *Example B (no shared factor).* $I = (\tau, \xi - 1)$: $g = 1$, $J = I$, $V(J) = \{(0, 1)\}$, finite.

Step 7: applied to u. Define $G_u = \gcd \mathrm{Ann}(u)$ and call the remaining ideal J_u, so $\mathrm{Ann}(u) = G_u J_u$. For $u = e^x$ in $\mathbb{C}[\partial_t, \partial_x]$ we found $\mathrm{Ann}(e^x) = (\partial_t, \partial_x - 1)$; the generators have no common nonconstant factor, so

$$G_u = 1, \qquad J_u = (\partial_t, \partial_x - 1), \qquad V(J_u) = \{(0,1)\},$$

the single point corresponding to the exponential $e^{0\cdot t+1\cdot x} = e^x$. A point of the characteristic picture corresponds to a possible exponential mode. *Qualification:* the reading of G_u as the "genuinely infinite-dimensional part" and of J_u as the "finitely many exponentials present" is an intuition for the later theory, not a universal identification for arbitrary functions without further hypotheses.

D.3 The finite dimension criterion for exponential-polynomials

(E3: after Lemma 2.7 (EP), before the Takeaway of Section 2.3) The lemma says: *a function is a finite sum of polynomial-times-exponential terms if and only if all its derivatives span a finite-dimensional vector space.*

Step 1: what is EP**?** The set of functions $w(t,x) = \sum_{j=1}^{N} p_j(t,x) e^{\tau_j t + \xi_j x}$, with p_j polynomials and $\tau_j, \xi_j \in \mathbb{C}$. Examples: $e^{2t+x} + 3e^{-t+4x}$ and $t^2 e^x + (t+x)e^{t-x}$. The second has polynomial prefactors, so EP is more general than sums of pure exponentials. Polynomial prefactors appear because differentiating such a function gives another function of the same form: $\partial_x(te^{2t+x}) = te^{2t+x}$ and $\partial_t(te^{2t+x}) = (1+2t)e^{2t+x}$. The polynomial factor changes, the exponent stays the same.

Step 2: what is $\mathcal{R}w$? $\mathcal{R}w = \mathrm{span}_{\mathbb{C}}\{\partial_t^j \partial_x^k w : j,k \geq 0\}$, the vector space spanned by *all* partial derivatives of w (including w itself): all finite complex linear combinations of them.

- *Pure exponential.* $w = e^{2t+3x}$: $\partial_t^j \partial_x^k w = 2^j 3^k w$, so $\mathcal{R}w = \mathrm{span}\{w\}$ and $\dim_{\mathbb{C}} \mathcal{R}w = 1$.
- *Polynomial times exponential.* $w = te^x$: $\partial_t w = e^x$, $\partial_t^2 w = 0$, $\partial_x w = te^x$; every higher derivative is a combination of te^x and e^x, so $\mathcal{R}w = \mathrm{span}\{te^x, e^x\}$ and $\dim_{\mathbb{C}} \mathcal{R}w = 2$. Although w has infinitely many derivatives, two independent functions span them all.

Step 3 ($\Rightarrow$). Let $f = p\, e^{\tau t + \xi x}$. Then $\partial_t f = (\partial_t p + \tau p)e^{\tau t+\xi x}$ and $\partial_x f = (\partial_x p + \xi p)e^{\tau t+\xi x}$. Differentiating a polynomial never raises its degree, so every derivative of f is $q\, e^{\tau t+\xi x}$ with $\deg q \leq \deg p$. The polynomials of degree $\leq d$ in two variables form a space of dimension $(d+1)(d+2)/2$, so the derivatives of each term span a finite-dimensional space; w has finitely many terms, so the same holds for w.

Step 4 ($\Leftarrow$). Suppose $W = \mathcal{R}w$ is finite-dimensional. Differentiation maps derivatives to derivatives, so $\partial_t W \subseteq W$ and $\partial_x W \subseteq W$: they define linear operators $T = \partial_t|_W$ and $X = \partial_x|_W$. They commute ($TX = XT$, mixed partials commute). Commuting complex matrices are simultaneously triangularizable, so in a suitable basis both are upper triangular. Their diagonal entries give pairs of joint eigenvalues (τ_j, ξ_j); the joint generalized eigenspaces produce functions $p_j(t,x)e^{\tau_j t+\xi_j x}$: the diagonal part of differentiation produces the exponential, the nilpotent parts produce the polynomial factors (they vanish when raised to a high enough power, since the space is finite-dimensional). There are finitely many pairs, hence $w \in$ EP.

Step 5: $\mathcal{R}w \cong \mathcal{R}/\mathrm{Ann}(w)$. Define $\Phi : \mathcal{R} \to \mathcal{R}w$, $Q \mapsto Qw$. It is surjective (by definition of $\mathcal{R}w$) and its kernel is exactly $\mathrm{Ann}(w)$ (since $\Phi(Q) = 0$ means $Qw = 0$). By the first

isomorphism theorem $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$. The quotient identifies two operators exactly when they act identically on w: $P \sim Q \iff (P-Q)w = 0$. Hence

$$w \in \mathrm{EP} \iff \dim_{\mathbb{C}} \mathcal{R}/\operatorname{Ann}(w) < \infty.$$

Step 6: $V(\operatorname{Ann} w)$ is the set of exponents. For $w = te^x$, $\operatorname{Ann}(w) = (\partial_t^2, \partial_x - 1)$, with symbol ideal $(\tau^2, \xi - 1)$. Its zero set is $\{\tau^2 = 0,\ \xi = 1\} = \{(0,1)\}$ (over $\mathbb{C}$, $\tau^2 = 0$ forces $\tau = 0$), exactly the exponent pair of te^{0t+1x}. For a general exponential-polynomial the variety gives the *distinct* exponent pairs that occur. It does *not* record multiplicities or the full polynomial prefactors: those affect the structure of the annihilator but introduce no new exponent pairs.

Step 7: why it is computationally important. The criterion converts a question about the *form* of a function into finite-dimensional linear algebra. If one can establish relations such as $\partial_t^2 w = 2\partial_t w + w$ and $\partial_x w = 3w$, then every mixed derivative reduces to a combination of w and $\partial_t w$, so $\dim \mathcal{R}w \le 2$ and $w \in \mathrm{EP}$. The big picture: the annihilator encodes all differential equations satisfied by w; the quotient $\mathcal{R}/\operatorname{Ann}(w)$ counts the independent derivatives; the joint eigenvalues of differentiation identify the exponential modes.

D.4 The ODE prototype, step by step

(E4: end of Section 3, before the keypoint) This section is the one-dimensional version of the whole theory. It starts with an ODE because in one variable the algebra is completely explicit: roots of a polynomial determine the solutions, and the roots lead naturally to the minimal annihilator, the companion matrix and the spectral projectors.

Step 1: the ODE. $p(D)u = 0$ with $D = d/dx$ and p of degree m, factored over $\mathbb{C}$ as $p(\lambda) = \prod_{j=1}^{r} (\lambda - \lambda_j)^{m_j}$, $\sum_j m_j = m$: the λ_j are the distinct roots, m_j their multiplicities, m the total degree. For $p(\lambda) = (\lambda-1)^2(\lambda+2)^3$ the equation $(D-1)^2(D+2)^3 u = 0$ has order five.

Step 2: why solutions are $q(x)e^{\lambda x}$. First, $(D-\lambda)e^{\lambda x} = 0$. If the root has multiplicity greater than one, further solutions appear: $(D-\lambda)(xe^{\lambda x}) = e^{\lambda x}$, hence $(D-\lambda)^2(xe^{\lambda x}) = 0$; in general $(D-\lambda)^m(x^k e^{\lambda x}) = 0$ for $0 \le k < m$. The reason is the identity

$$(D-\lambda)(q(x)e^{\lambda x}) = q'(x)e^{\lambda x}\ :$$

each application of $D-\lambda$ differentiates the polynomial prefactor once, and after m applications a polynomial of degree $< m$ has vanished. A root of multiplicity m_j therefore contributes the m_j solutions $e^{\lambda_j x}, xe^{\lambda_j x}, \ldots, x^{m_j-1}e^{\lambda_j x}$, and the complete solution space is

$$\ker p(D) = \Big\{ \sum_{j=1}^{r} q_j(x) e^{\lambda_j x} : \deg q_j < m_j \Big\}.$$

Worked example. For $(\lambda-1)^2(\lambda+2)^3$: root 1 (multiplicity 2) gives e^x, xe^x; root -2 (multiplicity 3) gives $e^{-2x}, xe^{-2x}, x^2e^{-2x}$. Five independent solutions, so the solution space has dimension five $= m$: the number of natural basis functions is finite and equals the degree of p.

Step 3: the minimal annihilator. For a particular function u, $\operatorname{Ann}(u) = \{Q \in \mathbb{C}[D] : Q(D)u = 0\}$. In one variable the annihilator of a nonzero u is generated by a single polynomial

of smallest degree; normalized to be monic it is the *minimal annihilating polynomial* of u. Example: $u = xe^x$ has $(D-1)u = e^x$ and $(D-1)^2u = 0$, so its minimal annihilator is $p_u(D) = (D-1)^2$; the multiplicity two tells that the solution has a polynomial prefactor of degree one. The polynomial p defining the ODE need not be minimal for a particular solution: $u = e^x$ solves $(D-1)^2u = 0$, but its minimal annihilator is only $D-1$.

Step 4: the companion matrix. Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$. Then $u^{(m)} = -a_{m-1}u^{(m-1)} - \cdots - a_0u$. With the state vector $Y = (u, u', \ldots, u^{(m-1)})^\top$ the ODE becomes $Y' = CY$, with C the companion matrix displayed in the text. Its characteristic polynomial is $p(\lambda)$: the roots of the polynomial become the eigenvalues of a matrix, and the multiplicities determine the possible Jordan blocks.

Step 5: spectral projectors. Suppose first the roots are distinct. The Lagrange polynomials $\ell_j(z) = \prod_{k\neq j} \frac{z-\lambda_k}{\lambda_j-\lambda_k}$ satisfy $\ell_j(\lambda_i) = 1$ if $i = j$ and 0 if $i \neq j$. Substituting the matrix gives the projectors $\Pi_j = \ell_j(C)$; each projects onto the eigenspace of λ_j, and $\Pi_i\Pi_j = 0$ for $i \neq j$, $\sum_j \Pi_j = \mathrm{Id}$. Consequently $e^{xC} = \sum_j e^{\lambda_j x}\Pi_j$ when C is diagonalizable with distinct eigenvalues. When roots are repeated, eigenspace projectors alone do not capture the polynomial factors: one needs the generalized eigenspaces and the nilpotent parts of the Jordan blocks, which produce $xe^{\lambda x}$, $x^2e^{\lambda x}$, ...

Step 6: why this is the prototype.

One-dimensional ODE	Later multivariable theory
polynomial $p(\lambda)$	polynomial symbol $P(\tau, \xi)$
roots λ_j	exponent pairs (τ_j, ξ_j)
minimal annihilating polynomial	annihilator ideal and its gcd
companion matrix	commuting differentiation operators on a finite-dimensional space
Lagrange spectral projectors	projectors isolating the individual exponential modes

In one variable the roots are isolated points and polynomial factorization gives the whole picture. In two variables the common zero set of polynomials can contain entire curves, so the geometry and the ideal decomposition become more subtle.

D.5 What Theorem 5.1 asks, and why

(E5: Section 4.1, after the easy-direction proof / before the takeaway) **Step 1: the problem.** Given an open connected $\Omega \subset \mathbb{R}^2$, a distribution $u \in \mathcal{D}'(\Omega)$ and a constant-coefficient operator with $P(\partial)u = 0$, does u *uniquely identify* P, in the sense $\mathrm{Ann}(u) = (P)$? That equality says that every constant-coefficient operator annihilating u is a multiple of P: there is no genuinely smaller equation. A single function can satisfy many different PDEs, so the answer is not automatically yes.

Counterexample. $P = \partial_x^2 - 1 = (\partial_x - 1)(\partial_x + 1)$ and $u = e^x$: $P(\partial)u = 0$, but also $(\partial_x - 1)u = 0$, so $\mathrm{Ann}(u) \neq (P)$ (in $\mathbb{C}[\partial_x]$ it equals $(\partial_x - 1)$; in $\mathcal{R}$ it is $(\partial_t, \partial_x - 1)$). Knowing $u = e^x$ does not tell us that the second-order equation is the minimal one.

Step 2: factor P. $P = \prod_{i=1}^r P_i^{e_i}$ with distinct irreducible P_i. Example: $P = (\tau-\xi)^2(\tau+\xi)$ has $P_1 = \tau - \xi$ ($e_1 = 2$) and $P_2 = \tau + \xi$ ($e_2 = 1$). Each factor is a potential piece of the equation that might be unnecessary to explain this particular u.

Step 3: the key construction. For each i put $w_i = (P/P_i)(\partial)u$. Because $P(\partial)u = 0$,

$$P_i(\partial)w_i = P_i(\partial)\tfrac{P}{P_i}(\partial)u = P(\partial)u = 0, \qquad \text{i.e.} \qquad w_i \in \ker P_i(\partial).$$

After removing one factor, the resulting function satisfies the PDE of that factor alone. The theorem asks whether w_i is an exponential-polynomial:

- $w_i \notin$ EP: it has infinitely many independent derivatives, which prevents the factor P_i from being removed from the annihilator;
- $w_i \in$ EP: it lives in a finite-dimensional derivative space, which signals a possible ambiguity in identifying the equation.

Step 4: part (a). $\operatorname{Ann}(u) = (P) \iff (P/P_i)(\partial)u \notin \mathrm{EP}(\Omega)$ for every i. If another annihilator Q were not a multiple of P, the annihilator would contain more information than (P); the theorem rules this out precisely by the non-EP tests. *Check on the counterexample:* for $P_1 = \partial_x - 1$, $P/P_1 = \partial_x + 1$ and $w_1 = (\partial_x + 1)e^x = 2e^x \in$ EP; for $P_2 = \partial_x + 1$, $w_2 = (\partial_x - 1)e^x = 0 \in$ EP. The tests say EP, so u does not identify P, as it should. Note the theorem concerns the *full annihilator ideal*, not merely whether P is satisfied.

Step 5: part (b). $V(\operatorname{Ann} u) = V(P) \iff (P/P_i^{e_i})(\partial)u \notin \mathrm{EP}$ for all i. In (a) one removes one copy of P_i, in (b) the whole power $P_i^{e_i}$, because the characteristic variety sees zeros, not multiplicities. For instance $(\tau - \xi)^2$ and $\tau - \xi$ have the same zero set $\{\tau = \xi\}$ but are different polynomials. Identifying the variety is therefore weaker than identifying the annihilator.

Step 6: part (c). If P is *reduced* (all $e_i = 1$), removing one copy of P_i equals removing its whole power, so (a) and (b) coincide. If P is monic in τ of degree k in τ, the test can be done on the finite t-jet $u, \partial_t u, \ldots, \partial_t^{k-1}u$: a jet is just a finite list of derivatives, so the test needs only finitely many derivative components.

The logical chain. Given $P(\partial)u = 0 \to$ factor $P \to$ remove factors, $w_i = (P/P_i)(\partial)u \to$ test membership in EP (finite-dimensional derivative span, Lemma 2.7) $\to$ decide whether u determines the full annihilator or only its characteristic variety. One precision: the theorem does *not* say that an exponential-polynomial solution can never identify any PDE; it tests the particular functions w_i obtained after removing factors, not whether u itself is in EP.

D.6 The point spectrum and the intrinsic operator G_u

(E6: Section 4.2, after the aside "Why G_u and not the full variety") **Step 1: the decomposition.** By Lemma 2.4, $\operatorname{Ann}(u) = G_u J_u$ with $G_u = \gcd \operatorname{Ann}(u)$ and $V(J_u)$ finite. G_u captures the common polynomial factor of all annihilating equations; J_u captures the residual finite structure. The paper calls $\sigma_p(u) = V(J_u)$ the *point spectrum* of u: a point (τ_0, ξ_0) corresponds to the exponential $e^{\tau_0 t + \xi_0 x}$. "Spectrum" is used in an algebraic sense: a finite set of exponent pairs, not a set of Fourier frequencies.

Step 2: why $\operatorname{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset$ and $G_u \sim P$. Here $G_u \sim P$ means they differ by a nonzero constant.

- Suppose $\sigma_p(u) = \emptyset$, i.e. $V(J_u) = \emptyset$. Over $\mathbb{C}$ the Nullstellensatz says an ideal without common zeros is the whole ring, so $J_u = \mathcal{R}$ and $\operatorname{Ann}(u) = G_u\mathcal{R} = (G_u)$: with empty point spectrum the annihilator is generated by its gcd. If moreover $G_u \sim P$, then $\operatorname{Ann}(u) = (P)$.
- Conversely if $\operatorname{Ann}(u) = (P)$, the gcd is P up to a constant, the cofactor ideal is $\mathcal{R}$ and its zero set is empty.

Interpretation: the equation is fully identifiable when there is no residual finite point spectrum and the common gcd agrees with the proposed operator.

Step 3: a single exponential. $u = e^{\tau_0 t+\xi_0 x}$ has $\partial_t u = \tau_0 u$, $\partial_x u = \xi_0 u$, so $\mathrm{Ann}(u) = (\tau-\tau_0, \xi-\xi_0)$. The generators have no common nonconstant factor: $G_u = 1$, $J_u = (\tau-\tau_0, \xi-\xi_0)$, and

$$\sigma_p(u) = \{(\tau_0, \xi_0)\} :$$

one point, exactly the exponential in u. Since $G_u = 1$ and the spectrum is nonempty, this u does not identify any nonconstant operator as the generator of its full annihilator: $\partial_t - \tau_0$ and $\partial_x - \xi_0$ both annihilate u, and neither alone generates $\mathrm{Ann}(u)$.

Step 4: the meaning of $\mathcal{R}u \cap \mathrm{EP} \cong \mathcal{R}/J_u$. $\mathcal{R}u$ is the span of all derivatives of u; $\mathcal{R}u \cap \mathrm{EP}$ consists of the exponential-polynomials obtainable from u by constant-coefficient operators and linear combinations; the quotient $\mathcal{R}/J_u$ describes this part algebraically. For the single exponential, $\mathcal{R}/J_u \cong \mathbb{C}$, matching $\mathcal{R}u = \mathrm{span}\{u\}$, one-dimensional and entirely inside EP. In general the finite-dimensional quotient also records *multiplicity* information that the set $\sigma_p(u)$ alone loses.

Step 5: why G_u is the intrinsic operator. Knowing some P with $P(\partial)u = 0$, one wants the operator intrinsic to u rather than P itself, which may contain unnecessary factors. Strategy: (1) factor $P = \prod P_i^{e_i}$; (2) remove factors and test whether the resulting functions are in EP; (3) recover G_u and J_u; (4) if $J_u = \mathcal{R}$, the point spectrum is empty and the annihilator is generated by G_u. At most $\sum_i e_i$ membership tests are needed.

Step 6: the repeated-factor example. The variety $V(\mathrm{Ann}\, u)$ can replace G_u only for reduced P, because zero sets do not see multiplicities. The aside above uses $P = \tau^2$ and $u = e^{x^2}$: here $\mathrm{Ann}(u) = (\tau)$, so $G_u = \tau \not\sim P$ while $V(\mathrm{Ann}\, u) = V(P) = \{\tau = 0\}$. The variety criterion would wrongly say that u identifies τ^2. (An earlier draft used $P = \tau$ for this example, which is reduced and so shows nothing about repeated factors; the revised text uses τ^2.)

D.7 The d'Alembert function does *not* split by terms (Proposition 3.3)

(E7a: Section 5.1 (after the heat example, before 5.2)) **Enunciation.** If $L = L_1 + L_2$ (constant coefficients), then

$$\ker L \cap (\ker L_1 + \ker L_2) \;\subseteq\; \ker(L,\; L_1 L_2),$$

and this space is finite-dimensional as soon as $V(L_1) \cap V(L_2)$ is finite, because $V(L, L_1L_2) = V(L_1) \cap V(L_2)$. In words: only a finite-dimensional family of solutions of L can be written as (solution of the first term) + (solution of the second term).

Applying it to the wave equation. Write the wave operator as a sum of its two *terms*, $L = \partial_t^2 - c^2\partial_x^2 = L_1 + L_2$ with $L_1 = \partial_t^2$ and $L_2 = -c^2\partial_x^2$. Then

$$\ker L_1 = \{a(x) + t\, b(x)\}, \qquad \ker L_2 = \{\alpha(t) + x\, \beta(t)\},$$

so every element of $\ker L_1 + \ker L_2$ is killed by $L_1L_2 = -c^2\partial_t^2\partial_x^2$ (the operators commute: $L_1L_2u = L_2L_1a + L_1L_2b = 0$).

Step 1: a d'Alembert solution. Take $u = f(x + ct) + g(x - ct)$, which solves $Lu = 0$. Computing,

$$\partial_t^2\partial_x^2 f(x + ct) = c^2 f''''(x + ct), \qquad \partial_t^2\partial_x^2 g(x - ct) = c^2 g''''(x - ct),$$

so $L_1L_2u = -c^4(f''''(x+ct) + g''''(x-ct))$.

Step 2: when is it zero? $f''''(s) + g''''(r) = 0$ for all independent $s = x+ct$, $r = x-ct$ forces $f'''' \equiv k$ and $g'''' \equiv -k$ for a constant k. So f and g must be polynomials (of degree ≤ 4, with matching quartic coefficients). The d'Alembert solutions that lie in $\ker L_1 + \ker L_2$ are polynomials only; a generic one, such as $f(s) = e^s$ ($u = e^{x+ct}$ for $g = 0$), gives $L_1L_2u = -c^4u \neq 0$, so it is *not* a sum of a solution of $\partial_t^2 u = 0$ and one of $\partial_x^2 u = 0$.

Step 3: the finite dimension. Here $V(L_1) \cap V(L_2) = \{\tau^2 = 0\} \cap \{\xi^2 = 0\} = \{(0,0)\}$, a point, so the space $\ker(L, L_1L_2)$ is finite-dimensional. It is the polynomial family above: f, g cubic polynomials plus the shared quartic term $\pm k\, s^4/24$, modulo the constant overlap $(f+\kappa, g-\kappa)$; this gives $4+4-1+1 = 8$ parameters, which agrees with the intersection multiplicity $\dim \mathcal{R}/(\tau^2 - c^2\xi^2, \tau^2\xi^2) = 4+4 = 8$.

Conclusion. The wave solution $u = f(x+ct) + g(x-ct)$ is the sum of *factor* components (next subsection) but, in general, not of *term* components. This is the structural reason why any attempt to split by terms leaves a remainder (Section 4).

D.8 The d'Alembert function proves Theorem 3.1, step by step

(E7b: Section 5.2, after the Takeaway of Theorem 5.3) **Enunciation (Factor decomposition).** Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and $P = P_1^{m_1} \cdots P_N^{m_N}$ with $Q_j = P_j^{m_j}$ pairwise coprime. Then every $u \in \ker P$ can be written $u = \sum_{j=1}^N u_j$ with $u_j \in \ker Q_j(\partial)$, and the ambiguity of the decomposition is an exponential-polynomial space of dimension exactly $\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j)$.

The case. Wave equation, $N = 2$, $m_1 = m_2 = 1$:

$$P = \tau^2 - c^2\xi^2 = Q_1Q_2, \qquad Q_1 = \partial_t - c\partial_x, \quad Q_2 = \partial_t + c\partial_x,$$

and the d'Alembert solution $u = f(x+ct) + g(x-ct)$. We check each claim of the theorem.

Step 1: the hypotheses. Q_1, Q_2 are distinct linear polynomials, hence coprime (no common factor). Ω is any bounded simply connected region, e.g. a disc.

Step 2: the kernels. $Q_1 f(x+ct) = cf' - cf' = 0$ and $Q_2 g(x-ct) = -cg' + cg' = 0$, so $\ker Q_1 = \{f(x+ct)\}$ (a profile moving left) and $\ker Q_2 = \{g(x-ct)\}$ (a profile moving right).

Step 3: existence of the splitting. Take any $u \in \ker P$. Then $w_2 = Q_2u \in \ker Q_1$ and $w_1 = Q_1u \in \ker Q_2$ (because $Q_1Q_2u = 0$). Since $Q_2 - Q_1 = 2c\,\partial_x$,

$$\partial_x u = \tfrac{1}{2c}(Q_2u - Q_1u).$$

Integrating in x at fixed t, define $u_1 = \frac{1}{2c}\int^x Q_2u\,\mathrm{d}x'$ and $u_2 = -\frac{1}{2c}\int^x Q_1u\,\mathrm{d}x'$. If $Q_2u = h(x+ct)$, then $u_1 = \frac{1}{2c}H(x+ct)$ with $H' = h$, and $Q_1u_1 = 0$; likewise $u_2 \in \ker Q_2$. By construction $\partial_x(u - u_1 - u_2) = 0$, so $u - u_1 - u_2 = k(t)$. Applying P gives $k''(t) = 0$, so $k(t) = a + bt$; and $t = \frac{1}{2c}((x+ct) - (x-ct))$ is itself a sum of an element of $\ker Q_1$ and one of $\ker Q_2$. Hence $u = u_1 + u_2$ with $u_j \in \ker Q_j$ after absorbing k: the splitting exists.

Step 4: the explicit example. With $c = 1$, $u = \dfrac{1}{1+(x+t)^2} + e^{-(x-t)^2}$: $Q_2u = 2f'(x+t)$ and $Q_1u = -2g'(x-t)$, so $u_1 = \frac{1}{2}\int^x 2f'(x'+t)\,\mathrm{d}x' = f(x+t)$ and $u_2 = -\frac{1}{2}\int^x(-2g')\,\mathrm{d}x' = g(x-t)$, with $k = 0$: the recovered components are exactly the two moving profiles.

Step 5: the ambiguity has dimension $\delta(P) = 1$. $Q_2 - Q_1 = 2c\,\xi$ and $Q_1 + Q_2 = 2\tau$, so $(Q_1, Q_2) = (\tau, \xi)$, which is not the unit ideal (the two lines meet at the origin), and $\mathcal{R}/(\tau, \xi) \cong \mathbb{C}$. So $\delta(P) = \dim \mathcal{R}/(Q_1, Q_2) = 1$. Directly: $f(x+ct)+g(x-ct) = 0$ for all (t, x) forces $f \equiv \kappa$, $g \equiv -\kappa$ for a constant κ. The only freedom is $(f, g) \mapsto (f + \kappa, g - \kappa)$: a one-dimensional space of constants, which are exponential-polynomials – exactly the dimension the theorem predicts.

Step 6: where the theorem's tools appear. The finite quotient $\mathcal{R}/(Q_1, Q_2)$ measures the failure of the one-variable picture. In one variable coprime polynomials satisfy a Bézout identity $aQ_1 + bQ_2 = 1$ and $u = bQ_2u + aQ_1u$ splits explicitly with a direct sum; in two variables only $(Q_1, Q_2) \supseteq (\tau, \xi)$ holds, and that is what produces the constant ambiguity. The wave case is the simplest example where the proof by exact sequence $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ can be followed by hand.

Contrast with 5.1. The same function is a sum of *factor* components for every datum (f, g), but a sum of *term* components only when f, g are polynomials. This is the difference between a factor and a term.

D.9 The Lagrange projector for evolution factors

(E8: Section 5.3, after Eq. (lagrange)) **Step 1: the evolution operator.** $P(\tau, \xi) = \prod_{j=1}^{N}(\tau - a_j(\xi))$ with a_j distinct polynomials in ξ. Replacing τ, ξ by ∂_t, ∂_x the equation is $\prod_j(\partial_t - a_j(\partial_x))u = 0$. Example: $a_1(\xi) = \xi$, $a_2(\xi) = -\xi$ gives $(\partial_t - \partial_x)(\partial_t + \partial_x)u = 0$, the one-dimensional wave equation $u_{tt} - u_{xx} = 0$. Because the factors are distinct, a solution decomposes as $u = \sum_j u_j$ where each component satisfies its own first-order law $\partial_t u_j = a_j(\partial_x)u_j$, i.e. $(\partial_t - a_j(\partial_x))u_j = 0$.

Step 2: isolating a component. Fix j and apply $Q_j = \prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$. Constant-coefficient operators commute, so $Q_j u = \sum_l Q_j u_l$.

- *Case $l \neq j$.* Q_j contains the factor $\partial_t - a_l(\partial_x)$, which kills u_l: $Q_j u_l = 0$. Every component except u_j disappears.
- *Case $l = j$.* No factor of Q_j kills u_j; use $\partial_t u_j = a_j(\partial_x)u_j$: for each $k \neq j$, $(\partial_t - a_k(\partial_x))u_j = (a_j - a_k)(\partial_x)u_j$. Applying all factors, $Q_j u = \prod_{k\neq j}(a_j - a_k)(\partial_x)\, u_j$.

Hence

$$u_j = \Big[\prod_{k\neq j}(a_j - a_k)(\partial_x)\Big]^{-1} \prod_{k\neq j}(\partial_t - a_k(\partial_x))\, u,$$

which is Eq. (8), provided the inverse operator exists on the function space considered. It is not automatically a differential operator: it is in general a Fourier multiplier, possibly singular at frequencies where $a_j = a_k$.

Step 3: why this is Lagrange interpolation. The Lagrange polynomials $L_j(z) = \prod_{k\neq j} \frac{z-a_k}{a_j-a_k}$ satisfy $L_j(a_l) = 1$ for $l = j$, 0 otherwise; for a matrix C diagonalized in eigenvectors, $L_j(C)$ acts as the identity on the j-th eigenspace and as zero on the others. Here the role of C is played by ∂_t and the eigenvalues a_j are replaced by the operators $a_j(\partial_x)$:

$$\Pi_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}, \qquad \Pi_j u = u_j.$$

The analogy is exact at the algebraic level: Π_j is the projector that extracts the j-th evolution component.

Step 4: rational in ξ. With the Fourier convention $\partial_x \mapsto i\kappa$ (κ real frequency), $a_j(\partial_x)$ becomes multiplication by $a_j(i\kappa)$ and the multiplier is

$$\ell_j(\kappa, \partial_t) = \prod_{k \neq j} \frac{\partial_t - a_k(i\kappa)}{a_j(i\kappa) - a_k(i\kappa)}.$$

The denominators vanish exactly where $a_j(i\kappa) = a_k(i\kappa)$: these are the potential poles of the projector. Strictly, the rationality concerns the dependence on the spatial frequency; the projector still involves ∂_t before it is restricted to a solution component, on which ∂_t acts as multiplication by $a_j(i\kappa)$.

Step 5: the wave equation. $(\partial_t - \partial_x)(\partial_t + \partial_x)u = 0$, $u = u_1 + u_2$ with $\partial_t u_1 = \partial_x u_1$ and $\partial_t u_2 = -\partial_x u_2$. Applying $\partial_t + \partial_x$: the u_2 term vanishes and $(\partial_t + \partial_x)u_1 = 2\partial_x u_1$, so

$$u_1 = \tfrac{1}{2}\,\partial_x^{-1}(\partial_t + \partial_x)u, \qquad u_2 = -\tfrac{1}{2}\,\partial_x^{-1}(\partial_t - \partial_x)u.$$

In Fourier variables ∂_x^{-1} is multiplication by $1/(i\kappa)$: a pole at $\kappa = 0$. This does not mean the wave equation has no decomposition (Section 3 proves it does), only that this particular formula needs a choice of inverse (an integration constant) or extra assumptions at zero frequency.

Step 6: what "coprime" means. $\tau - a_j(\xi)$ and $\tau - a_k(\xi)$ are coprime in $\mathbb{C}[\tau, \xi]$ exactly when $a_j - a_k$ is not the zero polynomial; for distinct a_j this holds, though $a_j - a_k$ may still vanish at particular frequencies, creating poles. Three levels must be kept apart: *coprime factors* (the algebraic decomposition exists in the localized, rational-operator sense); *no poles at any frequency* (stronger, not guaranteed by coprimality); *a bounded or well-defined projector on a chosen function space* (needs analytic conditions near frequencies where branches meet). The message: distinct evolution branches admit algebraic projectors, like distinct eigenvalues of a matrix; in spatial Fourier variables the projectors are rational multipliers whose denominators reveal where the branches intersect.

D.10 Regularity, the sum map S, γ_N and the Friedrichs angle

(E9: Section 5.5, just before the final Takeaway of Section 5) **The guiding idea.** γ_N measures *how reliably a solution can be separated into components belonging to different kernel spaces, and how much errors can be amplified when this is done.* Being able to decompose algebraically does not guarantee that the pieces have the regularity of the original solution; the obstruction is geometric: the component spaces may be too close to one another, which makes the decomposition unstable.

Step 1: why a component can lose a derivative. $Pu = 0$, $u = \sum_j u_j$, $Q_j u_j = 0$. The pieces exist, but $u \in H^m$ does not imply $u_j \in H^m$, because the formulas that extract u_j may involve inverse differential operators, which can lose regularity. In the wave example $u_1 = \frac{1}{2}\partial_x^{-1}(\partial_t + \partial_x)u$: even if u has m Sobolev derivatives the formula alone does not guarantee that u_1 has m, because of the inverse and its behaviour at problematic frequencies. This is the content of Proposition 3.6.

Step 2: the component spaces and the sum operator. $M_j = \ker_{H^m} Q_j$ (admissible components); for two components $M = \ker_{H^m} Q_1$, $N = \ker_{H^m} Q_2$. The decomposition question is packaged in

$$S : M_1 \times \cdots \times M_N \to H^m(\Omega), \qquad S(u_1, \ldots, u_N) = \sum_j u_j, \qquad \|(u_1, \ldots, u_N)\|^2 = \sum_j \|u_j\|_{H^m}^2.$$

Recovering the components means inverting S on the relevant solution space. Two things can go wrong: *non-uniqueness* (different tuples with the same sum: $\ker S = \{(u_j) : \sum u_j = 0\}$; if $v \in M \cap N$ then $(v, -v) \in \ker S$, so overlapping spaces give non-uniqueness) and *instability* (even if unique, a tiny change in the sum may require a large change in the pieces).

Step 3: what γ_N is.

$$\gamma_N = \inf \sigma\big(S^*S|_{(\ker S)^\perp}\big) = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|^2_{H^m}}{\|v\|^2}.$$

S^* is the adjoint, S^*S is positive self-adjoint, and $(\ker S)^\perp$ removes the directions that change the pieces without changing the sum (without this removal the ratio would be 0 regardless of the stability of the genuinely recoverable part). If $\gamma_N > 0$ then $\|Sv\|^2 \geq \gamma_N \|v\|^2$ on $(\ker S)^\perp$, i.e.

$$\|v\| \leq \gamma_N^{-1/2} \|Sv\| :$$

this is the stability estimate; the smaller γ_N, the larger the possible amplification. *Technical caveat:* if the restriction has continuous spectrum accumulating at 0, "smallest nonzero eigenvalue" (used loosely in the text after Figure 3) is not literally correct; the robust quantity is the *infimum of the spectrum* on $(\ker S)^\perp$, as in the theorem.

Step 4: two components and the Friedrichs angle. Let $I = M \cap N$. The Friedrichs cosine is $c_F = \sup\{|\langle x, y\rangle| : x \in M \cap I^\perp,\ y \in N \cap I^\perp,\ \|x\| = \|y\| = 1\}$, and $\theta \in [0, \pi/2]$ is defined by $c_F = \cos\theta$. If c_F is close to 1 the spaces have almost parallel directions; if it is clearly below 1 they are well separated; $c_F < 1$ means the sum $M + N$ is closed. A finite-dimensional analogy: decomposing a vector along two nearly parallel directions gives large components that nearly cancel, leaving a small sum; recovering them from the small sum is numerically unstable. The paper proves $\gamma_2 = 1 - c_F$, so $c_F < 1 \iff \gamma_2 > 0$.

Step 5: why pairwise angles do not suffice for $N \geq 3$. Three vectors in a plane pointing in three different directions can satisfy $v_1 + v_2 + v_3 = 0$: the instability depends on combinations of all components, not only on pairs. S captures these collective effects; its smallest nonzero singular value is $\sqrt{\gamma_N}$, so $1/\sqrt{\gamma_N}$ is the natural condition-number-like factor for recovering components from their sum, as in a least-squares problem where a small singular value makes inversion sensitive to errors.

Step 6: density. The theorem needs $M_1 + \cdots + M_N$ to be both closed and dense in $\ker_{H^m} P$. Dense plus closed gives $M_1 + \cdots + M_N = \ker_{H^m} P$, i.e. a decomposition for every solution, with stability from the positive-angle condition. The text states density on bounded convex domains for every P and m (Hörmander, Stein; see its status label), so closedness is the decisive remaining condition there.

The chain in six lines. (1) Algebra produces components attached to the factors. (2) Sobolev asks whether they remain in H^m. (3) This is controlled by S. (4) $\gamma_N > 0$ makes S bounded below after removing its kernel: stable recovery. (5) For $N = 2$ this is $c_F < 1$. (6) For $N \geq 3$ the collective γ_N, not the pairwise angles, controls the conditioning. One distinction: $\gamma_N > 0$ is about the stability of the decomposition map; it is not an algebraic statement about factorization. The factorization says which spaces to consider, their geometry in H^m says whether the components can be recovered with the desired regularity.

D.11 The additive interaction remainder and why the terms commute

(E11: Section 6.2, after the proof of Theorem 6.1 (before the concrete case)) **Step 1: the setup.** $\partial_t u = Au$, $u(0) = u_0$, with $A = \sum_{n=1}^{N} A_n$. The solution is formally $u(t) = e^{tA}u_0$. If term n acted alone it would produce $S_n(t) = e^{tA_n}$, with change $\Delta_n u_0 = (S_n(t) - \mathrm{Id})u_0$ (Δ_n is the *increment operator*). The additive interaction remainder is

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \Delta_n u_0,$$

that is, *interaction = full evolution − initial state − sum of the individual increments.* Note that this is an additive decomposition, *not* a decomposition into kernel spaces as in Section 3.

Step 2: why commutation gives an exact formula. If $A_i A_j = A_j A_i$ then (for suitable operators and a well-defined evolution) $e^{t\sum_n A_n} = \prod_n e^{tA_n} = \prod_n S_n(t)$. Since $S_n = \mathrm{Id} + \Delta_n$,

$$u(t) = \prod_{n=1}^{N} (\mathrm{Id} + \Delta_n)u_0, \qquad E_{\mathrm{int}}^{\mathrm{add}}(t) = \Big[\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n \Big] u_0.$$

This is pure algebra: multiply the increments, then subtract the constant term and all terms involving a single increment.

Step 3: $N = 2$. $u(t) = S_1 S_2 u_0 = (\mathrm{Id} + \Delta_1)(\mathrm{Id} + \Delta_2)u_0 = (\mathrm{Id} + \Delta_1 + \Delta_2 + \Delta_1\Delta_2)u_0$. Subtracting u_0 and the two single increments leaves

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \Delta_1\Delta_2 u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})u_0.$$

It is an interaction term because it contains the effect of both increments acting together; if either increment is zero on the relevant state, the interaction vanishes.

Step 4: $N = 3$ shows the general structure.

$$(\mathrm{Id} + \Delta_1)(\mathrm{Id} + \Delta_2)(\mathrm{Id} + \Delta_3) = \mathrm{Id} + \Delta_1 + \Delta_2 + \Delta_3 + \Delta_1\Delta_2 + \Delta_1\Delta_3 + \Delta_2\Delta_3 + \Delta_1\Delta_2\Delta_3,$$

so $E_{\mathrm{int}}^{\mathrm{add}} = (\Delta_1\Delta_2 + \Delta_1\Delta_3 + \Delta_2\Delta_3 + \Delta_1\Delta_2\Delta_3)u_0$: three *pairwise* interactions and one *three-way* interaction. In general $E_{\mathrm{int}}^{\mathrm{add}} = \sum_{J \subseteq \{1,\dots,N\}, |J| \geq 2} \big(\prod_{j \in J} \Delta_j\big) u_0$: the empty subset gives the initial state, subsets of size one the individual increments, everything left involves at least two operators.

Step 5: the Fourier formula. If each A_n is a constant-coefficient operator it becomes multiplication by its symbol $a_n(\kappa)$, so $\widehat{S_n(t)u_0} = e^{ta_n(\kappa)}\hat{u}_0$ and $\widehat{\Delta_n u_0} = (e^{ta_n(\kappa)} - 1)\hat{u}_0$. Then

$$\widehat{E_{\mathrm{int}}^{\mathrm{add}}}(t, \kappa) = \Big[\prod_n e^{ta_n(\kappa)} - 1 - \sum_n \big(e^{ta_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa).$$

At each frequency this is ordinary scalar algebra; the operators commute because their Fourier symbols are multiplications. *Notation note:* here κ is the real frequency and $a_n(\kappa)$ the symbol of A_n in the variable $i\kappa$ (e.g. $a(\kappa) = -ic\kappa - D\kappa^2$ for advection–diffusion).

Step 6: small-time reading. $\Delta_n = e^{tA_n} - \mathrm{Id} = tA_n + O(t^2)$. For two operators, $E_{\mathrm{int}}^{\mathrm{add}} = \Delta_1\Delta_2 u_0 = t^2 A_1 A_2 u_0 + O(t^3)$ (given enough regularity). Each single increment starts

at order t, their interaction at order t^2: it needs both effects to contribute. For N operators the interaction involving all N increments starts at t^N, while the full remainder also contains pairwise interactions, generally starting at t^2.

Step 7: why the terms commute. Two operators that are polynomials in the *same* ∂_x, e.g. $A_1 = a_0 + a_1\partial_x + a_2\partial_x^2$ and $A_2 = b_0 + b_1\partial_x + b_2\partial_x^2$, commute because powers of ∂_x commute: $[A_1, A_2] = 0$. More generally any two constant-coefficient operators, polynomials in $\partial_{x_1}, \dots, \partial_{x_d}$, commute; for instance $A_1 = \partial_x^2 + \partial_y$ and $A_2 = 3\partial_x\partial_y - \partial_y^2$, because partial derivatives commute on sufficiently smooth functions (and on distributions). So for constant coefficients commutation is not a restrictive hypothesis: it comes almost automatically from the structure of the operators.

The conceptual contrast. Section 3 asked whether a solution can be stably separated into pieces belonging to different kernels. Here the question is different: if the generator is a sum of operators, how much of the resulting evolution cannot be explained by adding their individual changes? When the operators commute the answer is exact: expand the product of the evolution operators, remove the initial state and the single-operator contributions, and what remains is exactly the mixed interaction terms.

D.12 Burgers: why the shock forms and what happens to the interaction

(E13: Section 7.2, after the paragraph on the regimes / table) **Step 1: why inviscid Burgers forms a shock.** $u_t + uu_x = 0$ is nonlinear transport: the velocity at each point is u itself. Along a characteristic starting at $x = \xi$, $dx/dt = u$ and $du/dt = 0$, so u stays constant, $u(x(t), t) = u_0(\xi)$. For $u_0 = \sin\xi$:

$$x = \xi + t\sin\xi, \qquad u = \sin\xi.$$

Differentiating, $\partial x/\partial\xi = 1 + t\cos\xi$; as long as it is positive everywhere the map $\xi \mapsto x$ is invertible and the solution is smooth. It first vanishes when $1 + t\cos\xi = 0$; since $\min\cos = -1$, the breaking time is $t^* = 1$ at $\xi^* = \pi$. The gradient $u_x = \dfrac{\cos\xi}{1 + t\cos\xi}$ has numerator $\to -1$ and denominator $\to 0$ at $(t, \xi) = (1, \pi)$, so it blows up. *Before* $t = 1$ characteristics do not cross and there is one value of u per x. *At and after* $t = 1$ they compress and cross, the parametric curve folds, no single-valued classical solution exists, and the entropy solution replaces the fold with a shock (for this symmetric datum at $x = \pi$).

Step 2: where the interaction enters. Viscous Burgers $u_t + uu_x = \nu u_{xx}$ is split into nonlinear transport $A_{\mathrm{tr}}u = -uu_x$ and diffusion $A_{\mathrm{diff}}u = \nu u_{xx}$. With $S(t)$ the full flow and S_1, S_2 the single-term flows,

$$E_{\mathrm{int}}(t) = S(t)u_0 - S_1(t)u_0 - S_2(t)u_0 + u_0.$$

Before the shock all flows can be compared with smooth expansions; after it, the transport flow develops a discontinuity while the viscous solution smooths the transition over a thin layer. That difference drives everything below.

Step 3: why the smooth expansion breaks down. Before breaking there is a first-order expansion in ν with remainder $O(\nu^2\varepsilon^2 t^3)$ (the exact coefficient depends on the definitions of the paper; the parameter ε is not defined in the text and is flagged in the author's notes).

A perturbative expansion is useful while the relevant solutions stay smooth. At t^* the inviscid gradient is unbounded and estimates relying on bounded derivatives fail. *Precision:* what blows up at t^* is the perturbative expression or its estimates, not the viscous solution itself, which stays finite because positive viscosity regularizes.

Step 4: a jump that survives $\nu \to 0$. After the shock the transport flow is discontinuous while the viscous flow has a thin smooth layer. Even if the viscous profile converges to the entropy solution, the individual pieces of the difference need not converge so that their pointwise jumps cancel; hence E_{int} can have a jump of size independent of ν. This does not contradict smallness in L^1: a function of amplitude $O(1)$ supported on an interval of width $O(\nu)$ has $\|E\|_{L^\infty} = O(1)$ but $\|E\|_{L^1} = O(\nu)$. The integral measures the area under $|E|$, not the maximum height.

Step 5: the three estimates.

Quantity	Claimed scale	Interpretation
$\|E_{\text{int}}(t)\|_{L^1}$ at fixed t	$\Theta(\nu)$	small in total absolute magnitude at a fixed post-shock time
$\sup_{t\le T} \|E_{\text{int}}(t)\|_{L^1}$	$\nu\|\log\nu\| + O(\nu)$	the worst case over the time interval has a logarithmic enhancement
$\|S - S_1S_2\|_{L^1}$	$\Theta(\sqrt{\nu})$	a different, larger scale: the error of replacing the full evolution by a composition of separate flows

$\Theta(\nu)$ means bounded above and below by positive multiples of ν as $\nu \to 0$. The third quantity is a different object: it concerns $S - S_1S_2$, not the additive remainder, and must not be conflated with it. Exact constants and rates depend on the definitions of the split flows, the domain and the theorem's hypotheses; the status labels in the text say what was proved and what is conditional.

Step 6: the broader lesson. Three regimes: before the shock, smoothness permits perturbative expansions and derivative-based estimates; at breaking, the inviscid gradient blows up and smoothness-based estimates fail; after the shock, pointwise behaviour can be discontinuous, so L^1 estimates become more informative than pointwise bounds. A nonlinear interaction can remain small in an integral norm even when its pointwise behaviour changes dramatically at shock formation.

D.13 Identifying a nonlinear equation relative to a dictionary

(E14: Section 7.3, after the dictionary criterion) **Step 1: why $\mathrm{Ann}(u)$ is no longer enough.** For a linear equation, e.g. $u = e^{x-t}$, $(\partial_t + \partial_x)u = 0$ is the action of a linear constant-coefficient operator. In $u_t + uu_x = 0$ the term uu_x multiplies two quantities depending on u: it is not the action of a linear constant-coefficient operator on u, so we need another language: polynomial expressions in the jet variables.

Step 2: jets and the dictionary. Fix a maximal order k and take the jet variables $u, u_t, u_x, u_{tt}, u_{tx}, u_{xx}, \dots$ up to order k. A polynomial in them is a candidate differential relation; Burgers is $F(u, u_t, u_x) = u_t + uu_x$. A *dictionary* $\mathcal{D}$ is a finite list of candidate monomials, e.g. $\{u_t, u_x, u^2, uu_x, u_{xx}, \dots\}$, and $V = \operatorname{span}\mathcal{D}$ contains all linear combinations. A candidate equation is $c_1u_t + c_2uu_x + c_3u_{xx} = 0$; identifying the equation from data means finding the

coefficients for which the relation holds on the observed solution. V is a finite-dimensional vector space of candidates, *not* necessarily an ideal: the dictionary limits what one is willing to consider.

Step 3: $I(u)$ versus $[F]$. $I(u)$ is the differential ideal of *all* polynomial differential relations satisfied by u. If $F = u_t + uu_x = 0$ holds identically, so do its derivatives, e.g. $D_xF = u_{tx} + u_x^2 + uu_{xx}$ (total derivative D_x). $[F]$, the differential ideal generated by F, contains F, its differential consequences $D_xF, D_tF, \ldots$ and polynomial combinations of them. If u solves $F = 0$ then $[F] \subseteq I(u)$, but the inclusion can be strict: a particular solution may satisfy additional relations that do not follow from the equation alone. That is the core of identifiability.

Step 4: why "$I(u) \cap V = \langle F \rangle$" is wrong. If $\mathcal{D}$ contains the monomials of both F and D_xF, both lie in V. If $\langle F \rangle$ meant only scalar multiples of F, it would not contain D_xF, which is certainly a relation satisfied by the solution. The correct comparison is

$$I(u) \cap V = [F] \cap V :$$

every relation of the dictionary that the data satisfy is already a differential consequence of the proposed equation. It does *not* require the data to satisfy only the original equation and nothing derived from it.

Step 5: the exact sequence.

$$0 \to [F] \cap V \to I(u) \cap V \to \mathcal{J}_u \cap \psi(V) \to 0.$$

$\mathcal{J}_u$ is the jet-level obstruction (distinct from the cofactor J_u of Section 2) and ψ the map from candidate expressions to jet-level objects (its precise definition is flagged in the author's notes). The first term holds the dictionary relations explained by the proposed equation, the middle term every dictionary relation the data satisfy, the last the extra relations left after quotienting out the consequences of F. Exactness gives $I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = \{0\}$: the dictionary contains no detectable obstruction. It is a *finite-dictionary* criterion: it does not say the data identify the equation among all conceivable nonlinear PDEs, only relative to the chosen candidate space.

Step 6: shocks and solitons. Identifiability depends on the solution, not only on the equation. *Burgers with N shocks:* in the paper's evolutionary dictionary the equation is identified iff the maximal polynomial degree satisfies $d \leq N$; shock structure gives information that distinguishes the equation from competing polynomial relations, depending on the number of shocks and on the allowed degree of candidate terms (specific to the dictionary and assumptions of the cited result). *N-soliton KdV:* the identifying polynomial has total degree $(N+2)2^{N-1}$; for $N = 2$ this is $(2+2) \cdot 2^1 = 8$. A special solution can satisfy many polynomial relations, so a sufficiently rich dictionary is needed to reveal which relation actually identifies the equation rather than describing that particular solution.

Step 7: why a closed form can fail to identify. $u = e^{x+t}$ satisfies $u_t - u = 0$, $u_x - u = 0$, and consequences such as $u_{tx} - u = 0$, $u_{xx} - u = 0$. The solution alone does not say which relation is "the" governing equation: several equations explain the same special function. That is the intuition behind "closed form $\Rightarrow$ does not identify"; the exact meaning of "closed form" in the theorem is the one fixed in the cited result, but the broad lesson is that an explicit solution may be too special to separate the true equation from competitors.

The big picture. The question changes from "which linear differential operators annihilate u?" to "which polynomial differential relations does u satisfy, and which genuinely identify its equation rather than merely describing a special solution?". The dictionary makes the question finite-dimensional; $[F]$ accounts for all consequences of the proposed equation; $\mathcal{J}_u \cap \psi(V)$ detects extra relations. The crucial distinction is between *fitting* a solution and *identifying* its equation.

D.14 Future comments: the linear and the nonlinear approach compared

(E15: Section 7.4, comment on the two approaches) You are not identifying the *functions*; you are identifying the *equations or relations* that the observed functions satisfy. Both approaches share this philosophy but use different mathematical objects.

Linear case: look for an annihilating operator. Given an observed $u(t, x)$ one searches for a constant-coefficient operator with $P(\partial_t, \partial_x)u = 0$. For $u = e^{x+t}$: $(\partial_t - 1)u = 0$ and $(\partial_x - 1)u = 0$. One studies $\mathrm{Ann}(u) = \{P \in \mathbb{C}[\partial_t, \partial_x] : Pu = 0\}$ and wants to characterize all the linear operators that annihilate u.

Nonlinear case: search inside a finite dictionary. For Burgers' $u_t + uu_x = 0$ the product uu_x is not a linear constant-coefficient operator applied to u. One chooses a finite dictionary, e.g. $\mathcal{D} = \{u_t, u_x, uu_x, u_{xx}, u^2\}$, and searches for coefficients, not all zero, such that $c_1u_t + c_2u_x + c_3uu_x + c_4u_{xx} + c_5u^2 = 0$. This is a *linear-algebra* problem in the unknown coefficients even though the candidate terms are nonlinear functions of u. If the data satisfy Burgers the coefficient vector $(1, 0, 1, 0, 0)$ works. The dictionary can be enlarged (higher-degree products, higher-order derivatives) but its limits must be decided in advance.

What stays and what changes.

	Linear approach	Nonlinear approach
Object observed	the function u	the function u
Candidate relations	linear differential operators	differential polynomials
Search space	polynomial ring of constant-coefficient operators	a chosen finite dictionary of jet monomials
Main question	which operators annihilate u?	which candidate relations does u satisfy?
Main limitation	structure of the annihilator	the dictionary may omit the true equation or contain too few informative terms

The common principle is *identifiability*: does the observed data contain enough information to distinguish the governing equation from alternative candidates?

Where components and interactions enter. For $u = u_1 + u_2$ one can look at the components, the total, and an interaction term such as $E_{\text{int}} = u - u_0 - (u_1 - u_0) - (u_2 - u_0)$ (the additive form of Section 4). *Decomposition and interaction analysis* ask how the solution is assembled and how its components influence one another; *equation identification* asks which differential relation governs the observed solution or its components. A linear annihilator may be used for a component when the assumptions allow, and a nonlinear dictionary to identify

a nonlinear equation governing the total solution: complementary analyses, not automatically the same procedure.

Key point. The nonlinear dictionary approach generalizes the identification philosophy; it is not a direct replacement for the annihilator in every mathematical sense. The dictionary makes the search computationally manageable, but it also limits what can be discovered.

D.15 The last section in plain words

(E16: Section 9, simple explanation of the last part) This section answers three simple questions: what can we *never* do with this method, what do we *not know yet*, and how much can we trust what is written.

1. What is impossible. These are proved facts, not gaps:

- *No finite proof of identification.* If u is not an exponential-polynomial, no finite list of numbers (derivatives at a point, samples) can prove that $\mathrm{Ann}(u) = (P)$, because every finite list can also come from some exponential-polynomial. Example: you can refute "u is a sum of exponentials" by finding a derivative that breaks it, but you can never confirm the opposite from finite data.
- *No splitting by terms.* Writing the solution as one piece per term of the operator fails: for the wave equation $\partial_t^2 - c^2\partial_x^2$ only polynomial solutions split that way (Section 3).
- *No stable removal of the interaction.* The interaction term cannot be switched off by a penalty or a regularization.
- *No identification with variable coefficients from one solution.*
- *No splitting on domains with a hole* (the logarithm example).
- *No finite product of single-term flows* reproduces non-commuting dynamics.

2. What is open or conditional. These are questions the paper cannot answer: KdV data with nonzero reflection coefficient; shock estimates uniform in time with several shock births; density of the sum of kernels on non-convex domains; exact values of an invariant $\delta_N(K)$; nonlinear asymptotics with derivatives; closedness of the sum of kernels for three or more differential factors. "Open" means unknown; "conditional" means true if an extra lemma holds.

3. How reliable is it. The paper is honest about this: many ingredients are classical (Nullstellensatz, Malgrange–Ehrenpreis, Cole–Hopf); the novelty is the *package* of sharp statements, negative theorems and counterexamples, each with a status label. The reviewers were AI agents, not human referees; fifteen algebraic theorems were also machine-checked in Lean 4, while the analytic parts (distributions, Sobolev spaces, semigroups) were not, because the library is missing. Some citations were checked only on secondary sources, so before relying on a result one should trace it to its primary source.

Take-home message. Read the labels: *proved* can be used, *conditional* needs an extra lemma, *open* is unknown, *false* is a counterexample. The method is exact and has precise limits.

E Primer for readers new to the concepts

This appendix *adds* concrete explanations; nothing in the main text depends on reading it. It covers four notions in this order: ideals and annihilators (§B.1), exponential-polynomials

(§B.2), the Chinese remainder theorem (§B.3), and the companion matrix on a numerical example (§B.4).

B.1 Ideals, annihilators and the gcd, starting from integers

Integers. The multiples of 6 form an *ideal* $(6) = \{\dots, -6, 0, 6, 12, \dots\}$: it is closed under addition and under multiplication by any integer. The ideal generated by 4 and 6 is $(4, 6) = (2)$, since $2 = 6 - 4$ and every combination is even: the ideal "remembers" only the gcd.

Operators. Replace integers by operators and "multiple of 6" by "kills u". For $u = e^x$ every operator $Q(\partial_x)$ with $Q(1) = 0$ kills u, i.e. every multiple of $\partial_x - 1$. So

$$\begin{aligned} \operatorname{Ann}(e^x) &= (\partial_x - 1), \\ \operatorname{Ann}(\sin x) &= (\partial_x^2 + 1), \\ \operatorname{Ann}(xe^x) &= ((\partial_x - 1)^2), \\ \operatorname{Ann}(e^x + e^{2x}) &= ((\partial_x - 1)(\partial_x - 2)). \end{aligned}$$

In one variable every annihilator ideal has a single generator (the *minimal annihilator*), exactly as every ideal of integers is (n). In two variables an ideal can need several generators, as $(\tau - \tau_0, \xi - \xi_0)$ does for $u = e^{\tau_0 t + \xi_0 x}$.

The gcd. $\operatorname{Ann}(u) = G_u J_u$ separates the part of the ideal that is a single polynomial multiple (G_u, "infinite part") from the part J_u that has only finitely many common zeros (J_u, "finite debris"). Example: for $u = f(x+t)$ with f generic (not an exponential-polynomial), $\operatorname{Ann}(u) = (\partial_t - \partial_x)$ by Theorem 5.1, so $G_u = \tau - \xi$ and $J_u = \mathcal{R}$; for $u = e^{x+t}$, $\operatorname{Ann}(u) = (\tau - 1, \xi - 1)$, so $G_u = 1$ and $J_u = \operatorname{Ann}(u)$.

B.2 Exponential-polynomials: examples and a test

Function	In EP?	Reason (derivatives of u span ...)
xe^x	yes	$\operatorname{span}\{xe^x, e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite sum of polynomial $\times$ exponential ($\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$)
e^{-x^2} (Gaussian)	no	$\frac{d^k}{dx^k} e^{-x^2} = (\text{polynomial of degree } k)\, e^{-x^2}$: independent for all k, infinite dimension
$1/(1 + x^2)$	no	derivatives are rational functions with growing denominators: infinite dimension

This is Lemma 2.7 in action: "$u \in$ EP" means "its derivatives span a finite-dimensional space".

B.3 The Chinese remainder theorem, concretely

Integers. The system $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 3$ has the unique solution $x \equiv 5 \pmod 6$: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$. Coprimality of 2 and 3 is what makes the pair of remainders determine x.

One-variable ODE. Let $p(\partial) = (\partial - 1)(\partial - 2)$. Then $\ker p = \ker(\partial - 1) \oplus \ker(\partial - 2) = \operatorname{span}\{e^x, e^{2x}\}$, and the Lagrange-type projectors are explicit: if $u = ae^x + be^{2x}$,

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = \; a\, e^x,$$
$$u_2 = \frac{\partial - 1}{2 - 1}\, u = \; b\, e^{2x}.$$

This is Eq. (11) with ∂_t replaced by ∂.

Two variables. For the wave equation $Q_1 = \tau - \xi$, $Q_2 = \tau + \xi$ are coprime, but the ideal $(Q_1, Q_2) = (\tau, \xi)$ is not the unit ideal: the two lines meet at the origin. The quotient $\mathcal{R}/(\tau, \xi) = \mathbb{C}$ has dimension 1, which is the constant ambiguity $\delta(P) = 1$ of the wave equation. Theorem 3.1 says that this finite overlap is the *only* failure of the integer picture.

B.4 The companion matrix on a numerical example

For $u'' - 3u' + 2u = 0$ the companion matrix is $C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, with characteristic polynomial $\lambda^2 - 3\lambda + 2$ and eigenvalues $1, 2$. The projectors of Eq. (4) are $\Pi_1 = \dfrac{C - 2\mathrm{Id}}{1 - 2}$ and $\Pi_2 = \dfrac{C - \mathrm{Id}}{2 - 1}$, and $e^{xC} = e^x \Pi_1 + e^{2x} \Pi_2$.

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